

LUANGWA COMMUNITY FORESTS PROJECT



Document Prepared By Biocarbon Partners

Project Title	Luangwa Community Forests Project
Version	2
Date of Issue	15-July-2019
Project Location	Zambia
Project Proponent(s)	BioCarbon Partners (primary project proponent), contact name: Dr. Hassan Sachedina, email: info@biocarbonpartners.com , phone: +260 211 290 051
Prepared By	BioCarbon Partners
Validation Body	SCS Global Services, Francis Eaton, Headquarters, 2000 Powell Street, Suite 600, Emeryville, CA 94608, USA +1.510-452-8000, info@SCSglobalservices.com
Project Lifetime	01-May-2015 to 30-April-2045, 30-year lifetime
GHG Accounting Period	01-May-2015 to 30-April-2045; 30-year lifetime
History of CCB Status	First application
Gold Level Criteria	Community – benefits to the poorest of the poor Climate – climate change adaptation through income diversification and habitat corridors Biodiversity – presence of endangered species
Expected Verification Schedule	Sept, 2019

Table of Contents

1 Summary of Project Benefits.....	6
1.1 Unique Project Benefits.....	6
1.2 Standardized Benefit Metrics.....	6
2 General.....	11
2.1 Project Goals, Design and Long-Term Viability.....	11
2.2 Without-project Land Use Scenario and Additionality.....	73
2.3 Stakeholder Engagement.....	75
2.4 Management Capacity.....	93
2.5 Legal Status and Property Rights.....	97
3 Climate.....	109
3.1 Application of Methodology.....	109
3.2 Quantification of GHG Emission Reductions and Removals.....	124
3.3 Monitoring.....	151
3.4 Optional Criterion: Climate Change Adaptation Benefits.....	200
4 Community.....	205
4.1 Without-Project Community Scenario.....	205
4.2 Net Positive Community Impacts.....	208
4.3 Other Stakeholder Impacts.....	209
4.4 Community Impact Monitoring.....	211
4.5 Optional Criterion: Exceptional Community Benefits.....	215
5 Biodiversity.....	220
5.1 Without-Project Biodiversity Scenario.....	220
5.2 Net Positive Biodiversity Impacts.....	222
5.3 Biodiversity Impact Monitoring.....	226
5.4 Optional Criterion: Exceptional Biodiversity Benefits.....	226
6 Appendices.....	228

List of VM0009 3.0 PD requirements applicable to F-U3 baseline type (See VM0009 Appendix I).

PDR#	Category	Page Number
PDR.1	Applicability Conditions	109
PDR.2	Applicability Conditions	109
PDR.3	Applicability Conditions	109
PDR.4	Spatial Project Boundaries	19
PDR.5	Spatial Project Boundaries	106
PDR.6	Temporal Project Boundaries	65
PDR.7	Temporal Project Boundaries	65
PDR.8	Temporal Project Boundaries	64
PDR.9	Temporal Project Boundaries	64
PDR.10	Temporal Project Boundaries	64
PDR.11	Carbon Pools	110
PDR.12	Carbon Pools	110
PDR.13	Carbon Pools	110
PDR.14	Grouped Projects	20
PDR.15	Grouped Projects	40
PDR.16	Grouped Projects	41
PDR.17	Determining the Baseline Scenario	114
PDR.18	Agents and Drivers of Conversion	114
PDR.19	Agents and Drivers of Conversion	115
PDR.20	Agents and Drivers of Conversion	115
PDR.21	Agents and Drivers of Conversion	119
PDR.22	Project Accounting Areas	125
PDR.23	Project Accounting Areas	125
PDR.32	Identifying the Baseline Type Forest	125
PDR.35	Delineation of the Proxy Areas	134
PDR.36	Delineation of the Proxy Areas	134
PDR.37	Delineation of the Proxy Areas	134
PDR.38	Delineation of the Proxy Areas	135
PDR.39	Describing the Baseline Scenario for Selected Carbon Pools	136
PDR.40	Delineation of the Reference Area for Planned and Unplanned Types	136
PDR.41	Delineation of the Reference Area for Planned and Unplanned Types	137
PDR.42	Delineation of the Reference Area for Planned and Unplanned Types	137
PDR.43	Delineation of the Reference Area for Planned and Unplanned Types	137
PDR.44	Delineation of the Reference Area for Planned and Unplanned Types	137
PDR.48	Defining the Reference Period for Unplanned Types	140
PDR.49	Defining the Reference Period for Unplanned Types	140
PDR.50	Defining the Reference Period for Unplanned Types	141
PDR.51	Defining the Reference Period for Unplanned Types	141
PDR.52	Historic Imagery to Parameterize α , β and θ	141
PDR.53	Historic Imagery to Parameterize α , β and θ	141

PDR.54	Historic Imagery to Parameterize α , β and θ	141
PDR.55	Historic Imagery to Parameterize α , β and θ	141
PDR.56	Historic Imagery to Parameterize α , β and θ	141
PDR.57	Sampling Conversion to Parameterize α , β and θ	141
PDR.58	Sampling Conversion to Parameterize α , β and θ	141
PDR.59	Parameterizing α , β and θ	143
PDR.60	Parameterizing α , β and θ	143
PDR.61	Parameterizing α , β and θ	143
PDR.62	Parameterizing α , β and θ	143
PDR.63	Minimizing Uncertainty in Parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$	143
PDR.64	Minimizing Uncertainty in Parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$	144
PDR.65	Minimizing Uncertainty in Parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$	144
PDR.66	Estimating Uncertainty in Parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$	144
PDR.67	Estimating Uncertainty in Parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$	144
PDR.74	Determining x_0	144
PDR.75	Determining x_0	144
PDR.79	Parameterizing q	145
PDR.80	Parameterizing q	145
PDR.81	Parameterizing r_U	145
PDR.82	Parameterizing r_U	145
PDR.83	Parameterizing r_U	145
PDR.84	Empirically Estimating λ_{SOC}	146
PDR.85	Empirically Estimating λ_{SOC}	146
PDR.86	Empirically Estimating λ_{SOC}	146
PDR.87	Empirically Estimating λ_{SOC}	146
PDR.88	Empirically Estimating λ_{SOC}	146
PDR.89	Literature Estimates for λ_{SOC}	146
PDR.90	Literature Estimates for λ_{SOC}	146
PDR.91	Literature Estimates for λ_{SOC}	146
PDR.92	Literature Estimates for λ_{SOC}	146
PDR.93	Baseline Reevaluation	146
PDR.94	Baseline Reevaluation	146
PDR.95	Baseline Reevaluation	146
PDR.96	Reevaluation of the Reference Area and Period	146
PDR.97	Re-parameterization of α , β and θ	146
PDR.98	Re-parameterization of α , β and θ	146
PDR.99	Demonstration of Project Additionality	120
PDR.100	Demonstration of Project Additionality	121
PDR.101	Demonstration of Project Additionality	121
PDR.102	Demonstration of Project Additionality	122
PDR.103	Demonstration of Project Additionality	122
PDR.104	Leakage Mitigation Strategies	49
PDR.105	Delineation of the Activity-Shifting Leakage Area	147
PDR.106	Delineation of the Activity-Shifting Leakage Area	147

PDR.107	Delineation of the Activity-Shifting Leakage Area	147
PDR.108	Delineation of the Activity-Shifting Leakage Area	147
PDR.109	Delineation of the Activity-Shifting Leakage Area	147
PDR.110	Determining the Market Discount Factor	148
PDR.111	Determining the Market Discount Factor	148
PDR.112	Determining the Market Discount Factor	148
PDR.113	Delineation of the Market Leakage Area	148
PDR.114	Delineation of the Market Leakage Area	148
PDR.115	Delineation of the Market Leakage Area	148
PDR.116	Delineation of the Market Leakage Area	148
PDR.117	Delineation of the Market Leakage Area	148
PDR.118	Ex-Ante Estimation of NERs	150
PDR.119	Ex-Ante Estimation of NERs	150
PDR.120	Ex-Ante Estimation of NERs	150
PDR.121	Data and Parameters Available at Validation	151
PDR.122	Description of the Monitoring Plan	194
PDR.123	Description of the Monitoring Plan	197
PDR.124	Description of the Monitoring Plan	197

1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or Impact Estimated by the End of Project Lifetime	Section Reference
1) Conserving a large part of the catchment of one of the longest undammed rivers in Southern Africa, the Luangwa	2
2) Conserving Zambia's largest hippopotamus population	5
3) Conserving a near continuous wildlife corridor between five national parks	2,5
4) Benefiting over 100,000 local community members	2,4

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
GHG emission reductions or removals	Net estimated emission removals in the project area, measured against the without-project scenario	NA	
	Net estimated emission reductions in the project area, measured against the without-project scenario	83,598,204	3

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
Forest ¹ cover	For REDD ² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	361,060	3
	For ARR ³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario	NA	
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices are expected to occurred as a result of project activities, measured against the without-project scenario	NA	
	Number of hectares of non-forest land in which improved land management practices are expected to occurred as a result of project activities, measured against the without-project scenario	3219	2,4
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	1500	2,4

¹Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

²Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood and fuelwood (*VCS Program Definitions*)

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	585 representing 40 percent of the target, based on current participation of female in training	2,4
Employment	Total number of people expected to be employed in project activities, ⁵ expressed as number of full-time employees ⁶	86	2,4
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	13	2,4
Livelihoods	Total number of people expected to have improved livelihoods ⁷ or income generated as a result of project activities	1500	2,4
	Number of women expected to have improved livelihoods or income generated as a result of project activities	585, representing current female participation of 40 percent	2,4
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	1468	2,4

⁵Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	754	2,4
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	488	2,4
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	250	2,4
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	514	2,4
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	264	2,4
Well-being	Total number of community members whose well-being ⁸ is expected to improve as a result of project activities	165,955	2,4
	Number of women whose well-being is expected to improve as a result of project activities	89,308	2,4

⁸Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation, ⁹ measured against the without-project scenario	943,676	2,5
	Expected number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities, ¹¹ measured against the without-project scenario	1 African Wild Dog (<i>Lyceon pictus</i>)	5

⁹Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰Per IUCN's Red List of Threatened Species

¹¹In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 GENERAL

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (G1.2)

The Luangwa Community Forests Project is a large scale grouped REDD+ project implemented in Eastern and Lusaka Province, Zambia with an initial project area of 943,676 ha (Figure 1). At validation, the project is implemented on communal land in 12 chiefdoms falling within Game Management Areas (GMA) and two private ranches. Additional project instances may be added in the future. Implementation is in partnership with the traditional authorities and the government of the Republic of Zambia.

The project will generate emissions reductions through avoided deforestation, using the following mitigation activities:

- Direct conservation support - Forest monitoring will be done using remote sensing, aerial and ground monitoring. Encroachment prevention will be accomplished by training, funding, and helping to manage community scouts.
- Engagement and capacity building with key Government and community stakeholders.
- Performance-based payments to community stakeholders delivered through local institutions.
- Promoting alternative livelihood activities including: conservation agriculture, non-timber forest product livelihoods, and sustainable enterprise development.

The baseline scenario is unplanned deforestation driven primarily by the expansion of subsistence agriculture. Furthermore, communities remain in an impoverished state and biodiversity declines due to habitat loss and increased poaching.

The project's community objective is poverty alleviation for at least 10,000 households, specifically targeting vulnerable households and the poorest of the poor. The project also aims to promote infrastructure development and water provision. The biodiversity objectives is maintaining a massive wildlife corridor between five national parks in the catchment of Zambia's 4th largest river system. Conserving and maintaining vulnerable and endangered species through habitat protection and reduction in poaching. The climate objectives are to avoid deforestation in the project area and assist communities and biodiversity with climate change adaptation benefits through income diversification, improved farming techniques, crop diversification and maintenance of habitat corridors.

The estimated emissions reductions for the 30 year project span is 83,598,204 tCO₂e at an average 2,985,650 tCO₂e/yr.

2.1.2 Project Scale

Project Scale	
Project	
Large project	x

2.1.3 Project Proponent (G1.1)

Organization name	BioCarbon Partners
Contact person	Dr. Hassan Sachedina
Title	Chief Executive Officer
Address	BioCarbon Partners, 5th Floor, Barkly Wharf, Le Caudan Waterfront, Port Louis, Republic of Mauritius
Telephone	+260 211 290 051
Email	info@biocarbonpartners.com

Organization name	Luembe Community Resources Board
Contact person	Mr. Davison Mwanza
Title	Board Chairperson
Address	Luembe Community Resources Board, Luembe Chiefdom - Nyimba
Telephone	Cell: +260976555356
Email	

Organization name	Nyalugwe Community Resources Board
Contact person	Mr. Yulu Joseph Njobvu
Title	Board Chairperson
Address	Nyalugwe Community Resources Board, Nyalugwe Chiefdom - Nyimba
Telephone	Cell: +260976345724
Email	

Organization name	Mpanshya Community Resources Board
Contact person	Mr. Chrispin Chikopa
Title	Board Chairperson
Address	Mpanshya Community Resources Board, Mpanshya Chiefdom - Rufunsa
Telephone	Cell: +260979029568
Email	

Organization name	Shikabeta Community Resources Board
Contact person	Mr. Mathews Nyangu
Title	Board Chairperson
Address	Shikabeta Community Resources Board, Shikabeta Chiefdom - Rufunsa
Telephone	Cell: +260971938928
Email	

Organization name	Sandwe Community Resources Board
Contact person	Mr. George Tembo
Title	Board Chairperson
Address	Sandwe Community Resources Board, Sandwe Chiefdom - Petauke
Telephone	Cell: +260973414944
Email	

Organization name	Msoro Community Resources Board
Contact person	Mr. John Banda
Title	Board Chairperson
Address	Msoro Community Resources Board, Msoro Chiefdom - Mambwe
Telephone	Cell: +260974762844
Email	

Organization name	Malama Community Resources Board
Contact person	Mr. Edward Tembo
Title	Board Chairperson
Address	Malama Community Resources Board, Malama Chiefdom - Mambwe
Telephone	Cell: +260965697694
Email	

Organization name	Mnkhanya Community Resources Board
Contact person	Mr. Augustine Mulwangi
Title	Board Chairperson
Address	Mnkhanya Community Resources Board, Mnkhanya Chiefdom - Mambwe
Telephone	Cell: +260968037096
Email	

Organization name	Nsefu Community Resources Board
Contact person	Mr. Lovedale Lungu
Title	Board Chairperson
Address	Nsefu Community Resources Board, Nsefu Chiefdom - Mambwe
Telephone	Cell: +26097113087
Email	

Organization name	Jumbe Community Resources Board
Contact person	Mr. Musamba Phiri
Title	Board Chairperson
Address	Jumbe Community Resources Board, Jumbe Chiefdom - Mambwe
Telephone	Cell: +260976735005
Email	

Organization name	Mwanya Community Resources Board
Contact person	Mr. Ringstone Mwandila
Title	Board Chairperson
Address	Mwanya Community Resources Board, Mwanya Chiefdom - Lundazi
Telephone	Cell: +260978579302
Email	

Organization name	Chitungulu Community Resources Board
Contact person	Mr. Alex Banda
Title	Board Chairperson
Address	Chitungulu Community Resources Board, Chitungulu Chiefdom - Lundazi
Telephone	Cell: +260977672636
Email	

2.1.4 Other Entities Involved in the Project

Organization name	Munyamadzi Game Ranch
Contact person	Thor Kirchner
Title	General Manager
Address	Munyamadzi Game Ranch, Nyimba, Zambia
Telephone	+260 97 8157643
Email	munyamadzi@live.com

Organization name	United States Agency for International Development (USAID)
Contact person	Catherine Tembo
Title	Natural Resources Specialist
Address	Embassy of the United States of America Subdivision 694/Stand 100 Ibex Hill Road, PO Box 320373 Lusaka, Zambia 10101
Telephone	T +260 211 357 000
Email	ctembo@usaid.gov

Organization name	Forestry Department (FD)
Contact person	Mr. Ignatius N. Makumba
Title	Director
Address	Ministry of Lands and Natural Resources, Forestry Department. Kwacha House, Lusaka
Telephone	+260 211 226 131
Email	Inmakumba@gmail.com

Organization name	Department of National Parks and Wildlife (DNPW)
Contact person	Mr. Paul Zyambo
Title	Director General
Address	Ministry of Tourism and Arts, DNPW Chilanga, Lusaka.
Telephone	+260978290175
Email	paulzya@yahoo.com

Organization name	Ministry of Chiefs and Traditional Affairs Provincial Office
Contact person	Mr. Pythias Kakoma
Title	Senior Chief and Traditional Affairs Officer
Address	Provincial Office Chipata
Telephone	+260979314181 or 0966314181
Email	Pythiaskakoma@gmail.com

Organization name	Ministry of Chiefs and Traditional Affairs Provincial Office
Contact person	Mr. Pythias Kakoma
Title	Senior Chief and Traditional Affairs Officer
Address	Provincial Office Chipata
Telephone	+260979314181 or 0966314181
Email	Pythiaskakoma@gmail.com

Organization name	Pia Manzi Wildlife Limited and Luansemfwa Safaris Limited
Contact person	Diego Casilli
Title	Director
Address	P.O. Box33711, Lusaka, Zambia
Telephone	+260 211 258 057
Email	info@napoliproperty.com

Organization name	Bee Sweet
Contact person	Nathan Enright
Title	Chief Executive Officer
Address	Plot No.1 Jacopo Rd Luanshya turn off ,P.O. Box 70839,Ndola, Zambia
Telephone	+260961690322
Email	nate.enright@gmail.com

Organization name	Ministry of Agriculture
Contact person	Kennedy Kaputo
Title	Mambwe District Agriculture Coordinator
Address	P.O. Box 19, Mfuwe, Mambwe, Eastern Province, Zambia
Telephone	+260977969411
Email	kaputoken@yahoo.com

Organization name	Ministry of Agriculture
Contact person	Philimon Lungu
Title	Lundazi District Senior Agriculture Coordinator
Address	P.O. Box 530016, Lundazi, Eastern Province, Zambia
Telephone	+260974515013
Email	philimonl@yahoo.co.uk

Organization name	Ministry of Agriculture
Contact person	Moses Katundu
Title	Nyimba District Senior Agriculture Coordinator
Address	P.O. Box 570026, Nyimba, Eastern Province, Zambia
Telephone	+2601472500
Email	

Organization name	Ministry of Agriculture
Contact person	Stanley Mvula Njobvu
Title	District Agriculture Coordinator
Address	Rufunsa, Lusaka Province
Telephone	+260972967670
Email	stanmavula@gmail.com

2.1.5 Physical Parameters (G1.3)

The 14 first activity instances lie in the Eastern and Lusaka Provinces of Zambia and fall into the Luangwa Zambezi River ecological zone and the Central Southern and Eastern Plateau Zone. In addition all the activity instances are situated in the Game Management areas (GMA) within the Chiefdom areas. The project sites in Eastern Province border the South Luangwa National Park and Luangwa River in the north-western direction. The project area consist of two sites is Lusaka Province in Rufunsa District, one in each side of the great east road in Shikabeta and Mpanshya chiefdoms. The others nine project sites lies in eastern province of Zambia in Mambwe, Lundazi and Nyimba Districts. Project sites in Mambwe Districts are in Mnkhangya, Jumbe, Nsefu, Malama, Msoro. In Lundazi District there is one project site in Mwanya Chiefdom. The project area in Mwanya is bordered by Lukusuzi National Park and Luambe National Park area. There are two sites in Nyimba District one each in Nyalugwe and Luembe chiefdom.

For each project area, a detailed map and description are provided in Appendix A in the Community Forest Management Agreements. Figure 1 shows the location map of the project area. The geographic boundaries of the project area is provided in a separate kml file (Appendix I). Table 1 and Table 2 shows the chiefdom and ranch areas with project size.

PDR.4

For the VM0009 requirement PDR.4, please see Figure 1, Figure 2, Figure 3, and Figure 6

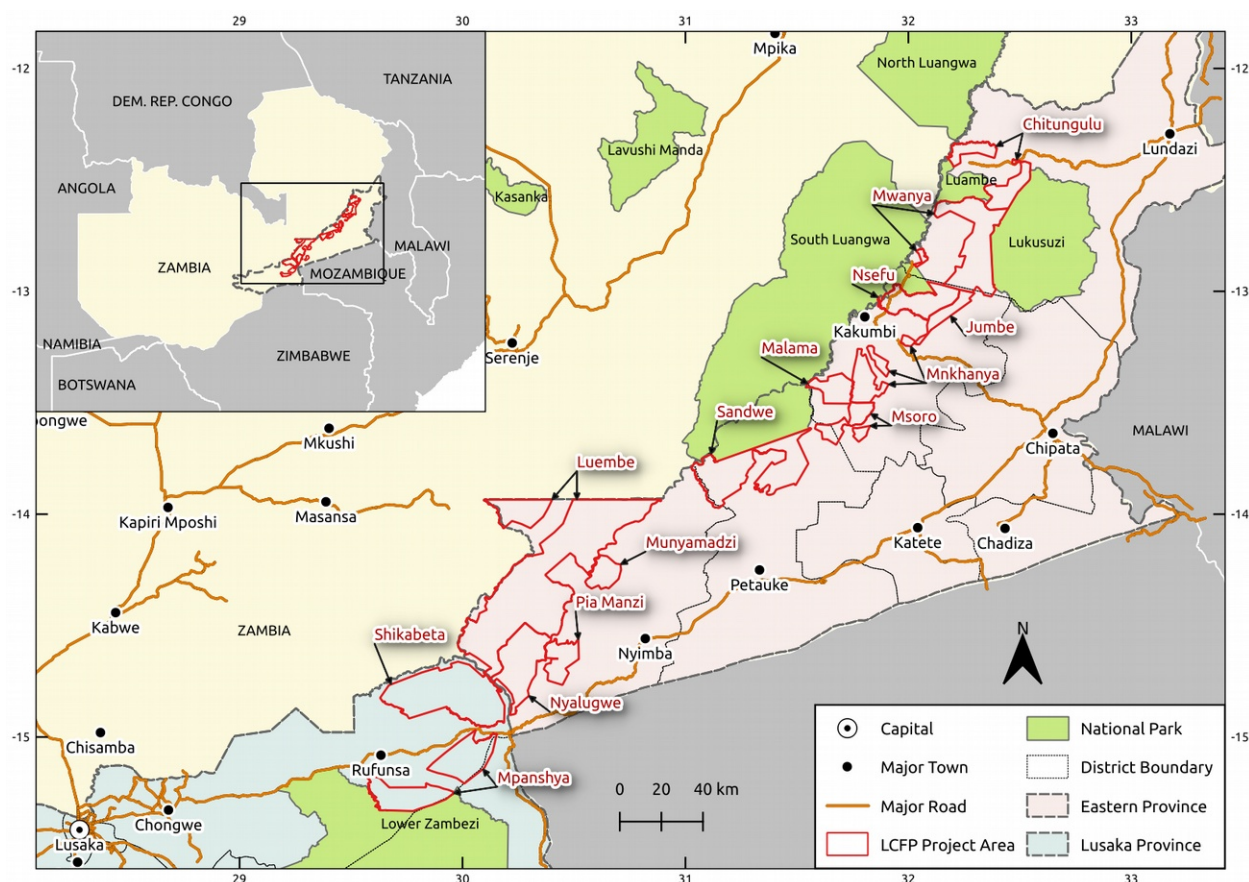


Figure 1: Project Area Location, Eastern and Lusaka Provinces, Zambia

PDR.14

Table 1. Summary of Project Area - REDD+ Project Zones under communal tenure, first activity instances.

Community Resource Board	Game Management Area	Chiefdom	Size (ha)	Forest Area (ha)
Mpanshya	Rufunsa	Mpanshya	84,530	82,183
Shikabeta	Luano	Shikabeta	118,102	111,968
Mwanya	Lower Lumimba	Mwanya	81,668	59,030
Luembe	West Petauke	Luembe	288,726	254,582

Nyalugwe	West Petauke	Nyalugwe	60,893	58,809
Msoro	Lower Lupande	Msoro	24,688	23,523
Malama	Lower Lupande	Malama	21,577	17,220
Mnkhanya	Upper Lupande	Mnkhanya	33,623	29,081
Jumbe	Upper Lupande	Jumbe	15,443	11,066
Nsefu	Upper Lupande	Nsefu	39,801	25,341
Chitungulu	Lumimba	Chitungulu	38,764	28,681
Sandwe	Sandwe	Sandwe	92,362	85,767
Total			900,177	787,251

Table 2. Summary of Project Area - REDD+ Project Zones on Private Land, first activity instances.

Private Game Ranch	Chiefdom	Size (ha)	Forest Area (ha)
Munyamadzi	Luembe	18,671	16,449
Pia Manzi	Nyalugwe	24,798	22,393
Total		43,469	38,842

Table 3. Summary of physical parameters.

Physical parameter	Description
Elevation	The elevation (Figure 2) in the project area ranges from 371 m to 1453 m above sea level. The median elevation is 669 m above sea level.
Topography	Project area slope (Figure 3) ranges from 0 to 182 percent with a median slope of 8%.
Geology	The Karroo system in the valley and escarpment and the Basement complex on the plateau with isolated patches of granite.

Hydrology	The Luangwa river runs through the project area (Figure 2), it is a major tributary of the Zambezi and is the longest undammed river in Southern Africa. The project area also borders the Lukusashi and Lunsemfwa rivers which drain into the Luangwa.
Agro-ecological zones	
Eco-region	The project area falls within the Miombo eco-region which lies within the Zambezian regional center of endemism (White 1983).
Soil	According to FAO, the dominate soil types in the project area (Figure 4) are Chromic Luvisols and Lithosols, which are both considered High Activity Clay Soils under the IPCC soil classification system.
Rainfall	Annual precipitation in the project area (Figure 5) ranges from 770 mm to 1001 mm with a median of 893 mm per year.
Temperature	Max ave 28-30°C Min ave 14-15°C
Vegetation	According to the Zambian natural vegetation classification system developed by Fanshawe, 1971, most the project area is within Miombo and Mopane woodlands with small areas of Munga woodland and Riparian forest (Figure 6). The total area of forest in the project area that is at least 10 years old and meets the Zambian definition of forest is 823,942 ha (Figure 7). A breakdown of the amount of project forest controlled by each land-holder is shown in Table 1 and Table 2.

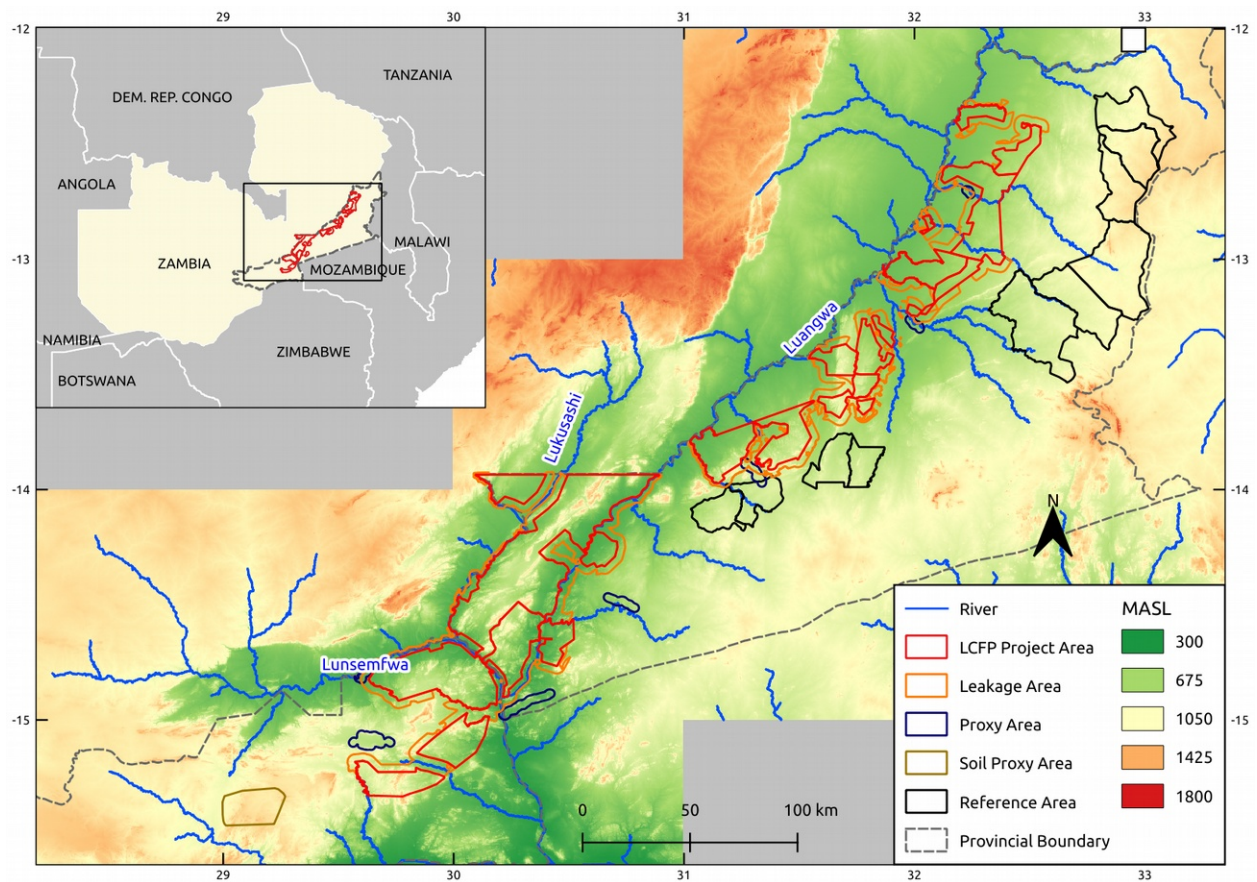


Figure 2: Elevation

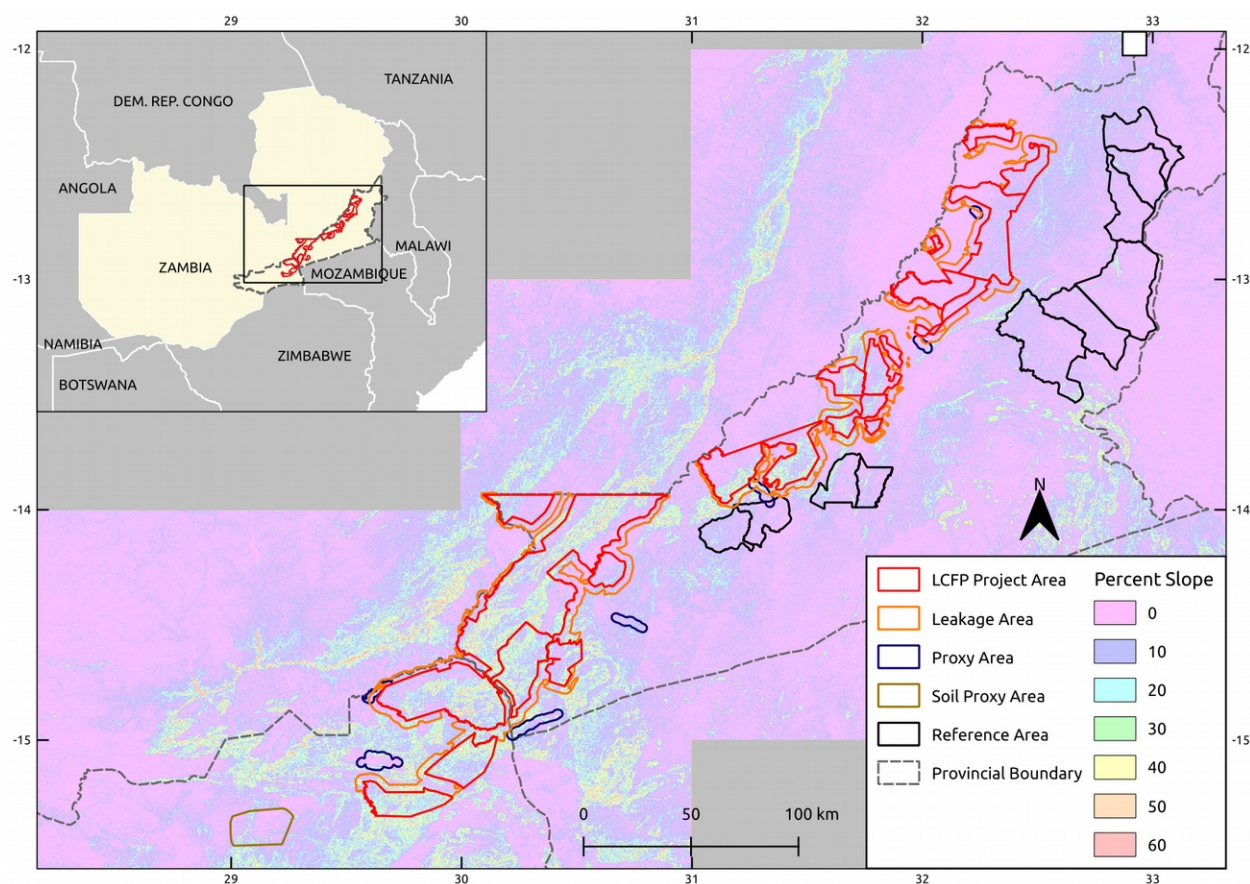


Figure 3: Slope

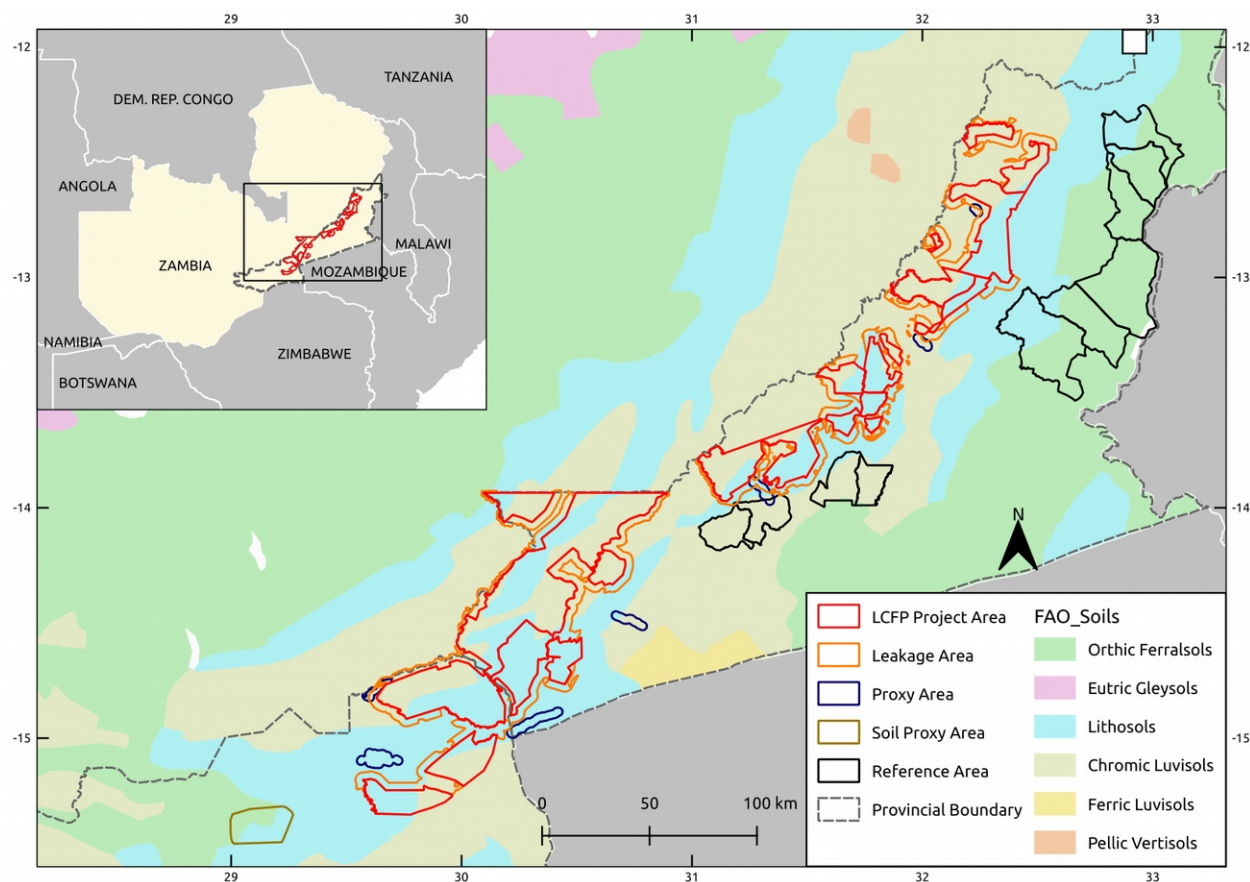


Figure 4: Dominant FAO Soil Types

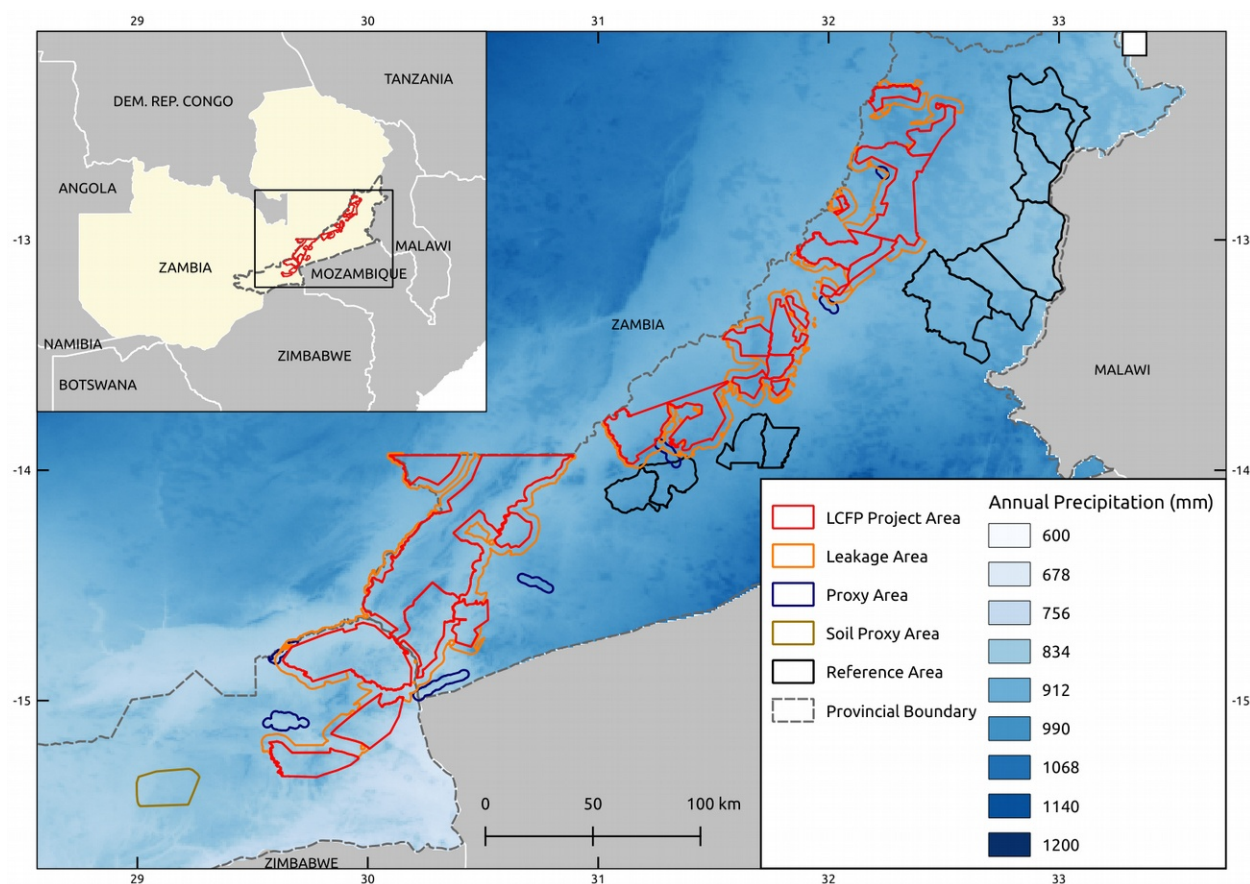


Figure 5: Average Annual Precipitation

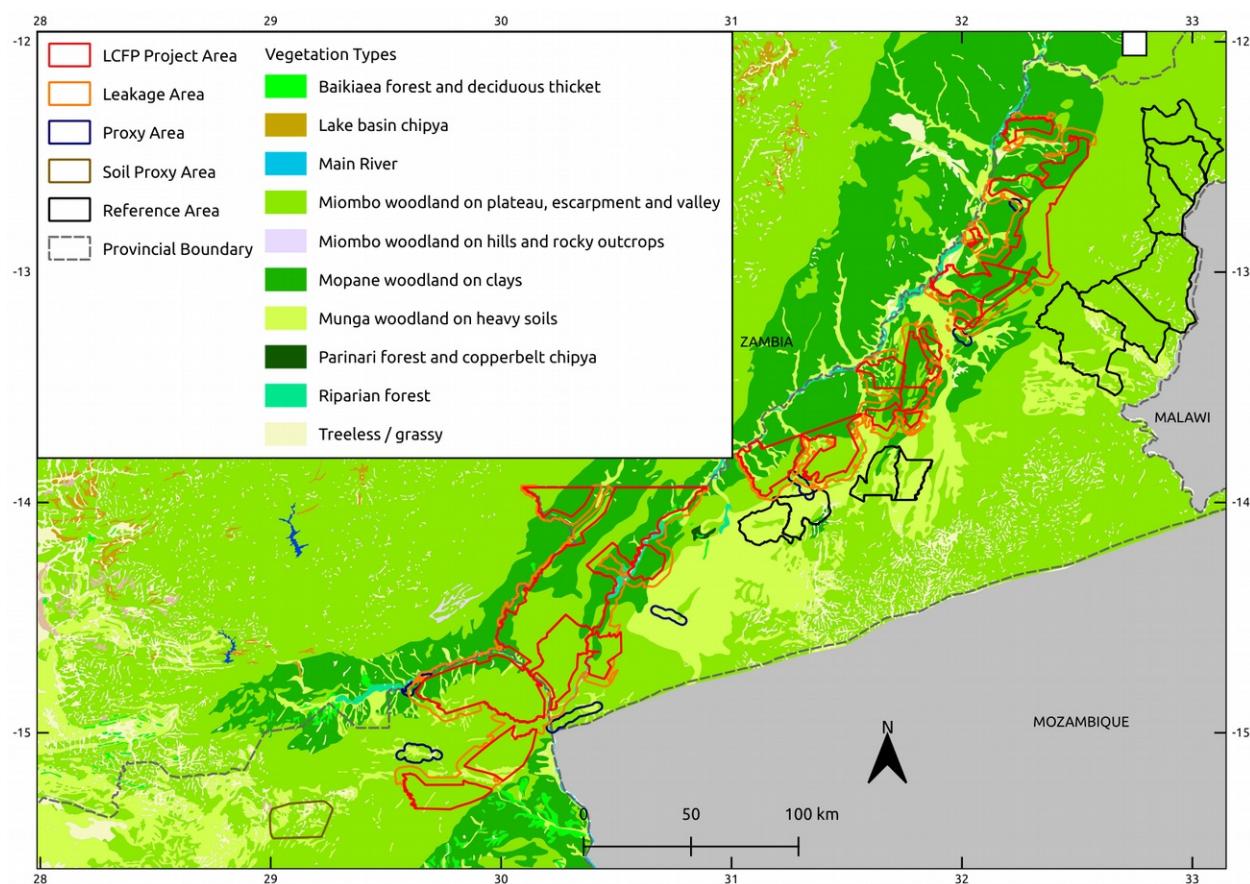


Figure 6: Natural Vegetation Types (see Fanshawe, 1971 in Appendix X).

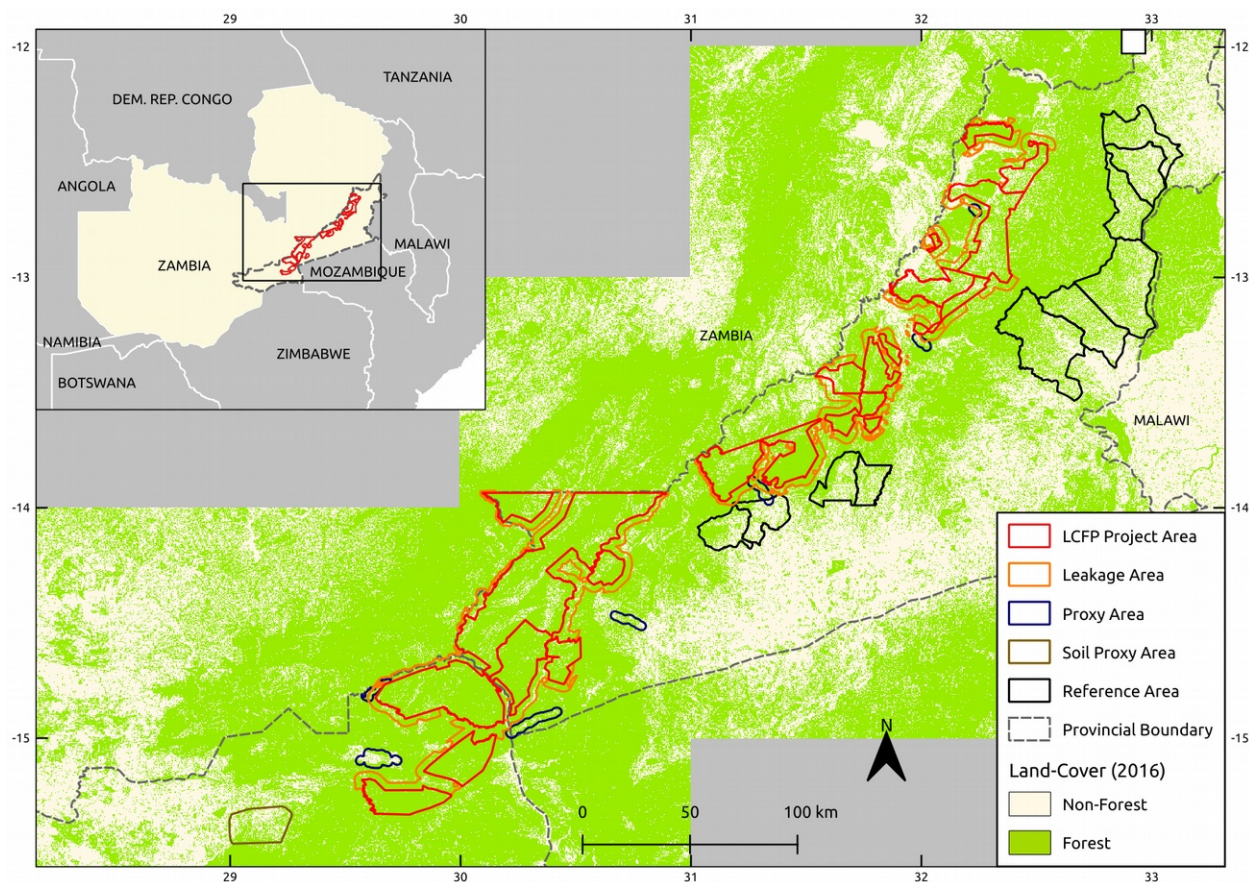


Figure 7: Land-Cover (2015)

2.1.6 Social Parameters (G1.3)

The LCFP is working with Government partners and community stakeholders from 12 Chiefdoms, located in 2 Provinces, in 4 Districts, in Zambia and two private game ranches (Table 1, Table 2). The LCFP is working with Community Resources Boards (CRBs) and Village Action Groups (VAGs) as community implementing partner institutions, to implement program activities. CRBs are Chiefdom-level institutions, and VAGs are village-level institutions, that represent groups of villages in specific geographic locations. The LCFP is working with a total of 12 implementing partner CRBs, and 62 VAGs (Table 1).

There are a total of 165 955 community stakeholders registered with the project, living in 28, 268 households, in 12 chiefdoms. 76, 648 of these are male, and 89,308 female. Community stakeholder information was collected during a Household Registry that took place under the LCFP in 2017.

Sources of information:

The LCFP has obtained information about local communities and patterns of forest use from the following sources:

- Approximately 3 years of community stakeholder engagement, including documented processes of Participatory Rural Appraisal (PRA) / Participatory Learning and Action (PLA), with the aim of understanding local community dynamics in terms of socio-economic indicators, development priorities, and forest use patterns;
- Specific studies from the LCFP, including a Baseline Survey for stakeholders in Eastern Province conducted in 2015 (Appendix F) and similar studies conducted in Rufunsa district for the LZRP in 2012 and 2017.
- Local studies of socio-economic conditions, land use, and natural resources, such as the Rural Agricultural Livelihoods Survey (RALS) and Sustainable Intensification on Landscapes and Livelihoods (SILL) study, that were conducted by separate, external entities in 2015;
- Specific studies of Agents and Drivers conducted by Unique (Forestry and Land Use), for REDD+ projects developed by both COMACO and BCP, in Zambia (Appendix F);
- Census information available through the Central Statistics Office for the Government of Zambia;
- Forest maps and biodiversity data collected by BCP, as well as Zambia's Forest Reference Emission Level (FREL) submitted to the UNFCCC.

Main settlements and land use:

Major settlements within the project area for the LCFP are concentrated around Mfuwe town, in Mambwe District, and Nyimba town, in Nyimba District. Major settlements and demand for resources are also driven by proximity to Chipata, in Eastern Province, and the capital city of Lusaka, in Lusaka Province, as larger urban areas. Chinyunyu and Rufunsa towns (located in Rufunsa District, in Lusaka Province), as well as Lundazi town (located in Lundazi District of Eastern Province) are also larger settlements. In general, settlements and farming activities are more common near to existing urban areas and settlements; these areas tend to be characterized by easier access (due to more roads, and shorter distances), more developed infrastructure and services (such as markets, shops, schools, clinics and/or government buildings), and less human-wildlife conflict, in areas that are farther from protected areas. Areas around Chipata and Lusaka are perceived to have more economic activities, fueled by existing businesses, as well as government or donor-funded programs.

Trophy hunting is practiced by professional hunters in certain sections of the Game Management Areas. The area around Mfuwe is also characterized by an established tourism industry, which has caused some in-migration into the area, as individuals move to the area, seeking employment in the tourism industry.

Socio-Economic and forest use information:

The stakeholder community for the LCFP is predominantly a rural population, characterized by the following dynamics:

- High population growth rates;
- High levels of poverty;
- Strong dependency on subsistence agriculture as a livelihood activity;
- Strong dependency on collection / use of forest resources, for consumption or as sources of income, including: timber, wood poles, fuel wood, and charcoal production;
- Strong beliefs in the power of traditional leadership structures (including: Chiefs, Headmen, Indunas), including to regulate forest resources, resolve conflicts, and have authority over decision-making pertaining to land-use.

In general, within the project area for the LCFP, settlements and agricultural activities are more extensive in areas nearer to larger settlements / urban areas, and near to main roads. In more rural areas, settlements and agricultural activities take place at a smaller scale in areas that are more remote, closer to the boundaries of protected areas (such as the South Luangwa National Park). These areas are often characterized by poor access (fewer, poor quality roads), less development (less or more difficult access to key social services, such as schools, clinics, water facilities, or other services / programs), and a remote nature that sometimes includes human-wildlife conflict, in areas with higher wildlife populations.

In areas with poor roads infrastructure, access can become very difficult / limited during the rainy season (from approximately November – April), which significantly limits development. Poor access and distance from markets tends to make agriculture less profitable in these areas, meaning that these areas are often characterized by higher dependencies on subsistence agriculture (as opposed to commercial agriculture), and more involvement in activities that can be undertaken in areas of intact forest, such as the extraction of forest resources. Remote, rural areas tend to be characterized by high levels of poverty.

In recent years, in-migration into more rural areas with intact forests has been driven by households seeking available land, for household expansion (due to growing populations), and agricultural purposes. The profitability of activities such as charcoal production, logging and/or timber extraction has also driven some individuals into these areas – although in general, these activities tend to be undertaken by members of households living nearby, or, by individual migrants. Charcoal production is more prevalent in areas that are characterized by the dual dynamic of available forest resources, combined with proximity to urban areas, where demand is high. Although the majority of households use wood-based fuel for cooking, charcoal production for rural household consumption takes place on a small scale, only, and is not identified as a major driver of deforestation in rural areas; it is the commercial charcoal trade, driven

by easy access to urban areas with high demand, that has become a rising threat to local forests, in recent years. Energy and economic policies have also contributed to the growing demand for charcoal, in recent years. In general, there is a strong link between agricultural clearing and charcoal production, whereby areas that are cleared for charcoal production are often quickly followed by agricultural expansion, as farmers take advantage of areas that have already been cleared. However, it should be noted that charcoal production is technically regulated by the government, and unless it takes place with a valid permit, charcoal production is considered to be illegal.

In Rufunsa District, the demand for charcoal is especially high, due to its proximity to Zambia's capital city of Lusaka. Studies show that a significant amount of charcoal produced in Rufunsa, and other areas, is produced illegally – in the absence of necessary permits. In Eastern Province, agricultural activities take place on a large scale, especially near to Chipata, where there are a significant amount of commercial agricultural businesses, and numerous donor-funded programs provide support to local farmers. Charcoal and other forest resources are also in demand, not only for Chipata, but also for consumption in nearby Malawi, where forest resources have been significantly depleted. The Luangwa River also flows through Eastern Province; with a significant amount of the project area for the LCFP falling within the Game Management Areas (GMAs) adjacent to the South Luangwa National Park. Within and near to these protected areas, local tour operators and Game Ranches provide employment opportunities and contribute to local economies. It is worth noting that both the photographic and hunting safari industries are dependent upon sustainable management of forest resources, and as such, many of these operators have expressed willingness to support the REDD+ activities of the LCFP, which are viewed to be compatible.

From an independent study to quantify the drivers and agents of deforestation (Appendix K), the main driver variables explaining the quantity and location of deforestation are population density and accessibility to forests, i.e., distance to roads, distance to settlements, and slope. With increase in population and expansion of road infrastructures, both population density and accessibility can be logically expected to increase in the future; while a key underlying cause of deforestation in the region, namely: extreme poverty and high dependency on forest and other natural resources is not expected to improve significantly anytime soon.

Poverty and lack of alternative livelihoods in rural areas cause rural population to depend principally on smallholder agriculture for food and income, and on firewood for cooking and heating. The agriculture practised involves poor farming practices such as burning of cropland and crop residues, and continual cultivation of fertility-mining crops. In urban areas, poverty and low incomes result in high dependence on charcoal for cooking and heating. With increasing population both in rural and urban areas, these situations translate into increased demand for agricultural land, firewood, and charcoal – resulting into forest exploitation and ultimately deforestation. These chains of events are illustrated in Figure 8 below. They are relevant now and for the foreseeable future.

The above deforestation occurs in a weak regulatory and institutional environment: The institutions responsible for protecting, managing, and monitoring forests such as FD, CRBs, and traditional authorities are ineffective due to their meagre staffing, technical/knowledge, and financial capacities. In addition, communities – where most of the forests are found – do not get any part of the revenues obtained from legal harvests of forest products. This breeds indignation, and demoralises communities from dutifully participating in forest conservation. The entire regulatory and institutional environment does not, therefore, effectively protect forest from exploitation for agricultural expansion, firewood extraction or charcoal production.

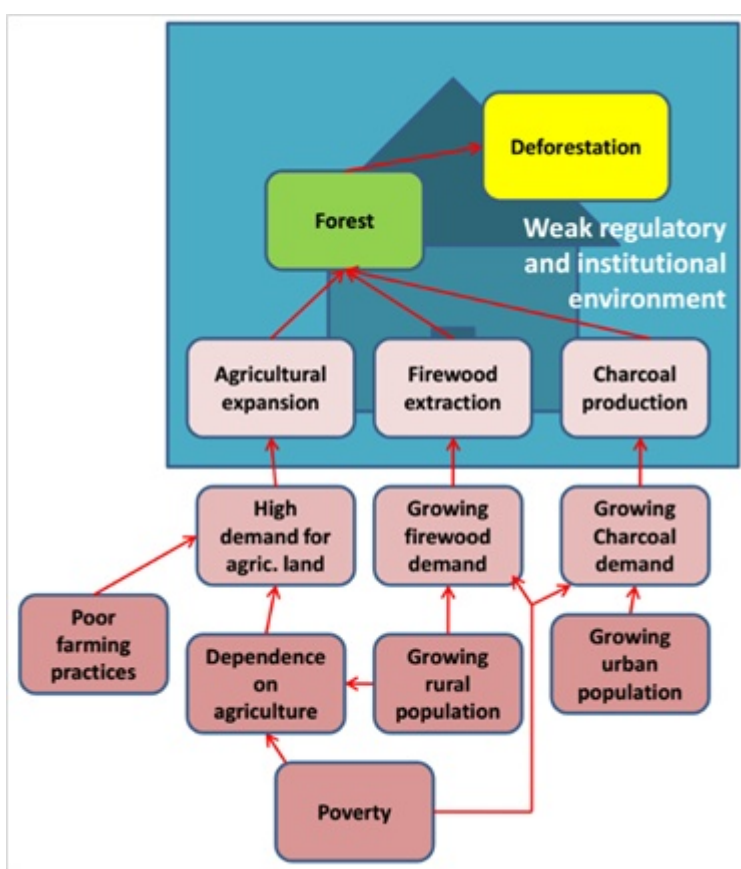


Figure 8. Chains of events leading to deforestation.

An initial baseline survey of stakeholder households for the LCFP in Eastern Province was completed in 2015, by an external firm that was contracted directly by USAID (Appendix F). The survey report compiles results from a representative sample of over 4,300 households surveyed in Mambwe and Nyimba

Districts. Key socio-economic statistics found by this survey include: findings of this baseline survey include:

The majority of households are poor. Households reported earning most of their living from small-scale agriculture and having limited income from other sources, including from forest-related activities. 66% (2,857) of households receive 2,000 ZMK (approximately \$200 USD) or less per year from off-farm income or remittances, and 18% (777) reported earning between 2,000 and 5,000 ZMK.

Households are highly dependent on forest resources. 84% (3,468 respondents) depend on the forest for consumption, and 19% (847) depend on the forest for income. The most commonly reported forest products for income are: charcoal, mushrooms, and fuel wood. The most commonly reported forest products for consumption are: fuel wood, wood for poles, fibers, followed closely by mushrooms.

The majority of households are local, but, there are high levels of local population movement and in-migration: 100% of surveyed respondents were Zambians – reflecting low levels of international migration into the area. Many (45%) household respondents reported they were born in the village where they currently live. Of those not from their village of residence, the results show a spread in point of origin: 26% are from another area in the same chiefdom, 29% are from another chiefdom in the same (Eastern) province, and 28% are from another province. The data showed that stakeholder villages tend to be built around a few family groups: 74% of households reported they were related to the headman. A smaller percentage (12%) reported they were related to the chief, whose authority extends over a significantly larger area than a village.

The **average reported household size** was 5.2.

The population reported minimal education: The average reported number of years of education in the study sample is 6.9; in 18% of households, adults reported no formal schooling. 17% of households reported that at least one member had completed Grade 7.

In terms of **ethnicity**: the majority of respondents were Nsenga (41%) followed by Kunda (20%), Chewa (16%) and Bemba (13%).

Additionally, other key findings include:

Recognition of climate change and deforestation as local community development problems:

- Over half (54%) of surveyed community respondents reported worsening forest conditions over the past 3 years; 25% ranked forest condition as degraded, and over half of respondents (55%) noted that the capacity of the forest to provide resources for the community has worsened over the past 3 years.
- Climate change is perceived to be a large threat to community health and development, although the link between forest management and climate change was not reportedly well understood.

79% headman surveyed ranked changes in rainfall as one of the top five problems facing their villages, and 21% ranked rainfall changes as their biggest problem – making poor rainfall the second-most commonly reported problem, after poor crop yields (and these problems are clearly linked). 31% of headmen believe climatic temperature changes to be one of the top five problems facing their community. 51% (2,229) of household respondents reported changes in rainfall patterns or temperature as the largest problems facing development in their villages; 19% said that deforestation or degradation is the biggest constraint to community development.

- Tenure and governance:
- Most people reported feeling secure in their rights to forests, although a significant minority report perceived tenure insecurity. Heads of households, wives, and headpersons appear relatively confident that their village's forested land will not be encroached upon by any party in the next three years. Over half of all headmen believe that encroaching upon their customary forestland in the next three years by investors, government, the chief, or elites, is "impossible."
- Conflicts about forest access or resources are currently extremely rare: 98% (4,246) of households reported no disputes over forest access or use in the past 12 months; 77% reported satisfaction with local leaders' abilities resolve conflicts and govern forest resources, as well as rules that govern forest access and use.

A slight majority of the sample (52%) reported that household land is passed down from the husband's family, whereas 32% of respondents reported that their land is passed down through the wife's family.

Gender differences:

Female respondents placed less value on forests and reported less satisfaction with forest rules and leadership. They reported they are less likely to attend forest meetings or participate in forest management groups/committees. Women were consistently reported to hold less decision-making power with regards to land management.

Please note that some of these key findings have been quoted directly from the Impact Evaluation Baseline Survey Report (Appendix F).

These survey findings for stakeholders from Eastern Province are complemented by the surveys of stakeholders to the LZRP in Rufunsa Province, which is considered to be representative for stakeholders from Rufunsa District (as the survey was conducted in the adjacent Bunda-Bunda Chiefdom). The Rufunsa surveys show similar dynamics of high levels of poverty and dependence on forest resources. Additionally, it is worth noting that the follow-up survey, conducted by an external party in 2017, documented significant positive impacts on local community stakeholders, as a result of REDD+ interventions, as well as significant levels of understanding and support for the project.

History:

Data and studies show that the stakeholder community for the LCFP is characterized by a large and growing population, comprised of mostly local families, that have grown, over time, to expand into new areas. Expansion into more rural / remote areas of intact forest, including near to protected areas, has taken place over time. In recent years, this is now taking place at a larger scale, and more rapidly, as a result of the larger population, and now that areas closer to large settlements, towns and urban areas are perceived to have become “full,” forcing new families to move further into new areas. Populations in remote, rural Nyimba and Lundazi remain smaller and lower density; areas around Nyimba town, Mambwe town, Mfuwe, Chipata and Rufunsa have larger populations, higher densities, and are growing more quickly. The area around Mfuwe, in particular, is marked by documented rapid expansion as a result of the development of the tourism industry, in recent years, and especially since the 1980s. Mambwe District is documented to have one of the highest population growth rates in Zambia, which is already a country with one of the highest population growth rates, and birth rates, in the world (according to CSO data, and the CIA World Fact book).

The LCFP is building on work that has been done by previous projects that were designed to promote local accountability and benefits for natural resources management, which took place during the 1980s, 90s and 2000s. Most notably, the Luangwa Integrated Resource Development Project (LIRDP) was a major local attempt to promote a Community Based Natural Resources Management (CBNRM) approach in the Luangwa Valley ecosystem, and channel pay-for-performance benefits to local community stakeholders. The LIRDP included work in many of the same communities that the LCFP is now working with, and also similarly combined donor-funding with collaboration with Government partners, and implementing partners. These projects met with a mixed record of successes and failures, which are documented in other studies (see, for example, Barry Dalal-Clayton and Brian Child, *Lessons from Luangwa: The Story of the Luangwa Integrated Resource Development Project, Zambia*. Wildlife and Development Series, 13, International Institute for Environment and Development, London, England). Importantly, the LIRDP pioneered attempts in Eastern Province, Zambia, to work with CRBs and VAGs as local community implementing partner institutions, to follow a CBNRM-type approach to the management of financial and natural resources on behalf of local communities. The LCFP is now following a similar approach, using the same community institutions, although we have updated and adjusted our model, to adapt and respond to “lessons learned” that have been highlighted both by previous professional and academic studies of these programs, as well as stakeholder inputs from local community and government partners. It is also worth noting that the LCFP is similarly building upon models that have previously been used to regulate and implement hunting activities taking place in GMAs in Zambia, whereby CRBs are empowered as local Chiefdom-level community institutions, responsible for the management of local resources, as well as receipt, and management, of performance-based payments, on behalf of the wider community, in close collaboration with local government.

2.1.7 Project Zone Map (G1.4-7, G1.13, CM1.2, B1.2)

A map of the project zone (Figure 9), shows areas within 5 km of the project area and 5 km of project communities. These are the areas where project activities outside the project area will be implemented. Additionally, the map shows the location of the high conservation value site Nyani Falls and field observations of high conservation value species (vulnerable or greater threatened status in IUCN redlist).

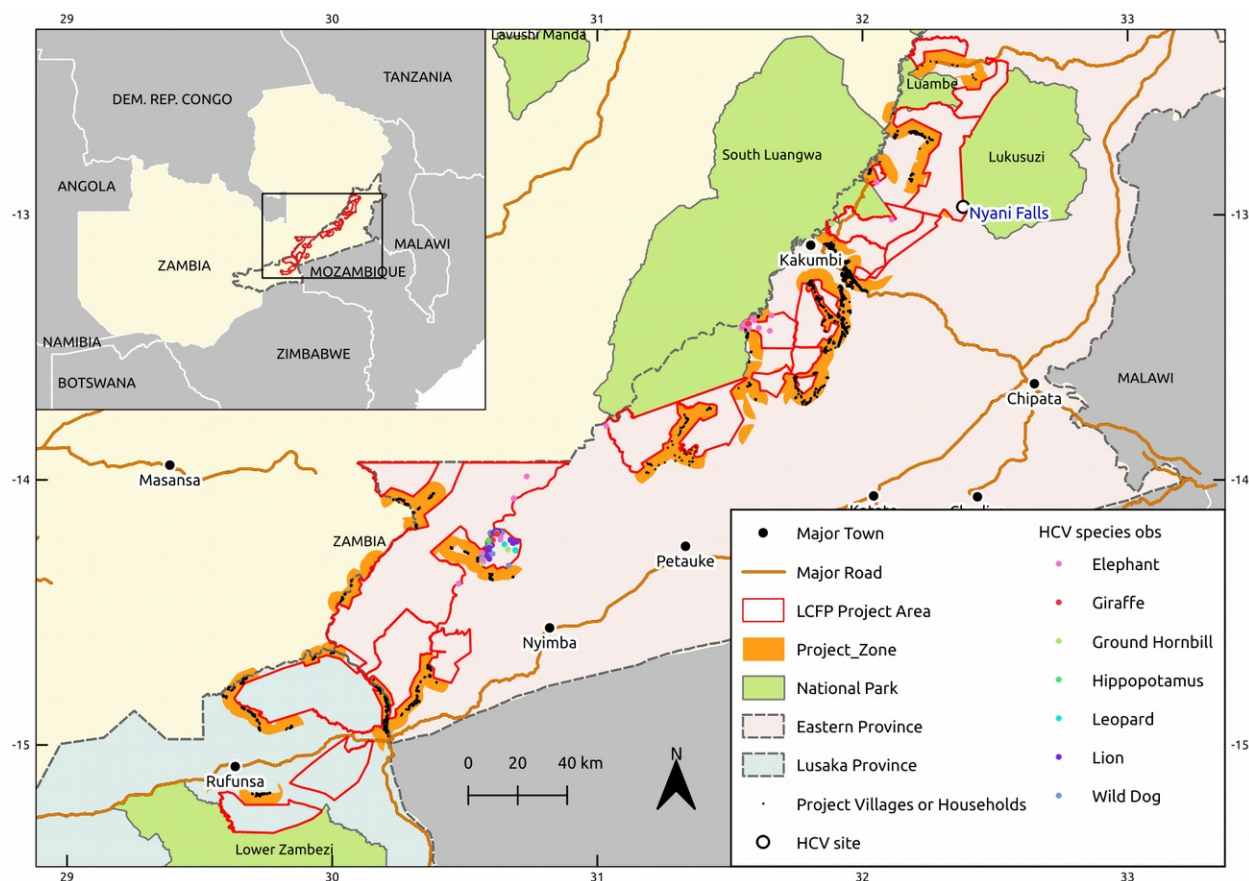


Figure 9. Map of project zone, project communities, project area, high conservation value species and sites within the project area.

Target areas for potential future project instances are shown in Figure 10 and listed in tables 4, 5, and 6. They include chiefdom land in game management areas in the greater Luangwa River watershed, including the Lukusashi and Lunsemfwa watersheds, and Chiawa GMA. The project has identified some other large forest areas on chiefdom land contiguous to the target GMAs that have potential for expansion. Finally, there are other private properties between the Sandwe GMA and Pia Manzi ranch that

may join the project. Together with national parks, the current and future project locations form one continuous region from the GMAs at the northern end of the Luangwa river, to Chiawa GMA to the southeast of Lusaka.

Table 4. Chiefdoms with potential GMA land for future project instances

Province	GMA	Chiefdom
Central	Chisomo	Chibale
Central	Chisomo	Chisomo
Central	Chisomo	Serenje
Central	North Luano	Chembe Mboshya
Central	North Luano	Mboroma
Central	South Luano	Chamuka
Central	South Luano	Chembe Mboshya
Eastern	Chanjuzi Lumimba	Kazembe
Eastern	Lower Lupande	Kakumbi
Eastern	Lower Lupande	Malama
Eastern	Lower Lupande	Mnkhanya
Eastern	Msoro	Jumbe
Eastern	Msoro	Mnkhanya
Eastern	Msoro	Msoro
Eastern	Mwanya Lumimba	Mwanya
Eastern	Nyamimnga Lumimba	Chitungulu
Eastern	Sandwe	Sandwe
Eastern	Upper Lupande	Jumbe
Eastern	Upper Lupande	Nsefu
Eastern	West Petauke	Luembe
Eastern / Muchinga	Chikwa Musalangu	Kazembe
Lusaka	Chiawa	Chiawa
Lusaka	Chiawa	Nkomeshya
Lusaka	Rufunsa	Mpuka
Lusaka	Rufunsa	Mpuka
Lusaka	Rufunsa	Mupanshya
Lusaka	South Luano	Mpanshya
Lusaka	South Luano	Shikabeta
Muchinga	Chikwa Musalangu	Chifunda

Muchinga	Chikwa Musalangu	Chikwa
Muchinga	Chikwa Musalangu	Tembwe
Muchinga	East Musalangu	Kambombo
Muchinga	East Musalangu	Mulilo (Senga)
Muchinga	East Musalangu	Muyombe
Muchinga	East Musalangu	Tembwe
Muchinga	Fulaza Musalangu	Chikwa
Muchinga	Fulaza Musalangu	Chiwale
Muchinga	Fulaza Musalangu	Lundu
Muchinga	Luwawata	Nabwalya
Muchinga	Mukungule	Mukungule
Muchinga	Nyampala	Nabwalya

Table 5: Chiefdoms with potential other lands for future project instances

Province	Chiefdom
Central	Chibale
Central	Chisomo
Central	Kanyenshya
Central	Kaunduia
Central	Mboroma
Central	Mukonchi
Central	Mulungwe
Central	Serenje
Eastern	Luembe
Eastern	Nyalugwe
Lusaka	Bunda Bunda
Lusaka	Mpanshya
Muchinga	Chikwanda
Muchinga	Mpumba

Muchinga	Mukungule
----------	-----------

Table 6: Private Ranches with potential lands for future project instances

Province	Private Ranch
Eastern	Mbizi
Eastern	Kazumba
Eastern	Nyalamu
Eastern	Mzanzara
Eastern	Nyakolwe

A map of the common reference area, proxy area, and activity-shifting leakage area for all project activity instances is shown in Figure 11. Activity-shifting leakage areas are subject to change in the future as per the guidance in section 8.3.2.1 of VM0009 v3.0 if the areas are no longer accessible to agents of conversion or new project instances require a new activity-shifting leakage area.

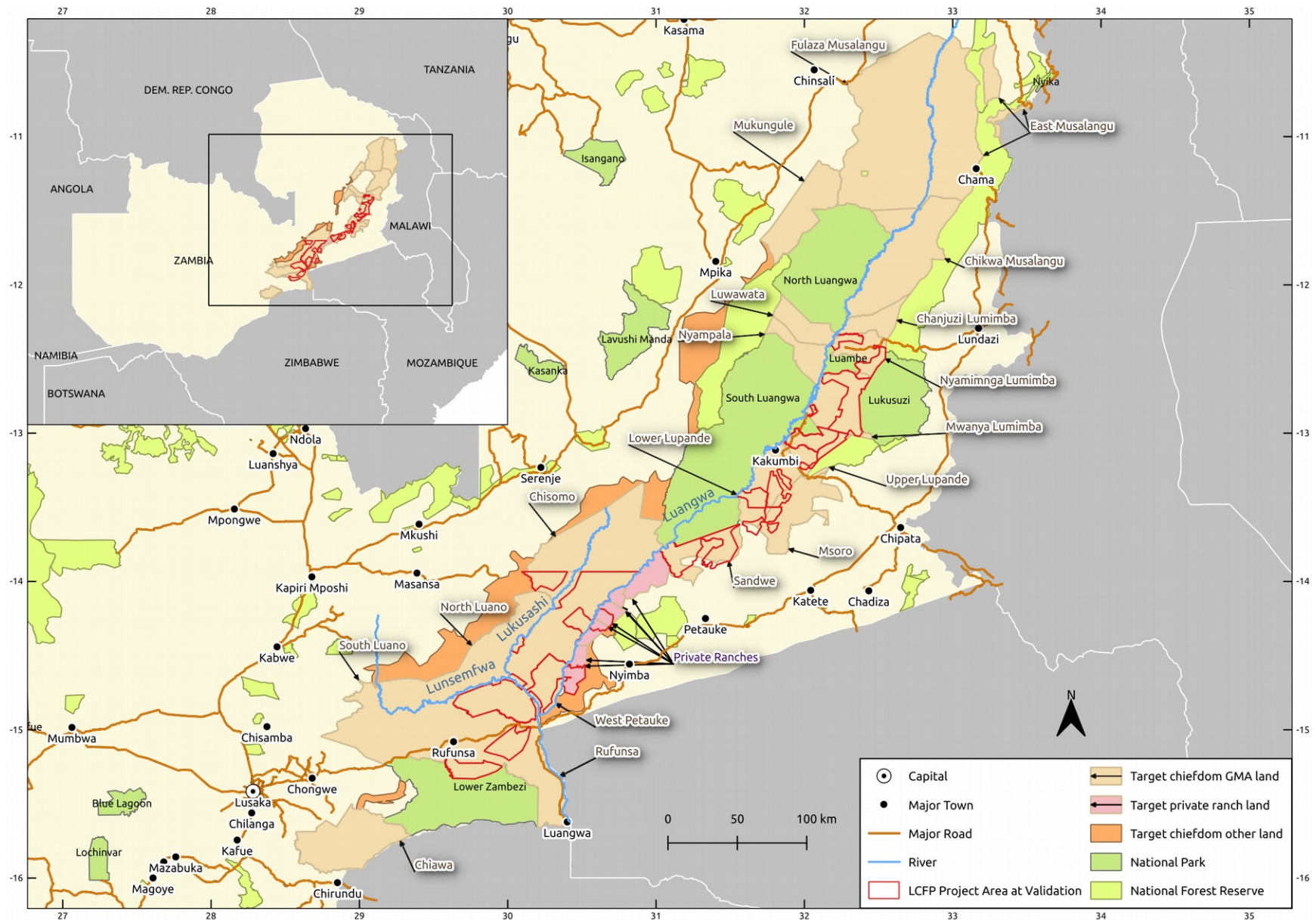


Figure 10: Target areas for potential future project instances (PDR.15)

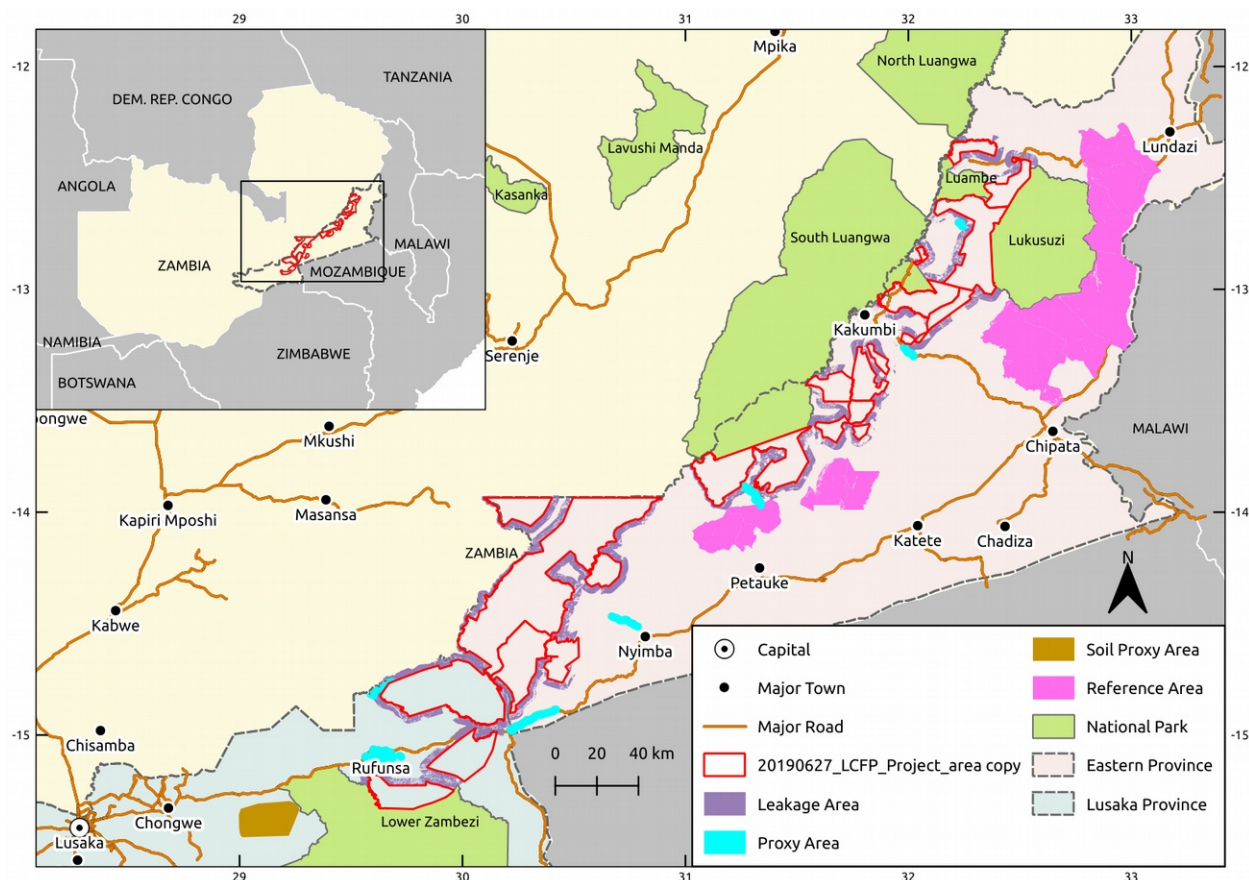


Figure 11: Common reference area, proxy area, and activity-shifting leakage area (PDR.16).

2.1.8 Stakeholder Identification (G1.5)

Stakeholder identification took place as part of the initial REDD+ viability assessment process that BCP undertook to identify new / potential REDD+ sites for the LCFP, beginning in 2014. The process followed can be broadly outlined as follows:

Workshops / Meetings with key Government and Community Authorities: Beginning at a national level, and then moving down to provincial, district and Chiefdom levels, BCP held a series of workshops and meetings, in which we outlined REDD+ and explained the REDD+ viability assessment process, including that two key criteria for REDD+ include:

- Selection of a viable forest area for protection;
- Documented consent from all relevant stakeholders / rights-holders.

- Permissions were then received from Government and Chiefdom authorities, to proceed with on-the-ground viability assessment activities, including participatory forest identification and stakeholder engagement in line with FPIC.

Initial “high level” stakeholder identification and mapping: During the initial District and Chiefdom-level workshops and meetings that were held, BCP solicited inputs from relevant Government and community authorities (namely: traditional authorities, including Chiefs / Chiefs’ representatives, and CRBs, as legally recognized, Chiefdom-level community institutions), about who could be counted as potential stakeholders to any proposed REDD+ project, and where they were/are located, in relation to the proposed forest areas under consideration for REDD+ protection. It was explained that identifying and consulting with any potential stakeholders is a verification requirement for REDD+ -- and more importantly, adequate stakeholder sensitization and consultation prior to project implementation is in everyone’s interest, to help ensure the sustainability of the project, by ensuring that all relevant stakeholders were consulted about, and consented to, participation in the project. It was explained that stakeholders could broadly be defined as anyone who could potentially be impacted by the proposed project (either positively or negatively), and/or anyone with potential decision-making power about whether or not to cut trees within the identified / proposed forest areas for protection. At the District/Chiefdom level, **Government and community authorities (traditional leaders) provided us with consistent feedback that, so long as a REDD+ project would be taking place on “community” land (including within a GMA), then all community members of the Chiefdom could be considered to be stakeholders, since traditional authorities recognize that community forests belong to the entire community within the Chiefdom**, and therefore, all recognized residents within the Chiefdom would have recognized rights, within the traditional leadership structure, to access and use forest resources within community forest areas. Initial community maps were drawn by these key stakeholders, and also confirmed and supplemented by satellite imagery, which confirmed the location of villages and settlements, near to proposed forest areas within each Chiefdom that had expressed willingness / interest to participate in the program.

Detailed Chiefdom/Village level stakeholder mapping - PRA/PLA FPIC activities: More detailed stakeholder identification and mapping took place during significant PRA/PLA techniques that were undertaken in each of the 12 partner Chiefdoms, as part of the FPIC processes that were undertaken by BCP during 2015-2017, at a community level. Our FPIC records show that these meetings were held at a Chiefdom level, in each of the 12 Chiefdoms, and that representatives from all 69 registered VAGs within these 12 Chiefdoms were involved in FPIC activities.

During the PRA/PLA activities that took place, community stakeholder identifications were confirmed, and maps were further refined, to ensure that significant community inputs were received to identify and include any potential stakeholder communities. Local (District) level Government officers were involved in this process, as well as representatives from Chiefdom level institutions (Chiefdom royal establishments; CRBs), as well as village-level representatives from VAGs and traditional leaders (Headmen). Meetings were also open to members of the wider community, who also participated and provided inputs. As part of

the stakeholder identification process, it was described, again, that “stakeholders” included anyone who could potentially be affected by the project, and/or anyone with a recognized “right,” or even ability, to influence decision-making about whether or not to cut trees within the identified forest area for protection. Again, stakeholders at all levels confirmed that community forests were/are viewed to be collectively “owned” community resources, and as such, “stakeholders” should include all Chiefdom inhabitants, in all villages within participating Chiefdoms, as their rights to access and use forest resources were/are recognized by members of the local community, as well as the Chiefdom authorities (traditional leaders) as well as Government – who recognize the authority of community institutions and traditional leaders, on community or “customary” land, even within GMAs. As such, it was agreed that a registry of Chiefdom inhabitants and villages could essentially function as a stakeholder registry. Initial lists of villages were received from Headmen, and were shared with BCP through VAG and CRB structures. These were used to inform data-collection plans, to complete an on-the-ground confirmation activity, to confirm, register and map all stakeholders for REDD+ projects within each of the 12 partner Chiefdoms of the LCFP.

Once FPIC processes were completed, community consent was received and documented in community consent letters for REDD+ implementation, signed by Chiefdom authorities with legally recognized authority to manage community resources (Chiefs, CRBs). Once these letters were received, BCP undertook a “**Stakeholder Household Registry**” process, with the aim of confirming, documenting, registering, mapping and counting community stakeholders to each REDD+ project, in every participating Chiefdom, in every VAG and village within that Chiefdom. This process was undertaken through CRBs and Village Action Groups (VAGs) as local-level community institutions, and involved significant inputs, guidance and participation of local traditional leaders (such as Headmen) to ensure that all members of their villages were registered as participants, stakeholders and beneficiaries to the project. The data collection process was led by locally hired staff, working hand-in-hand with local traditional leaders such as village headmen, as well as representatives from local VAGs, CRBs, and Government.

The LCFP “Household Registry” now provides a comprehensive database, and map, of all stakeholders to REDD+ projects supported under the LCFP. As the processes above describe, it has been informed by significant inputs from community authorities and government partners, solicited through a multitude of workshops, meetings, and PRA/PLA activities that have been undertaken as part of FPIC processes during the past three years, during which input was solicited, and received, concerning who should be counted as “stakeholders,” based on their recognized rights and abilities to inform decisions about forest protection within identified areas. The actual data collection process has taken approximately one year, and has involved dozens of data collectors across all 12 Chiefdoms.

Other stakeholders are hunting concession holders that operate within the REDD+ project areas. All hunting concessionaires have been consulted regarding the project and understand the benefits it will bring.

2.1.9 Stakeholder Descriptions (G1.6, G1.13)

Stakeholders to the REDD+ projects supported under the LCFP include the following:

Government agencies with legally recognized authority / mandates to manage resources within forest areas, including **Forestry Department (FD)** and, in Game Management Areas (GMAs), the **Department of National Parks (DNPW)**. **District and Provincial authorities** also have legally recognized rights and responsibilities related to natural resources management, and the enforcement of relevant laws, within Chiefdoms located within their jurisdiction. Relevant legislation confirming the authority, and therefore rights (and responsibilities) of these government agencies, is included within Section 2.5 of this document.

Chiefdom Traditional Authorities: In Zambia, the authority of Chiefs to manage community resources on behalf of “the community” is legally and culturally / traditionally recognized. Chiefs are legally recognized by the Government of Zambia, within the Laws of Zambia (meaning: they must be confirmed and formally designated by the Government, in order for their authority to be legally recognized). On customary / community land, this authority is clearly recognized, both legally and according to local culture, customs and traditions. In legally-designated Game Management Areas (GMAs), legally, resources are to be effectively co-managed by the Chiefdom community, in cooperation with DNPW. The authority of FD to regulate forest resources across Zambia, including in customary areas, is also legally recognized. As such, Chiefs and traditional authorities are recognized as stakeholders with key rights and decision-making abilities related to community forest resources, and therefore REDD+, in Zambia, and under the LCFP.

Community Resources Boards (CRBs) and Village Action Groups (VAGs): CRBs and VAGs are legally recognized community institutions, with legally recognized rights and responsibilities to effectively manage community resources. Under the LCFP, BCP is working with these institutions as implementing partners, to effectively manage natural / forest resources, and financial resources derived from them, on behalf of the wider community.

Community Members: Within a traditional authority structure (which is legally and culturally recognized within Zambia, as described above), members of the wider “community” are able to collectively possess rights to community resources, including forest resources. Community members are supposed to be documented in official village registers, which are maintained by Headmen. Under traditional authority structures, Headmen are recognized as holding rights and responsibility for the leadership of designated villages. Indunas may hold higher status, within traditional authority structures. All villages and their designated traditional leaders must be recognized, and fall under, the authority of the Chief / Chieftainess, within a Chiefdom. In the case of the LCFP, BCP has facilitated the process of confirming, documenting and mapping the villages within each Chiefdom, and registering community members within them, as part of the Stakeholder Household Registry.

As described in Section 2.1.8, above, throughout the stakeholder identification process, community and government implementing partners under the LCFP consistently confirmed that, according to customary / traditional rules, community resources are recognized as belonging to the members of the wider “community” within a Chiefdom, and therefore, all community members within a Chiefdom can be considered to be “stakeholders” to any REDD+ activities taking place on community land, involving community forests, within a given Chiefdom. However, the authority of the Chief and traditional leaders (working under the Chief) to develop and enforce rules for the management and regulation of community resources, is also clearly recognized, both legally and culturally/traditionally. The authority of CRBs and VAGs as community institutions with the right and responsibility to manage community resources, according to these rules (outlined by Chiefs/traditional leaders, and Government), is also legally recognized. This explains why BCP, under the LCFP, has ensured to receive documented permissions from relevant Government implementing entities with legally recognized rights / responsibilities related to the management of natural resources. However, in addition to these, we have *also* signed REDD+ implementation agreements with Chiefs/traditional leaders, as well as representatives from CRBs and VAGs, to confirm and document their commitments and entitlements, on behalf of the wider community, to be managed by these same institutions, on behalf of the community. In this particular context, both Government and community institutions have legally and culturally / traditionally recognized rights and responsibilities related to the management of community (natural) resources, as does the wider community, and as such, they are all considered to be stakeholders to the LCFP, in each Chiefdom.

Hunting Concession Holders: Game management Areas have hunting blocks where commercial trophy hunting takes place, mainly by foreign tourists. Professional Hunters bid for blocks and are awarded 10 year leases to hunt in a specific block. Some of the revenue is shared with local communities and the concessionaires support the local CRB in their duties as well. Concessionaires only have traversing rights and DNPW issues hunting licenses on an annual basis which is then purchased by the concession holder.

2.1.10 Sectoral Scope and Project Type

The project falls under scope 14 of the VCS, namely Agriculture, Forestry and Other Land Use (AFOLU). The AFOLU project category is Reducing Emissions from Deforestation and Forest Degradation.

This is a grouped project. All future activity instances meet the eligibility criteria of VCS Methodology VM0009 as described in section 2.1.21.

2.1.11 Project Activities and Theory of Change (G1.8)

The project activities and associated theory of change presented here are expected to generate benefits communities, wildlife and climate change mitigation. At a community level, BCP strategy aims to reduce poverty through social development, enhanced rights, governance empowerment and improved livelihoods. Improved livelihoods through conservation agriculture will achieve climate change benefits by reducing emissions from farming, reducing the need to shift farms through slash and burn soil nutrient depletion and increasing revenue from targeted farming surpluses. Furthermore, adaptation will be enhanced by diversifying income streams through non-timber forest products such as honey production. Biodiversity will benefit from reduction in habitat loss and improved wildlife corridors protected by increased monitoring and anti-poaching activities.

Early in 2015, LCFP, consulted widely with the local communities within the project area during the development of Participatory Forest Management Plans (PFMP), which have since transformed into Community Forest Management Agreements (CFMA). PLA tools, with FPIC at the core, was extensively used when conducting the FPIC meetings, this approach was key in helping the local communities develop community livelihood profiles, explore the drivers of deforestation, prioritize community needs, and design project interventions to maximize community, biodiversity and climate benefits.

The FPIC meeting reports, preliminary PFMP and CFMA processes not only described the proposed solutions to the drivers of deforestation, specific investments and activities that will be undertaken by LCFP throughout the duration of the project, but also outline how the revenue generated by the project that is to be provided to the local communities is to be managed, in addition to setting out how decisions related to direction and execution of project activities are to be determined. Given this framework, project activities continue to be defined and refined on an ongoing basis through FPIC consultations with the communities within the project area as well as other stakeholders and government regarding conservation fee management against the CFMA and before CFMAs, Conservation Fee Agreements. The community development aspects of the benefit sharing are designed to maximize community, wildlife and climate benefits.

Through the FPIC process mentioned above, local communities were led through the process of needs assessments that resulted in the selection of short term and long-term interventions aimed addressing the drivers of deforestation and improving community livelihoods. BCP provides tools to assist communities select interventions that maximize community, wildlife and climate benefits such as Project Selection Guidelines, and a 'Risk Mitigation' (RM) Tool. The RM tool is designed to allow communities to assess the 360 degree risks of a project implementation, and design mitigation. Not only is this empowering for communities and create more ownership, it ensures that projects are selected against principles which maximize community, biodiversity and climate benefits.

The social impact interventions can be broadly classified as follows:

- Support to improvement of social services through;

- Construction and rehabilitation of school infrastructure; classroom blocks and latrines,
- Construction and rehabilitation health care centers,
- Provision of hammer mills, maize and rice shellers,
- Provision of safe drinking water; through drilling of new boreholes, rehabilitation of existing boreholes and improvement of existing wells.
- Construction of culverts, drifts and provision of boats to facilitate safe river crossing
- Erection of solar fences to reduce human – wild animal conflicts.
- Support the improvement of agriculture, horticulture, apiculture and aquaculture production techniques and creation of market linkages
- Support to school environmental education programs
- Job creation through recruitment of employees from the local community

Project activities be broken down into two separate but connected activity areas, as described below. In some cases implementation of some of the specific activities has already commenced; these are listed as defined activities. Progress on these activities will be discussed in sections to follow. Other activities are under consideration and will be implemented after completion of appropriate due diligence and feasibility considerations, appropriate consideration is given by the local community through their respective Community Resources Boards, and additional funding is available; these are listed as additional activities undergoing investigation and further consultation.

Accompanying each set of activities is a list of positive impacts anticipated to result from their implementation. These impacts have been derived through the use of theory of change model detailed in the annex section. In addition to expected positive impacts, potential negative impacts resulting from project activities are also listed. It is important to note that the negative impacts listed are potential impacts commonly associated with similar projects and therefore, should not necessarily be considered as significant. Furthermore, some of the project activities, for example, some of those related to agricultural and apiculture improvement are focused toward mitigating potential negative impacts, rendering them a non-threat.

None of the project area is in a jurisdiction currently covered by a jurisdictional REDD+ program.

Activity Area 1: Infrastructure Development, Social Service Delivery;

Defined Activities

- Construction and rehabilitation of school infrastructure; classroom blocks and latrines,
- Provision of safe drinking water; through drilling of new boreholes, rehabilitation of existing boreholes and improvement of existing wells.
- Provision of boats to facilitate safe river crossing
- Support the delivery of health care services through provision of clinic furniture and fittings
- Provision of hammer mills, maize and rice shellers
- Construction and rehabilitation of health care centers.

Additional Activities Undergoing Investigation and Further Consultation

- Roof-based water collection (Self Supply)
- Micro-credit project(s)
- Construction and rehabilitation health care centers,
- Construction of culverts, drifts
- Erection of solar fences to reduce human – wild animal conflicts

Expected Positive Impacts

- Reduction in cost of living
- Increased level of education in communities
- Increased local capacities
- Increased life expectancy
- Reduced child mortality and disease transmission
- Reduced human-wildlife conflict
- Reduced degradation effects on forest from needing to boil water which does not come from a professional borehole.

Potential Negative Impacts

- Conflicts related to the equitable distribution of benefits

Activity Area 2: Apiculture, Agricultural Improvement, Diversification and Economic Opportunities (PDR.104 Leakage Mitigation Strategies)**Defined Activities**

- Develop agroforestry demonstration plots to test and demonstrate agricultural multi-cropping, diversification, and intensification.
- Train 84 lead farmers in conservation farming techniques to support 1 260 follower farmers through the farmer-to-farmer model.
- Develop apiary demonstration plots, and train 6 lead farmers (mentors) in hive construction, maintenance, and appropriate apiary and beekeeping techniques.
- Distribute 6 000 beehives to 1100 participating households within the project area.

Additional Activities Undergoing Investigation and Further Consultation

- Support development of horticulture production
- Support animal husbandry and facilitate the construction of animal enclosures to prevent spread of diseases and damage to crops
- Assistance with commercialization of agricultural products and Support in crop aggregation for easy access to markets outside the project area
- Land-based aquaculture
- Elephant-barrier beehive fences
- Fences to separate wildlife from crops.

Expected Positive Impacts

- Increased quantity, diversification, and value of crops
- Improved nutritional value of crops
- More sustainable practices (Conservation farming, Beekeeping and forest management training)

- Leakage mitigation due to redirecting peoples labor towards economic activities that do not involve cutting down forest.
- Improved resilience to crop failure and market fluctuations
- Increase employment opportunities (Lead Farmers)
- Improved quality of life and cash from food surplus sales
- Guaranteed market for honey
- Reduced deforestation from shifting slash and burn agriculture due to improved soil husbandry and moisture retention techniques.
- Reduced human-wildlife conflict and problem animal killing

Potential Negative Impacts

- Increased cost of living if value of crops increases
- Conflict related to unequal distribution of benefits and workload

Support to improved agriculture techniques through Conservation Farming

In 2017, LCFP began its conservation farming activities by supporting the work of the Ministry of Agriculture, specifically through the provision conservation farming extension services. Agreements were signed with the respective territorial Ministry of Agriculture offices within the project zone in the districts listed in Table 7. To date 319 lead farmers have been trained.

Table 7. District agreements with Department of Agriculture.

Location	Date Agreement signed with MoA
Mambwe District	21/11/2016
Nyimba District	15/03/2017
Lundazi District	21/01/2017
Rufunsa District	29/05/2017

Arising from the FPIC community consultative meeting; forests are one of the main sources of livelihood for the local communities with illegal charcoal production, illegal logging and unsustainable farming methods such as shifting cultivation (“slash and burn”) being the major activities conducted. These activities coupled with the increased in-migration of farmers from the plateau to the valleys, where soils are still fertile, has increased the demand for opening up new farm land within the project area. Conservation Farming was thus one of the activities identified during the needs assessment process as an intervention that would complement LCFP’s conservation activities.

For the 2017 – 2018 farming season, 2199 farmers from the target 3285 were enrolled by the Ministry of Agriculture to participate and receive extension services in Conservation Farming.

The main activities that are being supported under this intervention are;

- Promotion of planting of *Faidherbia albida* (Musangu) and other agro-forestry species.
- Training of “facilitators” “lead” farmers in participatory research and extension techniques and improved crop production technologies.
- Strengthening of existing farmers’ groups through training in group dynamics, business administration.
- Promote land management through farmer training in land preparation, soil fertility enhancement and moisture retention;

Expected Outcomes:

- Reduced encroachment into the project area.
- Increased production of higher value and higher nutrition crops (soya beans, cowpeas, sunflower).
- Improvements in the income levels of rural households through the sale of surpluses.
- Enhancement of the nutritional status of farm families, especially the children and women members through increased consumption of pulses.
- Confidence built amongst small-scale producers due to cultivating using sustainable methods and diversifying their food production, and entering into small-scale agri-business practices.
- Reduced dependence on the agriculture subsidies and use of artificial fertilizers.

- Enhancement of the knowledge base and skills of small-scale rural producers in modern production, processing and marketing practices.
- Access to the local, regional and national markets ensured small-scale rural producers through market linkages

Support to the development of Value Chains in Non- timber forest Products

Improved Apiculture techniques.

Whilst beekeeping is regarded as a cottage industry in Zambia, that is well suited for individuals and companies with limited resources for investment, its potential to contribute to the reduction of rural poverty cannot be ignored. According to the Central Statistics Office (2010), the incidence of rural poverty in Zambia is about 80%. Eastern Province, where the majority of LCFP occurs (10 of 12 Chiefdoms) has the highest overall poverty ranking according to the 2015 LCMS (of 10 Provinces). Honey production is a forest compatible, scalable livelihood diversification activity which BCP has invested meaningfully in. BCP funded a large pilot of 6,000 hives plus a support program costing approximately \$350,000. BCP chose to team with a private sector honey operator called BeeSweet who have export markets and core competencies in honey processing and sales. This strategy is designed to contribute towards poverty reduction as BeeSweet have markets, pay cash on delivery, and have a network of community 'mentors' to support bee farmers in each Chiefdom.

Although honey hunting and beekeeping have been practiced for a long time by almost all communities in Zambia, beekeeping is only now being promoted by Government as an alternative livelihood for rural communities as an alternative to charcoal production and illegal game hunting, as well as enhancing agricultural income.

Against this background, LCFP conducted a viability assessment in 2016, on the potential to establish honey production value chains within the LCFP project area. Based on the findings of this viability assessment, an agreement was signed with BeeSweet, to expand honey production to cover the LCFP project zone. Through this intervention, 12 from a target of 11 honey producer lead farmers (Mentors), received training from BeeSweet in hive construction and maintenance and apiary management (Table 8).

Table 8. Number of bee mentors trained in project zone.

District	Chiefdom	No. Mentors trained	Date of Completion of training

Mambwe	Mwanya	3	July 2016
	Jumbe	2	May 2017
	Mnkhanya	1	May 2017
	Msoro	3	August 2017
	Malama	3	August 2017

Through this intervention, 6 000 Kenya topbar beehives were distributed within the LCFP project area to 1,100 beneficiary households (Table 9). Honey sales from communities to BeeSweet have begun in the Project Zone. It is envisaged that the number of hives will scale up over time dependent on community demand to expand this activity.

Table 9. Beehive distribution in project zone.

District	Chiefdom	Target No, of hives to be distributed	Actual No. distributed	Date of last beehive delivery
Mambwe	Mwanya	1000	760	May 2018
Mambwe	Jumbe	1000	1000	August 2017
Mambwe	Mnkhanya	1000	1000	February 2018
Mambwe	Msoro	1000	900	May 2018
Mambwe	Malama	1000	1000	May 2018
Mambwe	Nsefu	1000	375	May 2018
Total		6000	5,035	May 2018

Expected Outcomes:

- Reduced encroachment into the forests under protection through REDD+ by local communities harvesting honey.
- Improvements in the income levels of rural households through the sale of honey.
- Reduction in use of traditional hives and falling of trees due to honey hunting.
- Enhancement of the knowledge base and skills of honey producer in modern honey production, processing and marketing practices.
- Direct market access for the honey produced – a critical component of any rural livelihood improvement program.
- Reduced leakage and deforestation in the reference area as hives are a disincentive to cut down trees.
- Reducing childhood stunting due to accessing honey as a nutrient supplement.
- Reduced human-elephant conflict in farm fields through the strategic placement of some hives around the farms.

Support to Infrastructure Development and Social Service Delivery

BCP's 'Beyond Subsistence' Livelihoods Strategy aims to improve community quality of life in a way that is forest-conservation compatible. A key principle is that projects should be demand driven from a Village Action Group level up. This grassroots approach is designed to mitigate against risks of elite capture. The table below captures the diversity of implemented projects. Given the very high poverty rates in Eastern Province, there has so far been a strong preference for social service projects related to education, health, water and sanitation, transport and agricultural processing equipment. Many of these investments are traditionally the responsibility of Government. However, as Government budgets are limited, communities choose to accelerate their own development path.

District	Latrine	Clean Water Supply	Class-room	Hammer Mill	Teachers House	Boat	Clinics	Shellers/ Dehuller	M'cycle or Bicycle	Laptop or Printer
Rufunsa	6	2	1	9			1	7	1	
Nyimba			1	1	5	1				
Mambwe		19	1		2			2	15	2
Lundazi			2			1	1			
Sub-total	6	21	5	10	7	2	2	9	16	2

Construction of School Infrastructure

In 2016 following the community needs assessment conducted in Luembe and Nyalugwe Chiefdoms, the project with the Luembe and Nyalugwe Chiefdoms and the Ministry of Education in Nyimba constructed a 1x2 classroom block at Mkoma Primary School and one teachers house, as Partnership Impact Projects intended to benefit the Village Action Groups of Mkoma and Mushalira.

In Luembe Chiefdom, the construction of a teachers house was completed in January 2017.

Construction of Clinic Infrastructure:

LCFP in Mwanya in consultation with Ministry of Health in Mwanya embarked on the construction of 1 maternity annex at Mkasanga Rural Health centre, as a Partnership Impact Project (PIP). The community insisted that local community member builders be engaged to construct the Maternity annex as a form of job creation. Construction works to the maternity annex which was completed by local builders were subsequently deemed unfit by certified engineers. This catalyzed BCP to develop a system of staged completion of infrastructure projects using reputable contracting firms, where each major stage of works have certificates of staged completion issued by licensed engineers to trigger the next release of payment to contractors. This has resulted in an improved quality of community project delivery as well as CRB capacity building in project management.

Support to delivery of Community Social Services:

As part of the community social services support through Partnership Impact Projects, in Mwanya Chiefdom, LCFP procure and supplied 1 fibreglass boat and hospital furniture to benefit the community of Mwanya Chiefdom. The boat provides safe river crossing at Lukusuzi for the community and especially the school children in Mkwela thereby increasing school attendance potential, Whilst the furniture and fittings provided was delivered to Mwanya Chiefdom for use at Yakobe Rural Health Post. The boat has a positive effect on wildlife by reducing the risk of human-wildlife conflict, especially with regards to crocodiles.

In Msoro Chiefdom, LCFP provided three maize shellers, as Partnership Impact Projects (PIPs) intended to benefit the entire community of Msoro. LCFP handed over the maize shellers to the community of Msoro Chiefdom through the community Resources Board, who have stationed each Maize sheller in specific zone within the Chiefdom for ease of access to the beneficiary local community. These maize shellers enhance the agricultural economy in Chiefdoms by reducing the time and labor required to process maize. This especially frees up women's time to invest in other activities of importance to the homestead such as child rearing.

Improved water supply:

Following the community needs assessment conducted during the FPIC meetings to develop the PFMP in October 2015, LCFP drilled six new boreholes to provide clean safe drinking water to the local communities in Jumbe and Kakumbi Chiefdoms. In addition, in February 2016, LCFP commenced works to improve 10 open community wells through the construction of well lining, covers and installation of handpumps as water lifting devices (Table 10).

Table 10. Infrastructure projects in the project zone.

Location (District)	Description of Intervention	Date started	Date Completed
Nyimba	Construction of 1 teachers house	23/07/2016	24/04/2017
Nyimba	Construction of 1x2 classroom block	25/11/2016	12/05/2017
Mambwe	Construction material support towards construction of Community Guest House	15/04/2016	On - going
Lundazi	Construction of Maternity Annex	27/09/2016	On – going
Mambwe	Drilling of 6 new boreholes	12/05/2016	28/11/2016
Lundazi	Provision of Furniture and fittings	15/04/2016	10/06/2016
Lundazi	Provision of fiberglass boat	13/04/2016	18/05/2016
Mambwe	Provision of 3 maize shellers	18/07/2017	25/09/2017
Mambwe	Improving of 10 open wells	18/02/2017	On -going

The provision of cleaner water has a positive impact on climate by reducing the need to boil water to sanitize it using wood or fossil fuels. Boreholes have an intended positive benefit for wildlife by reducing human-wildlife conflict over water by separating water sources for humans and wildlife, and ensuring wildlife have longer water supplies in the dry season. This is due to borehole offtake being sub-surface and usually close to human settlements. Clean water is key to improving quality of life by reducing water-borne diseases which can have a negative impact on communities, especially young children.

Expected Positive Impacts

- 1 Reduction in cost of living
- 1 Increased level of education in communities
- 1 Increased local capacities
- 1 Increased life expectancy
- 1 Increased institutional deliveries and improved antenatal and Postnatal services
- 1 Reduced child mortality and disease transmission
- 1 Reduced diarrheal diseases

Potential Negative Impacts

- 1 Conflicts related to the equitable distribution of benefits
- 1 Potential contamination of boreholes if communities place latrines too close to boreholes, or borehole drillers drill too close to latrines.

Our Theory of Change (Figure 14) around the provision of livelihoods support is that if we can help communities make long term investments in a shared vision for development and demonstrate that those investments are intrinsically linked to the successful long term conservation of forests, then livelihoods in those communities will improve demonstrably and be sustainable, while natural forests will remain intact.

Key risks to providing benefits to the communities include the following:

- 1 Risk of elite capture of benefits, either from REDD+ or from livelihoods activities themselves.
- 1 Communities will not grasp the connection between performance (REDD forest conservation) and the benefits they receive.
- 1 Households will not benefit enough from REDD+ to halt deforestation.
- 1 Households may “cheat” and take benefits while continuing to deforest.

To address some of these risks BCP is implementing forest and biodiversity protection measures, in addition, since these risks are complex and difficult to deal with we are also guided by a Beyond Subsistence Livelihoods approach that:

- 1 Builds the capacity of the Community Resource Boards (CRB) to manage financial resources, govern themselves effectively and build a shared vision for development.

- 1 Provides demonstrable, regular benefits that are used to invest in the shared vision for development that benefit the household level.
- 1 Ensures communities understand linkages between receiving REDD+ benefits and long-term forest conservation.

This combination of community based natural resource governance and sustainable livelihoods development will result in intended long term emissions reductions, biodiversity conservation and sustainable development impacts in these communities.

Our Beyond Subsistence Livelihoods Strategy activities are focused on achieving outcomes that will set the stage for achieving development impacts. CRB's need a vision for development implemented through effective and fair governance of the institution. Investments should lead to improved infrastructure and social services (education and health). The private sector should invest in these communities to build mutually beneficial partnerships. This approach requires that all members of the CRB "sign on" to and honor community covenants that spell out the responsibilities that go along with the benefits that come from REDD+.

If we can support our community partners to achieve these outcomes then these impacts will flow from our work:

- 1 Increased income at the household level from forest conservation friendly livelihoods.
- 1 CRB members will understand that deforestation leads to reduced benefits from REDD+.
- 1 CRB members will be empowered to govern effectively.
- 1 There will be no elite capture of REDD+ benefits.
- 1 Households will benefit from improved social services.

In combination with our efforts to build local capacity and responsibility for forest and biodiversity conservation these livelihoods efforts will lead to our ultimate goals of keeping natural forest ecosystems intact and ensuring that community livelihoods are profitable and sustainable.

Activity Area 3: Forest Encroachment and Wildlife Monitoring and Poaching Reduction

BCP provides support to the local CRB village scouts and the Department of National Parks and Wildlife (DNPW) to ensure protection of the project area from encroachment in the project area, monitoring of wildlife and reducing poaching. This activity is supported by aerial monitoring. The wildlife monitoring activities and theory of change is being developed with Lion Landscapes

(<https://www.lionlandscapes.org/>) who has partnered with BCP to develop a biodiversity conservation model using a theory of change centered around lions which are an HCV species (Figure 15).

Defined activities

- 1 Regular patrols by village scouts along project area boundaries identified as encroachment hotspots.
- 1 Regular aerial monitoring of encroachment along project boundaries, poaching sign and HCV presence
- 1 Monitoring of high conservation value species
- 1 Monthly anti-poaching activities including removal of snares, recording carcasses and other poaching signs, and apprehending poachers.

Expected Outcomes

- 1 Quick response to encroachment in project area
- 1 Rapid resolution of encroachment through consultation process by local governance structures notably the CRB and Chiefs.
- 1 Reduction of encroachment over time as project activities expand and awareness of monitoring activities take hold
- 1 Greater understanding of wildlife populations and threats to wildlife.
- 1 Adapt management to changing wildlife populations.
- 1 Increase in wildlife populations, especially high conservation value species.
- 1 Employment creation for village scouts.
- 1 Reduction in emissions.

Thus far monitoring has been rolled out in most of the project area by participating CRBs. BCP has developed a SOP for monitoring which outlines the interaction between aerial and ground based monitoring. Most encroachment is detected through aerial monitoring and recorded using a GPS. This information is then stored in a database and follow up ground monitoring is then actioned by BCP conservation staff. Scouts and/or CRBs and Traditional Authorities are given the location of encroachment, once confirmed on the ground and contact made with encroachers, the Community Engagement Team and CRB then follow up and investigate incidences further.

Monitoring has been rolled out in most of the project area and results of aerial monitoring can be seen in Figure 12 and ground monitoring in Figure 13.



Figure 12. Aerial monitoring of project area.

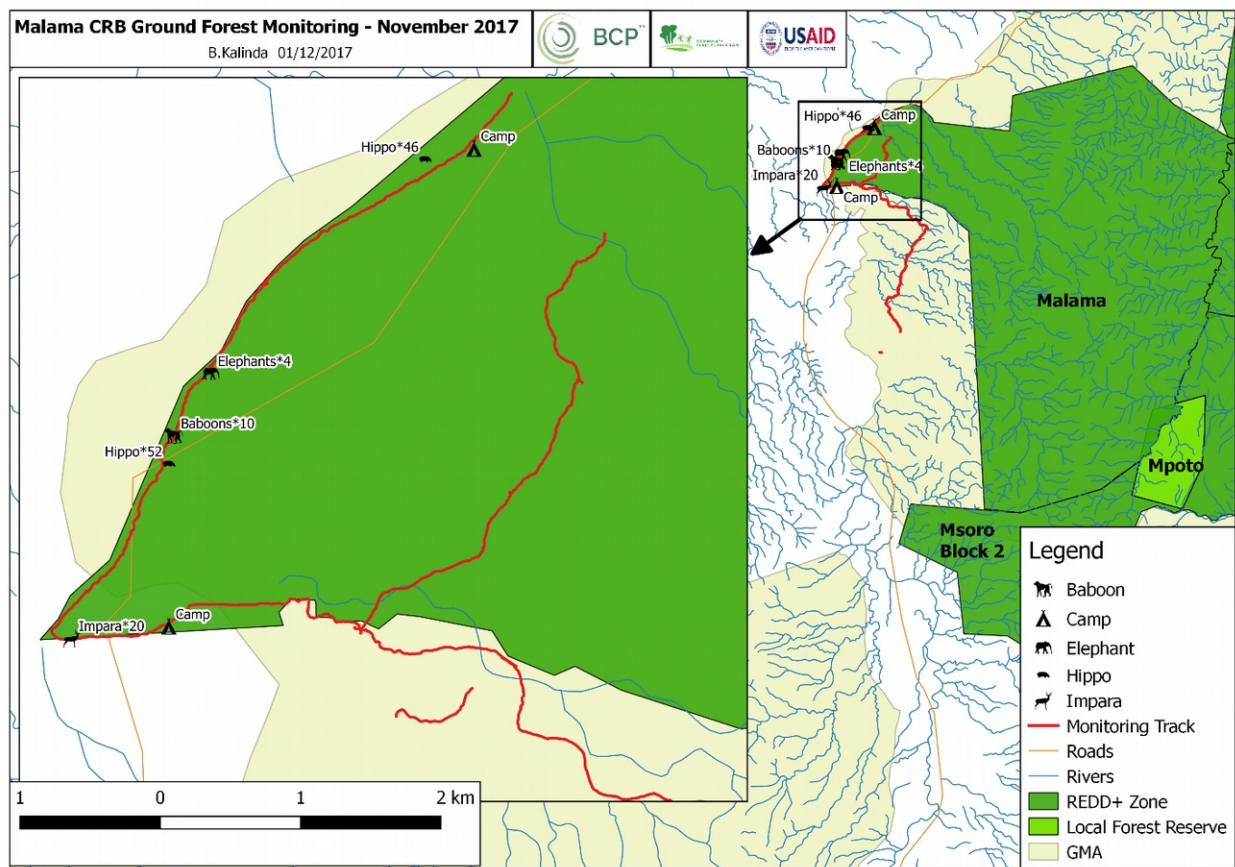


Figure 13. Ground monitoring of project area.

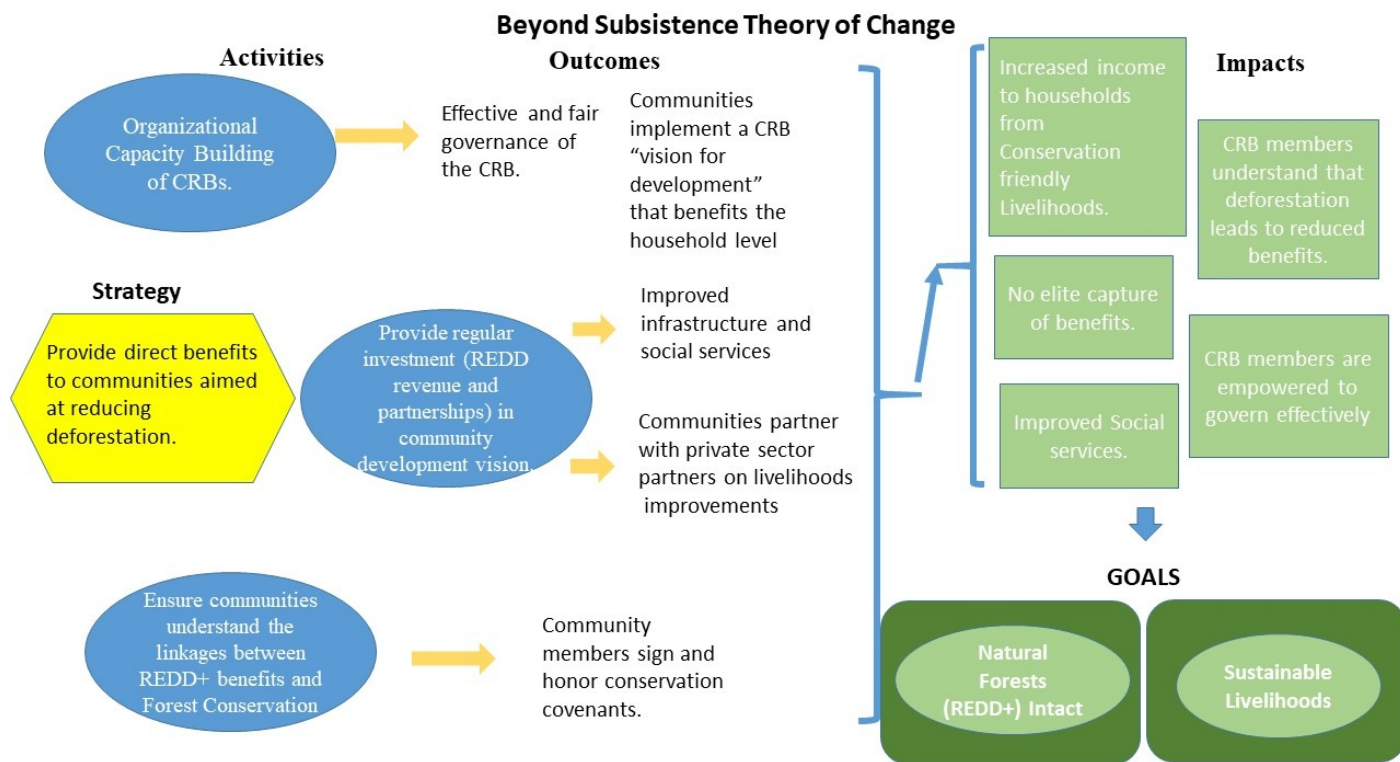


Figure 14. BCP Community Theory of Change

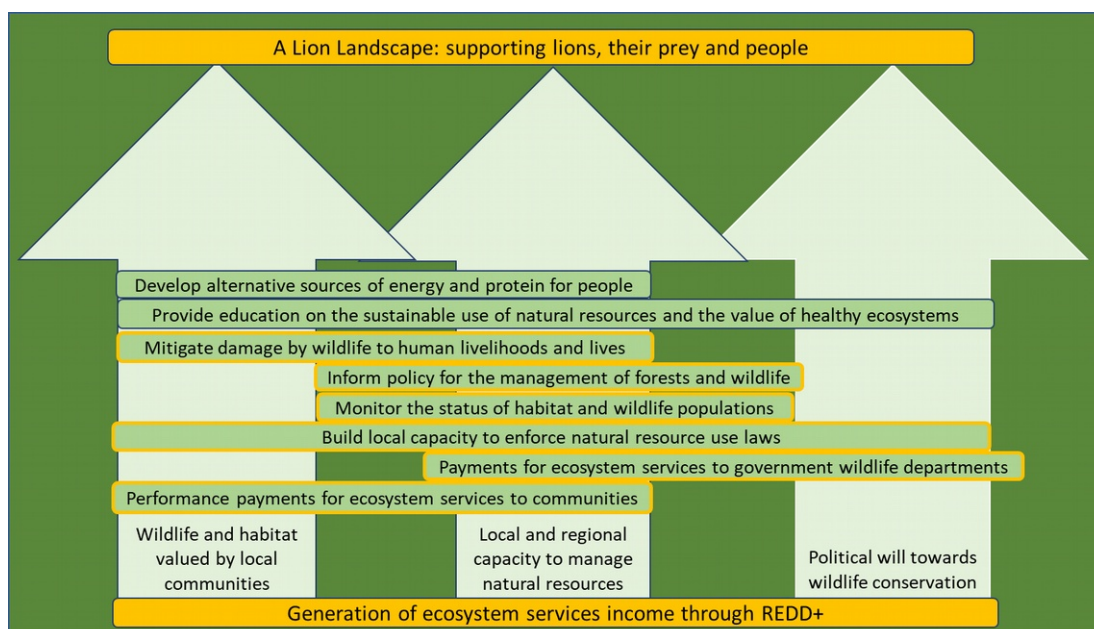


Figure 15. Biodiversity theory of change using lions as a key species developed by Lion Landscapes.

2.1.12 Sustainable Development

Nationally stated sustainable development priorities in Zambia are described and monitored through 5 year national development plans. The focus of the Government of Zambia's Five-Year 7th National Development Plan (2017-2021) is on targeting economic diversification and job creation; poverty and vulnerability reduction; reducing developmental inequalities; enhancing human development; and the creation of a conducive governance for a diversified economy. It sets targets for bringing additional areas of forest under sustainable management (500,000 hectares) and emphasizes measures to enhance the viability and diversification of agriculture co-operatives, farmer groups and community forest management groups.

The 7th NDP departs from sectoral-based planning to an integrated (multi-sectoral) development approach, which recognises the multi-faceted and interlinked nature of sustainable development. It calls for interventions to be tackled simultaneously through a coordinated approach to implementing development interventions, through clusters – involving all arms of Government, Cooperating Partners, Private Sectors, Civil Society Organisations, Faith Based Organisation and other interest groups. Further, the 7th NDP will be primarily implemented through programmes and projects that are devolved to the districts within the tenets of decentralised planning, monitoring and evaluation as determined by municipal and district councils. This will be done with a view to deal with district-specific development challenges.

The Districts design their own plans and monitoring and evaluation systems that will promote integration of development priorities at local level to provincial and national strategic areas of focus and outcomes.

In terms of supporting implementation of the 7th NDP – the five-year target of 500,000 hectares of forest under management increased will be exceeded just through the approval of the forest management agreements by the Director of Forestry under the initial BCP supported initiatives (943,676 ha) by just under a factor of 2. Reporting and monitoring on the LCFP contribution to the implementation of the 7th NDP is supporting at various levels. BCP is a member of the District Development Coordinating Committee which is chaired by the District Commissioner and Council Secretary, in each of the Districts LCFP operates. However, through our close partnership with Government Department and Agencies in each district sustainable development results and achievements are reported through their respective representatives. This includes: Department of National Parks and Wildlife, Forestry Department, Ministry of Chiefs and Traditional Affairs, Ministry of Agriculture, Ministry of Community Development and Social Welfare. At National level, BCP provides a presence on various national technical advisory committees and direct reporting as required to the line Ministries and their Departments. BCP contributes information and data to the National REDD+ Registry and Database Management System maintained by the Forestry Department. This allows Government of Zambia to monitor REDD+ activities covering information such as Project Documents, budgets, project area datasets, Certification Standards, emission reduction quantification and MRV, benefit sharing and safeguards.

2.1.13 Implementation Schedule (G1.9)

PDR.8-10

Date	Milestone(s) in the project's development and implementation
2014-02-01	BCP Awarded USAID Grant
2015-05-01	Beginning of forest demarcation by chiefdoms. Project start date and beginning of first monitoring (crediting) period.
2016-08-08	Department of National Parks and Wildlife approval
2016-11-01	Infrastructure Development starts
2016-12-01	Beekeeping starts
2017-05-01	Conservation farming starts
2017-07-25	Forestry Department Approval
2017-08-01	CRB-BCP Forest Monitoring kicks off

2018-12-31	End of first VCS monitoring period running from project start date. Thereafter monitoring periods will be annual for VCS and follow the calendar year (e.g. 2019-01-01 to 2019-12-31) except for the last monitoring period which may run from 2044-01-01 to 2045-05-01.
2019-04-16	Approval of the first 7 project area CFMAs by Director of Forestry
2019-9-30	Expected first verification date. Thereafter, verification for VCS will be expected to be completed within 6 months of the end of each 1 year monitoring period. Verification for CCB will be expected to be completed within 6 months of the end of each 5 year monitoring period.
2020-12-31	End of first CCB monitoring period running from project start date. Thereafter monitoring periods for CCB will be every 5 years and follow the calendar year.
2023-12-31	End of second CCB monitoring period.
2025-05-01	Mandatory baseline reevaluation as per current VCS AFOLU rules
2045-04-01	Project crediting period end date.

2.1.14 Project Start Date

PDR.6

Project boundary demarcation work, and activity that leads to emissions reductions, for 5 of the initial 12 chiefdoms participating in the project started in April, 2015 (Appendix A). Therefore, the project has chosen a start date of May 1st, 2015.

2.1.15 Benefits Assessment and Crediting Period (G1.9)

PDR.7

May 1st, 2015 to April 30th, 2015 a total of 30 years. Benefits assessment and crediting period are the same.

2.1.16 Differences in Assessment/Project Crediting Periods (G1.9)

Not Applicable

2.1.17 Estimated GHG Emission Reductions or Removals

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2015-2018	3,820,081
2019	1,339,517
2020	1,546,122
2021	1,705,230
2022	1,949,034
2023	2,083,752
2024	2,270,156
2025	2,431,957
2026	2,622,554
2027	2,775,802
2028	2,973,829
2029	3,019,096
2030	3,368,956
2031	3,266,165
2032	3,554,553
2033	3,420,344
2034	3,607,828
2035	3,778,205
2036	3,615,501
2037	3,647,599
2038	3,626,038
2039	3,950,695
2040	3,400,975
2041	3,639,073
2042	3,365,883
2043	3,001,994
2044	3,156,645
2045	2,660,620

Total estimated ERs	83,598,204
Total number of crediting years	30
Average annual ERs	2,985,650

2.1.18 Risks to the Project (G1.10)

Political Risk

Zambia is a stable and peaceful country with a well-established democracy. Human-induced political risks are therefore low and the project has the approval of central and local government, and traditional authorities. Zambia was one of the first nine UN-REDD Pilot Countries in the World and there is a high level of interest within government for REDD+ pilot projects. This project is significant in that it is the first REDD+ demonstration project in Zambia. It is therefore strategic as a sub-national project which aims to support and feed into the design of a national REDD+ program. The project benefits from solid government support and interest for learning reasons. The project proponent is investing significantly in ensuring that the GRZ is kept well informed of this project, and the project's goals to contribute to national development goals and the national REDD+ strategy.

It is thus reasonable to assume the longevity of the project from a political stability perspective. Zambia is a stable country and BCP has approval and support from all the relevant Zambian Government Departments to implement the project. It is thus reasonable to assume the longevity of the project from a political stability perspective. However, in order to mitigate political risks, BCP proactively consults, engages, coordinates and collaborates with various Government offices, in addition to keeping them regularly informed about the project to keep it in the spotlight.

Income Risk

Project financial viability and breakeven analysis is reported in detail in the non-permanence risk report. The project is set to break even in 2023 (Appendix H).

Community Risks

As described in Section 2.5.3, BCP has consulted extensively with community stakeholders for the LCFP, in order to understand the drivers of deforestation within the project zone and to formulate the most efficient mitigation strategy. BCP has a highly qualified and experienced team that have worked on conservation and natural resource management projects in Zambia and across Africa; this team has successfully led the LZRP as Zambia's first and only REDD+ pilot project for the past 5 years – which amounts to unparalleled experience and success with REDD+ in the Zambian context.

Only projects that have written approval from local residents are being implemented, under the LCFP. The link between project interventions and benefits, and deforestation mitigation efforts and conservation activities, is further cemented through the signing of Community Covenants and similar agreements / documents, that oblige both BCP and participants to comply with project guidelines and connect community benefits to forest protection efforts.

Substantial and repeated crop failure in surrounding communities could potentially lead to increased poaching and charcoal production for financial benefit. More erratic rainfall and pests make this likely, and all of our alternative economic development efforts are aimed at mitigating this risk, by promoting more resilient conservation agriculture techniques, and supporting alternative, sustainable livelihoods activities.

Although there are always (legitimate) concerns about corruption, in talking to local residents about the history of external involvement in their communities, BCP has learned that corruption is not the only risk that could potentially threaten projects: other difficult community dynamics (such as high alcoholism rates, internal jealousy and belief in witchcraft) have threatened or ruined other organizations' projects in the past.

However, BCP's extensive community engagement efforts and strong local presence have served as powerful tools against these problems. Local community members have explicitly told BCP representatives that we have already set ourselves apart from other organizations by conducting sensitization meetings and consulting local stakeholders prior to implementing the LCFP, and by closely involving the local community in the design and implementation of REDD+ activities. As a result, community members report that they feel greater ownership of the projects that we launch in partnership with them.

BCP has engaged in significant capacity-building efforts among our staff, local community members and local implementation partners, so that we can best avoid and respond to such problems. We are also managing the risk of corruption through documented partnerships with local community institutions, and other BCP-supported community consultation and implementation structures. In 2017, BCP facilitated two rounds of institutional capacity building training programs for the members of CRBs and VAGs as local community implementing partner institutions. These have been followed by an ongoing monitoring program. Under our field-based extension model, BCP has a consistent presence in local communities (thanks to a strong presence of locally-hired staff), regular site visits, monthly monitoring, and other activities that allow for regular engagement, support and monitoring to take place. Building on our experiences from the successful pilot LZRP, BCP has endeavored to create tight systems that do not leave room for manipulation or abuse, and, most importantly, by ensuring that our projects bring tangible community benefits, we have worked hard to reduce any incentives that individuals might feel to engage in destructive or corrupt activities.

BCP has signed numerous, enforceable agreements with community institutions, outlining clear commitments in terms of managing community forest and financial resources. These agreements provide

a standard against which performance can be monitored, measured and tracked, to reduce risks to the program.

Another key risk mitigation strategy is the development of substantial community benefit projects and systems (conservation community covenants).

Fire

The threat of fire to the project's carbon stocks – the Miombo woodlands and their associated biodiversity are discussed in detail in the non-permanence risk report (Appendix H).

In addition, BCP has agreed to implement fire management with communities (Appendix A).

The system has evolved with fire and if completely excluded from the conservancy as doing so would lead to the complete loss of the grass sward (Trapnell 1959), and would thus adversely impact the biodiversity through the loss of many grass species and large herbivores, as well as an entire suite of.

Drought

Drought poses another potential natural risk to the ecosystem. However, just as with fires, the woody vegetation is adapted to drought as the region is exposed to periodic droughts. Droughts are thus not considered a serious natural risk to the project.

An early burning fire management program will mitigate the risks that drought poses in contributing to late season fires that are potentially damaging to vegetation.

Elephant Damage

It is difficult to determine at what density elephants will start having a significant impact on woody vegetation as the impact is site-specific, however, it is safe to say that elephants do have some impact (Guldemond & Van Aarde 2008) on woody species density. What needs to be kept in consideration is that large tracts of miombo woodland remain intact, despite the presence of elephants (Ribeiro et al. 2008). It is unlikely that elephants will increase to such high numbers as to actually threaten the existence of the project.

Natural Disasters

The project area is geologically stable and there are thus no significant threats from seismic activities. Similarly, the risk of flooding is very low as the project area is situated on top of an escarpment and the soils are mostly sandy, thus drain easily.

With a view to mitigating the risk of natural disasters, BCP's community projects are designed to promote and build local stakeholder resilience to disasters, including the negative effects of climate change.

Projects such as crop diversification, income diversification, and our conservation farming training program—which is designed to introduce improved farming techniques—are intended to mitigate the negative effects that disasters, such as the army worm pest that was declared a national emergency in 2012 and again in 2016, do not destroy local livelihoods, and to ensure that households are more resilient in the face of changing rainfall patterns, including drought.

2.1.19 Benefit Permanence (G1.11)

There are long-term contractual arrangement in place with the rights holders and the GRZ to implement the project. Furthermore, the project has been carefully planned to become self-sustaining in the long run. BCP anticipates that deforestation mitigation activities will become self-sustainable in the long run.

There are a number of depleted areas in Zambia, yet few projects that aim to innovate products and strategies that are conservation compatible and rehabilitate areas with traditionally low value. LCFP business model incorporates innovative niche tourism development, the development of a breeding center for wildlife, and a center for educational and training opportunities related to conservation.

Deforestation mitigation activities, such as conservation farming initiatives and bee-keeping, have been designed from the outset to become self-sustainable (see Theory of Change). They build upon existing livelihood activities that are forest incompatible, but improve them, and make them forest compatible. Thus, the chance of success of these projects is higher as BCP does not need to catalyze significant attitude or behaviour changes.

2.1.20 Financial Sustainability (G1.12)

Financial sustainability of the project is discussed in (Appendix H). Breakeven point is predicted in 2023.

2.1.21 Grouped Projects

1) Eligibility Criteria for Grouped Projects (G1.14)

Future project instances (Table 4) will meet the applicability conditions specified in section 4 of VM0009 v3.0.

The relevant conditions include:

- The project type for future project instances must be avoided unplanned deforestation. Accordingly, the end land use in the baseline scenario must be non-forest.
- The baseline drivers and agents of deforestation in future project instances must be of the same type as the current project instances - clearing of forested land for expansion of small-holder farming and, or charcoal production.
- Future project accounting areas must have met the Zambian definition of forest for at least 10 years prior to the project start date.
- Some point within 2km of the perimeter of future accounting areas must have been converted from forest to farming.
- Future project accounting areas must be located in a tropical ecosystem and must not contain peat soil.
- For each new accounting area, a reference region meeting the requirements in terms of size and similarity to the accounting area as per section 6.8.1 of VM0009 v3.0 must be delineated.
- For any new reference regions that may be required, there is sufficient historical imagery meeting the requirements of section 6.8.4 of VM0009 v3.0.
- Project activities for new project instances are the same as those described in this PD.
- The project proponents can access any new leakage or proxy areas that may be required for monitoring related to new project instances as per sections 8.3.2.1 and 6.4 of VM0009 v3.0.

Future activity instances must also have the same governance and land tenure system and social parameters. They are subjected to the same biodiversity and community without project scenarios described in section 2.2.3.

2) Scalability Limits for the Grouped Projects (G1.15)

Scalability limits are primarily land use or geographical. In proximity to the LCFP are the Mozambique and Malawi borders. This project cannot scale into other countries, and we are restricted by Provincial Boundaries (Lusaka and Eastern) as we have not done a baseline analysis for other Chiefdoms.

Additional scalability limits include areas within Zambia by these two international borders. They are largely devoid of biodiversity due to human causes. The project expanding to areas south of the Great East Road may not generate net biodiversity benefits. To mitigate these risks, BCP ensure that we spatially know the locations of international boundaries. We also work hard to ensure that there are clear community expectations at a Chiefdom level of the areas that are targeted for positive intervention. This

avoids a case whereby Chiefdoms which are close to the Mozambique border have VAGs pressuring for benefit sharing from distant forest areas. To help mitigate this risk of high expectations, we refer to the CCBA Standard which provides a guide of working in communities up to 20 kms away from Project Area boundaries where communities impact the Project Area.

Project activities have a positive impact on biodiversity, so there are no thresholds beyond which there would be negative impacts on biodiversity.

In order to expand beyond the current project area, additional economic, capacity and managerial resources will be required. BCP has initiated the process of fundraising for what is informally referred to as “LCFP Expansion”. The project is well placed to expand and benefit from economies of scale as there are already teams, offices, vehicle fleet and project aircraft close to areas which are targeted for grouping. Therefore, the relative cost per hectare to expand is significantly lower than starting a new VCS project from scratch. We estimate that start up costs for LCFP Expansion could be 50% or less per hectare than LCFP as it stands today given these economies of scale and refinements in BCP’s systems and processes since LCFP development was launched in 2013.

3) Risk Mitigation Approach for Grouped Projects (G1.15)

The strategy for expanding capacity is both internal and external. Internally, BCP has invested significantly in managerial training for mid to upper level management. The structure through which this happens is called the “Leadership Development Team” (LDT). The LDT meets on average monthly for up to a work week to focus on internal training and capacity building. As the LCFP teams are dispersed, bringing managers together is important. The LDT is a group which mixes experienced managers with junior staff identified to have leadership potential. The goal of REDD+ Week is to build a management team with consistent processes, tools and behaviors.

Externally, the strategy for expanding capacity is to fund raise and seek strategic partners. By expanding our investment base, BCP is able to hire more skilled staff from the open marketplace to help us to scale up. BCP has a partnership with Maliasili Initiatives whose mission is to enable local conservation organizations to become more effective and achieve greater impact (www.maliasili.org). Maliasili assists BCP in planning for scaling and staff development.

At the senior management level, BCP has invested in external management training courses at Harvard Business School over the 2016-2018 period to expand capacity in anticipation of scaling LCFP beyond its current scale.

In order to mitigate risks to cash-flow, BCP has partnered with Peace Parks Foundation, an international conservation organization. This partnership is designed to seek offset buyer partnerships through PPF’s corporate network generate conservation finance for LCFP even as it scales. Scale is an important part

of the PPF and corporate partnerships targets as evidenced by the MoU shared with the VVB which specifically mentions the area north of LCFP as a target partnership area.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Land Use Scenarios without the Project (G2.1)

The baseline scenario is the same as the conditions existing prior to the project initiation – conversion of forest to non-forest through unplanned mosaic deforestation. Please refer to section 3.1.4.

2.2.2 Most-Likely Scenario Justification (G2.1)

Please see section 3.1.4 for evidence of the most likely land use scenario. Evidence was generated through PRA and a detailed study correlating the drivers with deforestation, remote sensing and consulting peer reviewed literature (Appendix K).

2.2.3 Community and Biodiversity Additionality (G2.2)

In addition to the description of the likely community and biodiversity outcomes in the absence of this project, the project activities leading to the community and biodiversity outcomes would not have been implemented without the revenue from greenhouse gas emissions reductions due to significant barriers as demonstrated by the VCS VT0001 Tool. See section 3.1.5 for details.

Community

As other sections in this document have described, the activities of this project are taking place in Game Management Areas (GMAs), where there are existing areas of intact forest, as well as areas of community settlements. These areas exist within a Chiefdom structure, governed by local traditional leaders. Clearing of forested areas takes place mostly near to existing settlements, where trees are cut to allow for expansion of crops (agriculture), household expansion, or other developments (e.g.: clearing for new roads or infrastructure). Members of local communities may access areas of forest near to existing settlements in order to cut trees for poles, or some commercial timber or charcoal production. Legally, land and other resources within these areas are to be co-managed by the community and Government entities. Within the legally-recognized traditional leadership structure, the authority to regulate land-use allocations is vested within the Chief. Local traditional leaders, such as Headmen and Indunas, are vested with the authority to make local land-use and resource-decisions on behalf of the Chief, at a local (e.g.:

village) level. Licenses are technically required to be issued through the local government (Forestry Department) in order to clear trees for charcoal or timber, and any new development activities (such as road or public infrastructure) are supposed to be regulated through local authorities, such as the District council.

Although land and resource allocations are technically supposed to be regulated by this combination of local legal and traditional structures, in practice, there are problems with enforcement, as well as unplanned allocations – meaning that experience shows that land and resource allocations do not always place in line with any sort of sustainable land-use plan, even where they exist. Practice shows that there are ongoing tensions between local communities, traditional leaders and Government agencies with regard to General Management Plans (GMPs) that are legally supposed to exist in each GMA, to regulate local land and resource allocations. Studies show increasing rates of deforestation, unplanned clearings and declining biodiversity in the areas that this project is intending to protect.

Should existing conditions continue, local forested areas would continue to be threatened by unplanned, unregulated, informal/undocumented and even technically “illegal” allocations of land and other natural resources. With the large and increasing population growth rates predicted for these same areas in the upcoming years, these threats are only expected to increase, as pressures to clear new areas for new homesteads and new villages, to provide more infrastructure and development for these new communities, and for expanding agricultural and other commercial activities, to sustain these growing populations, are only expected to increase.

In the absence of this project, as relevant studies have indicated, these local deforestation pressures are expected to continue, and increase, in upcoming years.

The LCFP is designed as a specific intervention to address these drivers of deforestation, primarily by providing funding and support to activities that promote increased governance, collaboration between Government and traditional authorities, land-use planning and enforcement mechanisms, biodiversity monitoring, community engagement, sustainable livelihoods activities and incentives-structures. All of these interventions are collectively intended to help reduce the current drivers of deforestation, as listed above. In the absence of this project, no other viable source of funding or support is predicted to become available that would comprehensively address this suite of drivers in the same way.

Biodiversity

The project area falls within and adjacent to Game Management Areas (Table 1, Table 2, Figure 10) which are designated buffer zones to national parks and human settlement and consumptive wildlife use is permitted within GMAs. The sustainability model of GMAs is based on generating revenue from trophy hunting and tourism activities and ploughing it back into local conservation efforts and community development through revenue sharing and employment of village scouts through local community resource boards.

This is all governed by the Zambia Wildlife Act. The unfortunate reality though is that the GMA system is not working well¹². Wildlife populations are severely depleted in most GMAs and benefits are simply not flowing to local communities. The DNPW does not generate enough income through leasing of hunting concessions and hunting licenses to efficiently manage wildlife populations and poaching is thus rampant. The result is that key game species populations are depleted to such low levels that growth rates are very low.

Lindsey *et al.* (2014) found that the primary cause of the situation is the lack of community buy in and ownership of the resource. The role of the CRBs and the DNPW are vague and income generated often gets misspent or go to a few local elites. There is no clarity who exactly constitutes the local community. The result is a negative feedback loop that sees massive in-migration to clear land, hunt meat and fell trees before others come in and do it. Resources within DNPW are so restricted that this situation is also taking place within national parks where no human settlements are allowed. Also, some parks have been heavily encroached.

Severe loss of wildlife and habitat destruction will thus in all likelihood continue under the current legal, policy and economic framework without interventions from the LCFP that will generate additional income to bolster conservation efforts.

2.2.4 Benefits to be used as Offsets (G2.2)

No climate (greenhouse gas emission reductions), community or biodiversity benefits will be used for any other offset projects.

2.3 Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

The draft PD for this project was shared through informal meetings with project stakeholders in each of the 12 participating Chiefdoms. Once the project is validated and the first monitoring report is verified, meetings will be held in each chiefdom to share the final PD and monitoring report. This process will be repeated after each monitoring period. Due to high levels of illiteracy, we combined the dissemination of these informational materials, with presentations and meetings led by local staff and implementing

¹²Lindsey, P.A., Nyirenda, V.R., Barnes, J.I., Becker, M.S., McRobb, R., Tambling, C.J., Taylor, W.A., Watson, F.G. and t'Sas-Rolfes, M., 2014. Underperformance of African protected area networks and the case for new conservation models: insights from Zambia. *PLoS one*, 9(5), p.e94109.

partners, who have been provided with training to allow them to accurately describe the content of the PD, to explain the verification process and the significance of these documents, and to respond to any questions or concerns that may be raised by community stakeholders to the project.

2.3.2 Dissemination of Summary Project Documents (G3.1)

In addition to a full draft of the PD, a summary of the PD was also shared in the same meetings described in section 2.3.2. Project documentation has been made available to stakeholders throughout the project development process, including Free, Prior and Informed Consent (FPIC) stakeholder sensitization activities that were initiated in 2014, and which have continued through 2017. Informational handouts clearly outlining the proposed project model, and accurately describing REDD+ activities, were developed and have been widely distributed to community (and Government) implementing partners, throughout this time. These handouts, and other informational materials (such as informational posters about REDD+, BCP and the CFP) have been distributed to each of the 12 partner Community Resources Boards (CRBs) that BCP is working with, to implement REDD+ activities under the LCFP. Throughout nearly 3 years, from 2015-2017, BCP has provided capacity building training to CRB members, as well as local Government implementing partners, and locally hired staff, to ensure their ability to accurately explain and disseminate information about REDD+, in line with FPIC requirements. BCP staff also developed and launched a standard “sensitization program,” that provided critical information about REDD+ and the proposed activities of the LCFP, and which made information available in a multitude of local languages, presented by locally hired and trained staff, who ensured that this information was disseminated in locally appropriate ways. In combination with these sensitization programs, BCP has ensured that informational materials have been made available to local community stakeholders in each Chiefdom, and that there are trained staff and implementing partners available at all times, in each Chiefdom, to undertake ongoing sensitization activities, to accurately disseminate information about REDD+, and to be available to answer any questions or concerns that stakeholders may have, at any given time.

2.3.3 Informational Meetings with Stakeholders (G3.1)

As described above, informational meetings have been held in each of the 12 partner Chiefdoms for the LCFP, with mechanisms for further information dissemination, and facilitation of community stakeholder feedback at village and VAG levels.

2.3.4 Community Costs, Risks, and Benefits (G3.2)

Information about the proposed REDD+ project model, including potential costs, risks and benefits to communities, has been widely disseminated throughout all 12 local partner Chiefdoms, throughout / during / as part of the FPIC stakeholder engagement processes that took place between 2014-2017. This process is well documented, and shows that meetings were held with stakeholders in all 12 partner Chiefdoms, through a combination of Chiefdom-level meetings (in all Chiefdoms), VAG-level meetings (in most VAGs), village-level meetings, as well as individual sensitization activities targeted at key stakeholders, for example, those living along or near to proposed protected forest area boundaries.

In particular, this information was presented, and stakeholder feedback was solicited, throughout a series of participatory meetings that took place following PRA/PLA techniques, in each of the 12 partner Chiefdoms, as part of the forest identification process.

The forest selection process, itself, was a stakeholder-led process, involving representatives from community institutions (CRBs, VAGs), as well as traditional leaders and local Government partners. The overall process was facilitated by trained staff members from BCP, who had received specific training in FPIC and PRA/PLA techniques. This process has also been well documented, with program records available to show the processes followed in each Chiefdom, to demonstrate that community leaders were sufficiently involved in this process, throughout the process -- and also showing that the program was fully described, and adequate information provided, to allow community members to ultimately make informed decisions about the areas of forests that they identified for protection through the implementation of REDD+ activities. These messages were reiterated at the time of signing community consent letters for REDD+ implementation (at the end of 2016, following the completion of nearly 2 years of documented stakeholder sensitization and engagement activities), at the time of signing Conservation Fees Guidelines (in early/mid 2017), and at the time of signing Conservation MOUs (at the end of 2017).

Throughout this time, stakeholders have received considerable amounts of information through series of meetings and regular interactions with BCP, as well as through informational materials that have been widely distributed, and made available. This has provided stakeholders with adequate information to make informed decisions, in the first place (regarding whether or not to consent to the REDD+ project, and then, to identify the areas where they would like REDD+ activities to be undertaken). Stakeholders have also, by now, experienced approximately one year of consistent activities and messaging that has further help them to understand, practically, the implications of undertaking REDD+ activities (including potential costs and benefits), which has further allowed them to make ongoing decisions about whether or not to consent to ongoing REDD+ implementation, in advance of project verification. The fact that community stakeholders continue to remain supportive of REDD+ activities being undertaken in their Chiefdoms, now that they have begun, is a strong testament to their ongoing understanding of, and support for, the project taking place.

2.3.5 Information to Stakeholders on Validation and Verification Process (G3.3)

Stakeholders have been informed of verification processes through a series of meetings that have been held at a Chiefdom-level, in each of the 12 partner Chiefdoms. These informational meetings have been held by trained members of staff, as well as implementing partners, including representatives from local Government and community institutions (such as CRBs). The meetings have followed a standard curriculum, with the aim of ensuring that key points are accurately and consistently communicated in a standardized way. Information is shared in local languages, in locally appropriate ways. Informational materials (handouts) outlining / explaining the verification process have also been developed and distributed widely through local community institutions. The curriculum and informational handouts (which complement each other) are available as supporting documents.

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

Similar to the process that we followed for the verification of the LZRP, BCP will ensure that the schedule for the auditor's site visit, and community meetings, will be widely announced -- both through verbal announcements disseminated through local staff, implementing partners and community institutions, as well as through hard-copy announcements (posters, flyers) that will be posted openly in public spaces, to ensure that announcements have been widely, openly, transparently and publicly made. This announcement process will be documented, including photographs taken of announcements being made publicly in communities, in advance of meetings and/or visits taking place.

2.3.7 Stakeholder Consultations (G3.4)

Project design has been significantly influenced by stakeholder inputs, including, especially, inputs from local community stakeholders, as well as local Government implementing partners. The PRA/PLA process that BCP followed in each Chiefdom, which has been well documented, was specifically designed and intended to solicit stakeholder feedback in terms of project design, and ensure that this has been used to inform and shape project design. Stakeholder feedback has been critical, especially with regard to the following aspects of project design:

- Identification of selected areas (boundaries) for **forest protection**, based on a full and accurate understanding of "what it means" to protect an area, under REDD+ (including: potential costs and benefits, as well as allowed / not-allowed activities);
- Identification of critical **community development** needs, to be supported using performance-based funding made available to communities, to help create locally appropriate "incentives" for conservation;

- Collaborative development of an appropriate **forest monitoring** model, to be undertaken by CRB members / community scouts, with support from BCP and through the REDD+ project;
- Identification of critical **capacity building** needs, to empower community and government stakeholders to implement REDD+ activities on their own -- this, too, required an in-depth awareness of stakeholder rights and responsibilities with regards to REDD+ implementation, in order to ensure that we received informed and relevant feedback about the activities that project stakeholders felt prepared, and unprepared, to implement, to allow us to ensure to build capacity in necessary areas, in order to improve the chances for success for the project.

BCP's REDD+ model, under the LCFP, is to empower and support local community and Government partners to implement REDD+ activities, in the long term. In order for us to do this, it has therefore been imperative that we have sought to ensure that our local community and Government implementing partners possess a full knowledge of what a REDD+ project entails, to allow them to provide us with accurate and useful information to inform the design of a project model that is most likely to work, to achieve our shared goals. If community stakeholders are unaware of what a REDD+ project entails, they cannot give us this necessary feedback. As such, we have sought to inform and solicit feedback from our stakeholders, to ensure that they helped us to identify viable areas for REDD+ protection, and viable forest management / monitoring models, as well as community development plans -- all of which they will play a leading role in implementing, with our support. At its heart, BCP's model for the LCFP is a community-led approach to REDD+, with community stakeholders taking the lead in terms of project design and implementation. We strongly feel this is the most appropriate model, and most likely to succeed, in the long term, if communities are aware of what it takes to implement a REDD+ project, and if they are actively involved in the development of plans and models to do so, for themselves, based on their extensive local knowledge, and their lead-taking in decision-making about whether or not to protect or cut trees, on their own local land.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

Ongoing communication and consultation will take place between BCP and local community and Government stakeholders, through the following mechanisms and channels:

- Locally hired BCP staff from each partner Chiefdom - provide 24/7 communication channel;
- Bi-weekly coordination visits under BCP's regular operational model, allowing for regular face-time between BCP team leaders and community institutions / implementing partners. Since these visits will take place regularly, according to a regular and communicated schedule, these will also provide opportunities for meetings with traditional leaders and/or members of the wider community, if they are requested by either side;

- Quarterly partners meetings taking place at a District level, at least, and potentially Chiefdom-level, involving representatives from each Chiefdom / community institution, traditional leaders, local Government partners, and any other local implementing partner organizations;
- Monthly reports from community institutions (CRBs), to be submitted to BCP (as part of supporting documentation to be submitted, in order to receive ongoing monthly payments for conservation and community engagement work, undertaken by the CRB);
- Monthly and quarterly updates / reports prepared by BCP, in collaboration with implementing partners, to be shared with local community and Government stakeholders, as well as shared externally.

All of the above are “internal” channels for communication / collaboration among REDD+ implementing partners for the LCFP, including BCP, and community and Government partners. In terms of communication with a wider range of external actors, the following additional mechanisms are in place:

- Members of the wider community may participate in ongoing sensitization meetings, and/or quarterly partners meetings, which are open to the public;
- Regular (quarterly) newsletters are distributed by BCP, outlining key highlights - anyone may request to join the mailing list;
- BCP regularly posts blogs with updates onto our website - these are able to be viewed by anyone. BCP aims to post at least 1 blog update per month;
- BCP uses social media like Facebook and Twitter to disseminate key messages and updates - anyone may choose to “follow” our work on social media;
- Information is regularly updated and available on BCP’s website;
- Informational materials such as handouts and flyers about REDD+, BCP, CFP (etc.) are available in hard and soft copy, upon request.

2.3.9 Stakeholder Consultation Channels (G3.5)

To have firm and legal consultation channels, communities are required to form community organisations in this case the Community Forest Management Groups (CFMG) to be eligible to enter into a Community Forest Management Agreement (CFMA) and gain access to Community Forest Management (CFM) as provided for under the Forest Act of 2015. The functions of these community organisations include:

- Providing a legal entity that can enter into agreements with other partners;

- Providing a point of entry to communities for the provision of management, capacity building and funds;
- Providing a forum for the resolution of disputes within the community, which are called Project Steering Committees;
- Ensuring equitable sharing of resources within the community following the signing of a CFMA; and
- Providing community members with capacity building in organisational, financial and enterprise management.

Starting with Government, the MOUs have provided for channels of communication and prescribed terms of engagement, defining roles of each party in the LCFP. With communities consultation is documented and channels of communication established from the start. FPIC was used as a tool to enhance information sharing and providing as much information as possible to communities.

- After approaching government at National, Provincial and District level
- Community request for viability assessment was written to the chiefdoms and upon authorisation that is when BCP moved in.
- Then a consent to REDD+ implementation was requested from the Chiefdom after the completion of the viability assessment
- This then led to participatory forest selection using PLA tools
- Start date letters were written and we moved into preparing the communities to enter into Community Forest Management Agreement.

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

BCP's FPIC Guidelines have been designed to ensure participation of all stakeholders on an equal basis (Appendix E). The consultation procedure ensures that all members of community have an opportunity to be present at meetings and that benefits are equally distributed. Records are kept of all meeting participants by gender and so to participation and will be made available to the VVB.

2.3.11 Anti-Discrimination Assurance (G3.7)

BCP has a strict HR Policy that follows Zambian labour laws and international standards. BCP policies do not allow any form of discrimination or abuse by any employees or project participants while participating in project activities. BCP has a policy entitled “WORKPLACE HARASSMENT, DISCRIMINATION AND BULLYING POLICY”. This policy extends to contractors.

Other entities involved in project design, such as CRBs, are Government recognized, and therefore need to comply with Government standards to mitigate discrimination and sexual harassment. For example, All community scout selections are open to men and women. BCP supports the employment of 8 female community scouts who passed shortlisting, selection, basic, advanced, and in service training as well as male community scouts. This means that 30% of BCP-supported community scouts are women. Community scouts are held to strict CRB codes of conduct which are overseen by Government, as Community Scouts are required to work with Government Wildlife Police Officers, who are also governed by strict uniformed services policies against discrimination and sexual harassment.

Therefore, organizations from Government involved in this project such as DNPW and Forestry Department are governed by set Government policies within the laws of Zambia to ensure that civil servants are not complicit in discrimination and sexual harassment. The same applies to CRB which fall under the Wildlife Act of 2015.

In addition, BCP’s agreements with communities consistently say that a definition of non-performance are threats, violence towards or blocking of access of BCP staff and partners to areas of work. In a form, this mitigates against possible discrimination against project staff and project partners.

2.3.12 Feedback and Grievance Redress Procedure (G3.8)

BCP has a well-established and publicly known grievance mechanism that has been designed to address and resolve any problems. All the CFMAs that have been signed by communities have a grievance procedure stipulated covering the contractual agreements regarding the project activity (Appendix A). In addition a Stakeholder Grievance Response SOP is in place, signed by lead members of the engagement team and signed for as received by each CRB and kept on file (Appendix E). The grievance mechanism is clearly outlined and discussed in community sensitization meetings, and contact information for BCP representatives is included on all public communications. Meeting minutes (written records) from all of the community sensitization meetings document that the grievance mechanism has been explained to the local community representatives in attendance—including their right to receive a response within 30 days, and their right to contact third party arbitrators directly should they have any ongoing concerns. In 2018, BCP updated the Grievance Mechanism from the 2013 version.

Grievances that are submitted to BCP will be documented, as will the response to any such grievances. These documentations are copied to the relevant GRZ office to ensure a third party involvement, to

prevent any conflict of interest. Project management will attempt to resolve all reasonable grievances raised, and we commit to providing a written response to grievances within 30 days. In order to track grievances, BCP has established a Grievance Tracker by Chiefdom (Appendix E) which keeps track of dates received and actions taken to close-out of a grievance.

BCP's internal grievance mechanism is clearly outlined in BCP's Human Resources (HR) Manual V.6 2019a (Appendix M). This is guided by BCP's 2016 Disciplinary Code and Grievance Procedure (Appendix M). The grievance procedure specifies the steps to resolution internally. It is approved and stamped by the Government of Zambia Department of Labor.

Our SOPs Concerning 1) Community Engagement; and 2) Social Monitoring reference the 2018 Grievance Mechanism as a tool that accompanies these community engagement processes. The formal process for handling grievances is outlined in these documents which will be made available to the VVB. The grievance mechanism is also written into all contracts (such as Community Covenants) that are signed between BCP and community representatives, such that BCP and local community members are able to hold each other to account with regard to the delivery of services as per the terms of agreements. The Community Engagement Teams are expected to serve as liaisons between stakeholder communities and the project team. In addition, they are expected to communicate any needs, questions or concerns that either side has.

In the event that community stakeholders have a serious concern or grievance about BCP's Community Engagement Strategy, or if there is an on-going or long-lasting grievance among community stakeholders that BCP has been unable to address, BCP's grievance mechanism allows for the involvement of a neutral third-party arbitrator (including legal counsel) that will be able to clarify or resolve the grievance for either or both sides in the dispute. The preferred third-party arbitrator, in the event of disputes, would be a representative from an appropriate Non-Governmental Organization with local presence in Zambia—for example, in the event of grievances concerning land, an organization that deals with land-access and/or land-rights will be involved. In the event that there is a grievance concerning food security, an organization such as the Conservation Farming Unit would be involved. In the event that the community prefers to involve local government, BCP has advised the communities that they are welcome to bring concerns to the District government which is headed by the District Commissioner. BCP has a working relationships with local governments in the Districts of the project zone, and local District government officials have been identified as a mutually-trusted third party for both the stakeholder communities and BCP.

Throughout BCP's community consultation and engagement processes, BCP informs community stakeholders of their right to seek legal counsel in the event that a serious grievance occurs, as a part of BCP's grievance mechanism.

Our relationships with other Implementing Partner organizations or companies will be governed by legally-binding signed contracts. All contracts with BCP employees are governed by the Laws of Zambia, and

any ongoing employment-related disputes or grievances will be resolved through the Zambian Ministry of Labour.

2.3.13 Accessibility of the Feedback and Grievance Redress Procedure (G3.8)

All stakeholders are made aware of grievance procedures at stakeholder meetings, the CFMAs (Appendix A) have a grievance procedure incorporated and feedback has to be given to all grievances raised.

Grievances are documented through the receipt of grievance letters, or meeting minutes. We encourage any verbal grievances to be channeled through Government-recognized local governance institutions such as Community Resource Boards, Traditional Authorities, Cooperatives or Village Action Groups. The Grievances received are filed digitally in a BCP's 'Grievance Register' Tool (Appendix E). This tool is designed to monitor the grievance close out process against BCP's '2018 Grievance Mechanism and Stakeholder Grievance Response SOP' (Appendix E). Grievance letters are signed for as received and responded to in writing within 30 days. Government and any other key stakeholders are copied in to ensure that a response is transparent and shared with community institutions. Where a Grievance is raised at a smaller scale, such as a VAG or Zone, the grievance response letter is followed up by a meeting to discuss the response and any follow up actions needed for closing out the grievance.

Copies of the Stakeholder Grievance Mechanism are shared with Cooperatives, CRBs, VAGs, and signed for as received by each CRB and kept on file.

In addition, all community meetings now have an agenda item titled 'Grievances' to ensure that at each meeting there is a public opportunity to air grievances in a formal, more regular and minuted forum. In order to continue to reinforce the importance of the Grievance Mechanism, training is specified in the BCP SOP on Stakeholder Grievance Responses for community institutions on an annual basis. BCP's goal is to build community relationships to a very strong level through this process of providing pathways for Grievances to flow more easily, as well as binding BCP to 30 day response periods.

2.3.14 Worker Training (G3.9)

BioCarbon Partners has a well-developed training programme for employees and community members and the documentation will be made available to the Validators. Areas of training include: Climate Change and REDD+ overview, biomass sampling and FPIC compliance. New staff members are also provided with Human Resources (HR) training, to ensure that they are aware of their rights and responsibilities as employees.

BCP provides training to all new staff, and we ensure that our project leaders are able to effectively provide training to anyone who is involved in project activities.

The following training courses have been presented to BCP staff either by in-house experts or experts from partner organisations:

A) Compliance with socio-economic standards of the Verified Carbon Standard (VCS) and Climate Community and Biodiversity Alliance (CCBA) Standards

Course Objective

To ensure field extension staff are familiar with the requirements of the above standards and collect the correct field data, as well as follow the correct protocols in their dealings with local stakeholders.

Main topics

- Understanding the principles of obtaining Free, Prior and Informed Consent (FPIC)
- Understanding project baselines and project activities
- Identifying deforestation agents and drivers
- Baseline data collection
- Delineating project areas and project zones
- Using technology, particularly GPS and GIS, to collect spatially referenced data
- Data capturing using electronic devices such as smart phones and tablet PCs
- Data storage and dissemination.

B) Sustainable forest management and charcoal production

Course Objective

To train local stakeholders to become implementers of key deforestation mitigation activities with particular reference to sustainable forest management and charcoal production

Main topics

- The principles of sustainable forest management
- Improved charcoal production using efficient kilns
- Identifying forests suitable for charcoal production and willing participants

- Calculating standing biomass and potential take off
- Mapping of forests using technology such as GPS, smart phones and basic GIS
- Data capturing, cleaning, storage and dissemination
- Monitoring harvesting and charcoal production and ensuring compliance
- Monitoring of transport and sales

C) Forest biomass data collection in compliance with carbon market standards

Course Objective

To train local community members to collect the required forest biomass data for baseline projection purposes as well as future compliance monitoring

Main topics

- Overview of the principles of REDD+
- Overview of the principles of forest biomass enumeration
- Compliance with carbon market standards
- Training in the use of field equipment including GPS, clinometers, tape measures and smart phones
- Plot based sampling – setting up permanent plots and the sampling procedure
- Locating plots using GPS
- Capturing data using smart phones and tablet PCs
- Data cleaning, storage and dissemination
- Soil carbon sampling principles
- Collecting and storing of soil samples
- Collecting and cleaning of soil data

D) Wildlife and forest law enforcement training

Course Objective

To train scouts operating within REDD+ project areas in the enforcement of forest and wildlife laws to ensure adequate protection of core project areas

Main topics

- Wildlife law in Zambia - overview
- Forestry law in Zambia – overview
- The roles and responsibilities of scouts on private property
- Positive community engagement and fostering positive relationships with neighbouring communities
- Engagement with offenders – arrest procedures of poachers and illegal charcoal makers
- Patrol strategies and poaching prevention
- Intelligence gathering

E) GIS, remote sensing and GPS use for forest carbon projects**Course Objective**

To train key BCP field staff in the use of Geographic Information Systems (GIS), Remote Sensing basics and Global Position System (GPS) use in order to streamline compliance of projects with carbon market standards.

Main topics

- Introduction to Global Positioning Systems
- How GPS works
- Navigating and capturing data
- Understanding accuracy and signal strength
- Waypoints, tracks and Metadata
- GPS data in QGIS

- Metadata and data management

F) Principles of Deforestation, Climate Change, REDD Mechanisms and BCP's Model

Course Objective

To train Community Engagement Team (CET) members in the principles of climate change, REDD mechanisms and BCP's model, and to empower them to accurately describe and promote BCP's work at a community level, facilitating sensitization, project support and decision-making. After undergoing this training and gaining relevant community experience, members of BCP's CET present an abbreviated version of this program to local communities, so as to raise their awareness and understanding of the significance and implications of a REDD+ project, in line with the principles of BCP's obtaining "free, prior and informed consent" (FPIC) from local communities. CET members are explicitly instructed to target and involve members of under-represented groups in their sensitization and consultation efforts.

Main topics

- Basic environmental principles – deforestation, emissions, the greenhouse effect, climate change
- Demonstrating how deforestation causes emissions that are linked to climate change
- Definition of REDD and what REDD projects entail
- How forest protection relates to REDD
- The main aims and objectives; including forest protection and community benefits
- The model, which focuses on partnerships with the local community
- Identifying causes and drivers of deforestation
- Forming partnerships with local communities to provide alternatives and incentives to prevent deforestation
- Education training: promoting environmental awareness, describing REDD and BCP to communities
- Sensitization and Consultation targets: We aims to inform and involve all community members, and CET staff must ensure to provide relevant training to under-represented people living within the community.

Training is provided to new BCP employees and project participants at the time of their recruitment/beginning of involvement with BCP projects. The BCP Guide to Community Engagement

handbook contains a detailed description of the training that members of BCP's Community Engagement Team receive, as well as an outline of BCP's overall Community Sensitization and Consultation process and Zone Development Committee formation and development process.

As BCP's Guide to Community Engagement and SOP Concerning Community Engagement describe, BCP staff and project participants receive regular and ongoing training that is intended to promote local involvement in and ownership of projects, as well as to build local capacity and promote self-sustaining projects that benefit from local knowledge and oversight.

New BCP staff members are provided with relevant training, which has been designed by relevant staff members. Members of BCP's community-based Plot Team, for example, have received training from senior managers (including BCP's Chief Technology Officer), and have access to handbooks and SOPs that have been specifically created to provide on-going information and training for employees. Similar hands-on trainings and specially designed training programs have been developed for BCP's Forest Protection Officers ("Scouts"), as well as our Community Engagement Team.

2.3.15 Community Employment Opportunities (G3.10)

BCP adheres to all employment equity laws and prioritizes employing local community members. Approximately 65% of BCP's entire workforce is from communities in which we work. Job opportunities are widely advertised and publicized amongst local community members and qualified community members are given priority. A goal of BCP is to create as many jobs, especially skilled, from the Chiefdoms in which we work. Where skills may not be present, we invest heavily in building them, such as the Forest Monitoring Teams, Community Scouts and Chiefdom Engagement Officers.

BCP is an equal opportunity employer and prides itself in giving people from all walks of life an opportunity for employment. Furthermore, we abide by the relevant Zambian labour laws that prescribe equity.

BCP employs people living within the project zone wherever possible. All staff employed has had training specific to their positions e.g. scout training. When jobs cannot be filled from local communities, then the recruitment net is widened up nationally through Zambian websites, social media and BCP's website. The entire Forest Monitoring Team, Community Scouts, some community engagement staff, majority support staff are all local project zone members. In an area that historically had very few sources of formal employment, BCP's policy of local employment first will likely result in the project becoming a key source of local employments and benefits.

BCP's job application process is straightforward and transparent. When announcing positions, BCP releases a job advertisement that outlines the position, expectations and qualifications that are being sought after. All job descriptions clearly state that BCP is an Equal Opportunity Employer, and that we

accept applications from members of all community groups. Job descriptions are physically distributed among communities, and public announcements are made in local languages at community meetings. To apply for a position, candidates are simply expected to submit a CV and cover letter/letter of interest to BCP. In the case of community applicants, BCP allows them to submit less formal lists of qualifications (in the place of a CV), and BCP staff members are instructed to notify relevant BCP managers in the event that an applicant has expressed serious interest but is unable to complete a written application (for example, if they are illiterate). In some cases, BCP will conduct a trial of several days for interested applicants for a more practical position (for example, an engagement position). This is to gauge skills and attitude fit with BCP's team, and enable the candidate to make an informed choice about committing to the field and community nature of BCP's work. BCP will clearly disseminate information about the place, date and time at which interested applicants are expected to convene, and the hiring process and expectations will be clearly communicated to all interested applicants at that time.

BCP staff members are required to ensure that information about employment opportunities is clearly disseminated to members of all community groups. BCP staff members who are involved in hiring processes are expected to provide equal consideration to all applicants for a position, so long as they meet relevant criteria, through a process which is very sensitive to mitigating nepotism.

As an example for how workers are selected, an example is Community Scout selection:

- Jobs are advertised through community institutions like CRBs and in public places.
- Applications are received by the CRB and shortlisted by a panel of CRB, DNPW and BCP staff based on minimum criteria like educational qualifications.
- Shortlisted candidates are interviewed by a panel and quantitatively scored.
- Shortlisted applicants are given a health check by Ministry of Health before embarking on an 8km timed run. The health check is to ensure no one puts themselves at risk on the run. Jobs are so scarce in communities that there are known cases of candidates pushing themselves so hard with health issues that they die on the timed run.
- The top 5 candidates per Chieftdom are selected to go for attrition training based on their physical aptitude and qualifications for the job.
- In the case of Community Scouts, a higher number is 'selected' for training than there are available jobs. Training lasts from 6-12 weeks and is used to weed out those who are not fit for the job. Through this method, the mentally strongest and most committed qualify to become Community Scouts.

The advertisement method through the CRB enables all VAGs to receive the information, as CRB's are comprised of VAG elected representatives. The transparency of the interview panel working as a group,

and the selectiveness of the timed run ensures that marginalized people have an equal chance to compete, and succeed.

2.3.16 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

There are various legislations that have been enacted to protect the workers in Zambia. Some of these laws are:

The Employment Act (CAP. 268):

Establishes guidelines on employment of an employee in Zambia to be read together with the Minimum Wages and Conditions of Employment (General) Orders. It provides for dismissal of employees and the grievance procedures to be followed as well as basic employment rights relating to leave, working hours, maternity leave, minimum contents to be included in written contracts of service; retirement and redundancy benefits etc.

Minimum Wages and Conditions of Employment Act (CAP. 276) (Statutory Instruments (SI) No. 2 & 3) SI No. 46 of 2012 and the General Order, 2011:

Establishes minimum guidelines for remuneration and conditions of service for employees in Zambia setting out a minimum wage for specific sectors and basic rights and conditions of service.

Industrial and Labour Relations Act (CAP. 269):

Establishes guidelines on employee and employer organisations and on the process of collective bargaining and dispute resolution. It provides for the formation of trade unions and employers representative organisations and recognises the right to strike under specified circumstances. It provides for the settlement of disputes, strikes and lockouts as well as for the use of the Industrial Relations court at which employees can have their grievances and cases heard and determined.

BCP has a professional human resources and administration management team in place in Zambia that is familiar with Zambian labour law and ensures strict compliance to these laws. The project meets or exceeds all applicable laws and regulations covering workers rights. An example of this compliance is related to minimum wage. In 2012, the GRZ increased base minimum wages by approximately 67%. Rural agricultural land units (such as game ranches) in other parts of Zambia are reported to pay some staff according to lower agricultural "rates". BCP met or exceeded urban minimum wage requirements for project area staff, many of whom are from the local community, in some cases doubling what staff made under previous employers. BCP has also introduced benefits such as staff rations (removing food costs from staff), a benefit which previously was not in place.

Carbon finance will enable the project to build out improved and more comprehensive incentive structures to attract and retain (not to mention, expand the size of the workforce) the best staff. Combined with professional management, training, equipment and investment in staff, a vision of the project is to become one of the most desirable places to work in Rufunsa District, Lusaka Province, the conservation sector more generally in Zambia, and even regionally.

BCP staff are informed of their rights through various ways. These include:

- Through discussions regarding contract signing and contract renewals. As BCP contracts are been legally reviewed by a Zambian law firm multiple times, and attested by the Zambian Government Department of Labor, they are confirmed as in conformance with the law.
- For staff with limited English, such as community employees, extra measures are taken to verbally translate contracts and address all queries and concerns.
- All staff are provided with a copy of BCP's HR Manual. In 2016, staff were given an opportunity to form a committee to review contract templates and the HR Manual and to provide inputs.
- BCP is in the process of sourcing for an outsourced HR function which provides for a mobile officer to travel to each of the 6 districts in which BCP currently works to provide a more regular, in person, mobile presence to consistently address information regarding employment laws.

2.3.17 Occupational Safety Assessment (G3.12)

Scouts will receive training in firearm safety and arrest techniques from government approved trainers. Furthermore, their intimate knowledge of animal behaviour (having all grown up in the bush) ensures that all scouts know very well how to avoid dangerous situations with animals such as elephant, lions and snakes.

In the unfortunate case of a crisis, BCP has a reaction plan in place. Additionally, ever since BCP took over management of Rufunsa Conservancy, scout teams have access to transport and communication, as do all other teams. All scouts are equipped with suitable boots and protective uniforms. Scouts have also been issued with strict official rules of engagement procedures to minimize risk and exposure to armed poachers.

Participants in other activities, such as eco-charcoal and conservation agriculture projects, are not employed by the REDD+ project, but their safety is nonetheless a concern. All participants in the eco-charcoal project are seasoned producers and are aware of the risks of cutting wood, packing kilns and removing charcoal. We will, however, equip association members with safety clothing, gloves and boots, and invest in additional training to ensure safe production practices.

Prior to participating in any company supported activity, the company requires participants to openly discuss any potential risks involved in the activity. Staff members are also expected to communicate any potential risks involved in the activity, and to provide participants with any necessary Personal Protective Equipment (PPE), such as work gloves or safety boots. Participants in BCP-supported projects are explicitly instructed not to take unnecessary risks with their personal safety. BCP staff members are expected to provide regular oversight of project activities, and they are required to communicate any concerns about safety to BCP's management team.

Vehicle accidents are a risk to worker safety. Risks will be minimized through maintaining a high standard of vehicle maintenance, and ensuring that drivers are current, well-trained and adequately monitored. BCP has had two intensive vehicle training sessions for all drivers of vehicles by two qualified instructors. The training involved intensive road and off road driver training, defensive driving and vehicle recovery. BCP drivers sign onto a comprehensive vehicle policy (Copy to be made available to Validator upon request) which strictly outlines company vehicle policy, intoxicants, limits of geographical use, speed limits set internally and an un-authorized passenger policy. In spite of the inherent risks of vehicle accidents in Zambia and on rural dirt roads, BCP is confident that we have taken as many measures as possible to reduce this risk. Our emergency response capacity was recently tested when an employee was involved in a serious motorcycle accident. Organizational management, Project-level management, partners, and our F&A team responded immediately to a life-threatening situation which resulted in the employee being checked out of a government hospital and checked into a private hospital (at company expense). The employee received professional healthcare and was fit for work a month after being discharged.

The Human Resources Manual contains a section concerning Occupational Health and Safety. Since all new employees receive an induction that includes HR orientation, it is expected that all BCP employees will be aware of their rights and responsibilities insofar as promoting workplace safety is concerned. We will, however, equip association members with safety clothing, gloves and boots, and invest in additional training to ensure safe production practices.

2.4 Management Capacity

2.4.1 Project Governance Structures (G4.1)

The project is being implemented by BCP, in collaboration with the Government of Zambia, and local communities.

Local community partners include 12 partner Chiefdoms, which are legally and traditionally recognized entities in Zambia. Each Chiefdom is governed by a Chief or Chieftainess, who is legally and traditionally recognized. The LCFP is being implemented through Community Resources Boards (CRBs) as local governance institutions with authority operating at a Chiefdom level, with the authority of the local

Chief/Chieftainess, and government, behind them. Chiefdoms are comprised of many villages, of varying sizes and populations. Villages and groups of villages are governed by traditional leaders known as “Headmen” and “Indunas,” who are recognized within traditional governance structures. Groups of villages are clustered into Village Action Groups (VAGs), that operate on a sub-Chiefdom governance level. CRBs are comprised by representatives from each VAG within a Chiefdom. While Headmen and Indunas are appointed positions recognized within local traditional leadership structures, VAG and CRB representatives are locally-elected positions.

The LCFP is taking place with the awareness and support of the Government of the Republic of Zambia, in line with the Laws of Zambia. Key Government implementing partners include the Forestry Department (FD) and Department of National Parks and Wildlife (DNPW). The LCFP also works through local Government structures at Provincial and District levels, including the Provincial authorities and District authorities, for each of the 2 Provinces and 4 Districts where project activities are taking place.

Key roles and responsibilities of each of these implementing partners can be summarized as follows:

- BCP: Responsible for overall project management, verification compliance and funding
- FD have the overall responsibility for policy formulation and coordination; development of standards and guidelines; law enforcement; forests inventories; and research and development. In addition, FD is responsible for the provision of technical guidance to all actors in forest protection.
- DNPW: Responsible for providing policy and technical guidance on Operations within the game Management Areas
- Provincial and District authorities: Responsible for formulation and enforcement of by-laws relating to Community Forest Management within their jurisdiction.
- Chiefs: Responsible for administration and management of forests within their areas of Jurisdiction
- CRBs: are key actors in planning and management of forest and investment in forestry at local levels and play a key role in participatory forest management

2.4.2 Required Technical Skills (G4.2)

Key technical skills required to implement the project successfully include:

- Community engagement skills, including: community mobilization, consultation, PLA/PRA techniques, sensitization, conflict-resolution, and awareness/respect for local practices.
- Biodiversity assessment and monitoring skills
- Carbon measurement and monitoring skills, including
- Government engagement skills, including: government engagement and consultation, knowledge of and respect for local laws, ability to ensure compliance.
- Business development and sales and marketing skills, relevant to the ability of the project to develop a sustainable financial model, based on the generation and sale of verified credits, to fund ongoing project activities.

2.4.3 Management Team Experience (G4.2)

The LCFP management team possesses significant and unique skills and experience relevant to the implementation of large-scale land management and carbon projects in Africa.

Since 2012, the BCP management team has been responsible for the development and implementation of the Lower Zambezi REDD+ Project (LZRP), which was the first verified REDD+ activity in Zambia. The LZRP was also the first REDD+ project in Africa, and only the second in the world, to achieve gold-level validation against all three criteria of the Climate, Community and Biodiversity (CCB) standard. The BCP management team has continued to successfully manage the LZRP throughout the past 6 years, and has achieved four verifications – a globally unique milestone.

Based on our demonstrated achievements, BCP was awarded a Cooperative Agreement from the United States Agency for International Development (USAID) to implement the Community Forests Program (CFP), in partnership with the Government of Zambia and local communities.

Since 2014, BCP has been working to implement the USAID-funded CFP, implementing activities across nearly 2 million hectares of forests and community land in eastern Zambia, relevant to biodiversity monitoring, community engagement, and overall REDD+ implementation.

The BCP management team possesses 14 years of collective experience in the implementation of REDD+, conservation, and carbon-related activities. Our team of core managers includes competent specialists in conservation, ecology, carbon, community engagement, business administration and sales and marketing. We have a solid team of Zambian leaders. We are supported by implementing partners from Government institutions with relevant authority and skills to assist in forest and biodiversity management and community engagement.

2.4.4 Project Management Partnerships/Team Development (G4.2)

Two key partners provide expertise critical to our livelihoods work under the LCFP. Bee Sweet is a for profit firm focused on selling honey on the international market. It uses an innovative approach of out sourcing hive management to farmers through bee keeping “mentors” who distribute and help farmers manage their hives. In this way Bee Sweet ensures the technical side of honey production, one of the key problem areas for farmers. The other common problem honey producers often have is access to markets. Bee Sweet also handles this side of the equation by guaranteeing farmers a market at a good price for the honey they produce. There is room for tremendous growth in honey production in Lusaka and Eastern Provinces.

The other key partner is the Ministry of Agriculture, through the District Agriculture Coordinating Officers referred to as DACOs. The DACOs provide training to lead farmers in Conservation Farming techniques. The government of Zambia has invested in training their agricultural extension agents in basic Conservation Farming techniques that help farmers intensify their agricultural activities, build in resilience to climate change and improve productivity per hectare. The LCFP works through an MOU with each District Agriculture office.

2.4.5 Financial Health of Implementing Organization(s) (G4.3)

The financial health of the main implementing partner (BCP) is presented as part of the non-permanence risk assessment (Appendix H). The appendix includes financial analyses for both BCP REDD projects.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (G4.3)

BCP adheres to the strictest local laws further bolstered by internal policies preventing corruption and unethical behavior. There is a zero tolerance policy around corruption. Furthermore BCP is an equal opportunities organization and our HR policy and recruitment process prevents nepotism and cronyism.

2.4.7 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

There is no commercially sensitive information that has been excluded.

2.5 Legal Status and Property Rights

2.5.1 Statutory and Customary Property Rights (G5.1)

The 1995 Lands Act vests all land in the President for and on behalf of all the Zambian people. It provides that land may be administered under two tenure systems: statutory and customary tenure. While statutory land is administered in accordance with written laws, by government officials, customary land is administered by traditional authorities based on unwritten and localised customary laws.

The resources in the Project Zone are guided by the 2015 Wildlife Act and 2015 Forest Act. the Project zone is in the Game Management area that is an area of land declared under section twenty-eight of the Wildlife Act 2015; “28. (1) The President may, after consultation with the Minister and the local community, by statutory order, declare an area of land within the Republic to be a Game Management Area for the sustainable utilisation of wildlife and for the purposes of this Act, and may, in like manner, define or alter or extend the limits of the area or order the area to cease to be a Game Management Area.

The 2015 Forest Act states that 3. The ownership of all trees standing on, and all forest produce derived from, customary areas, National Forests, Local Forests, State Land, botanical reserves and open areas is vested in the President, on behalf of the Republic, until lawfully transferred or assigned under this Act or any other written law.

To harmonise and safeguard these property rights and guarantee the project permanency the Department of National Parks and Wildlife (DNPW), Forestry Department and Traditional Authorities have been consulted. letter of support and Memorandums of Understanding were signed with the Government departments. Importantly, based on the Forest Act of 2015 there are signed agreements with all community groups and Community Forest Management Groups have formally been established.

2.5.2 Recognition of Property Rights (G5.1)

The evidence of right of use, is enshrined in the Chiefdom Community Forest Management Groups (CFMG) Agreements between BCP and the community structures in place the CFMG. In respect of a GHG emission reduction, the unconditional, undisputed and unencumbered ability to claim that the project will cause such reduction is given through the clauses pertaining to the project activities and monitoring arrangements in the Community Forest Management Agreements. There is no transfer of any property rights to Biocarbon Partners. For private ranches BCP has entered into a contractual arrangement and there will also be no transfer of property rights.

BCP pledges performance in term of what is provided for in the Community Forest Management Agreements on community Forests on customary land and agree to the monetisation of GHG benefits by BCP. REDD: Chiefdom agreements that stipulate the GHG REDD process and monetisation of Emission reductions by BCP. Therefore BCP has been established as project proponent and the one with the ability to claim that the project will generate removals, and therefore to the proceeds from the sale of VCU's once the project is validated and verified.

BCP has been instrumental in the operationalisation of the Forest Act 2015 by developing guidelines that secure statutory rights like the Forest Carbon Stock Management guidelines and also the Community Forest Management Statutory Instrument. these will guide and secure statutory rights.

2.5.3 Free, Prior and Informed Consent (G5.2)

In developing the REDD+ projects supported under the LCFP, BCP has adhered to strict standards for ensuring that adequate stakeholder consultations and engagements have taken place, in line with international standards and “best practice” for ensuring Free, Prior and Informed Consent (FPIC).

To start, in 2014, BCP facilitated a series of workshops involving numerous representatives from Government and civil society, to collaboratively develop a Standard Operating Procedure (SOP) for FPIC in Zambia. This FPIC SOP was finalized in early 2015, and BCP has ensured to adhere compliance with the terms of this SOP, throughout all of our FPIC activities and subsequent stakeholder engagements, up to now, in 2017.

Stakeholder consultations and engagements, in line with the principles of FPIC, were a central component of BCP's REDD+ viability assessment process that took place throughout 2015 and 2016, up until the point of receiving community consent letters for REDD+ implementation – as our Viability Assessment process handouts (available as supporting documents) show.

As Section 2.1.8 explains in further detail, stakeholders to the REDD+ projects supported under the LCFP include Government partners and community members from 12 Chiefdoms. These community members are represented by 10 CRBs and 62 VAGs, and fall under the jurisdiction of the 12 Royal Establishments (Chiefs / Chieftainesses, supported by traditional leaders such as Headmen and Indunas), for our 12 partner Chiefdoms. BCP possesses a full set of documentation, confirming that significant community consultations and engagements took place, over a period of more than 2 years, in line with the principles of FPIC, in all 10 partner Chiefdoms for the LCFP – *prior* to receiving any consent letters for REDD+ implementation, from these same community stakeholders.

In 2016, BCP received consent letters for REDD+ implementation from community stakeholders from all 12 of our partner Chiefdoms. These letters were / are signed by key representatives from each Chiefdom,

including: all 12 Chiefs/Chieftainesses, and representatives from each of the 12 implementing partner CRBs, from all 12 of our partner Chiefdoms, as well as representatives from each of the 69 VAGs, as local-level institutions representing all community stakeholders. These representatives are recognized to have the authority to make such commitments on behalf of the community, both according to local laws, as well as local customs/traditions.

The receipt of these community consent letters to REDD+ implementation marked the end of the viability assessment process, for BCP. Since then, BCP has signed two subsequent agreements with community stakeholders, further documenting specific commitments with regards to forest protection activities, and community benefit sharing. As of the end of 2017, we have additionally signed in-depth Community Forest Management Agreements (CMFAs) that function as Participatory Forest Management Plans.

BCP possesses a full set of supporting documentation to confirm that FPIC processes took place, according to the FPIC SOP, and in line with international standards and “best practice.” This set of records is available for review by an auditor, upon request. The set of supporting documentation collectively shows that:

- Local communities have been widely and appropriately consulted, prior to project implementation;
- Local communities are aware of, and freely and willingly consenting to the terms and boundaries of the REDD+ projects;
- Local community stakeholders were involved in participatory forest selection and project design, through a series of PRA/PLA activities that took place in each Chiefdom, which were led by trained facilitators, and fully documented;
- Stakeholders to proposed REDD+ activities were provided with accurate and adequate information concerning proposed REDD+ project activities, proposed REDD+ project location(s) and size(s), the nature of REDD+ activities to be supported, relevant timelines, scope, personnel involved, and reasons/purposes for REDD+ activities to be undertaken. Moreover, stakeholder feedback was solicited, and was used, significantly, to inform these very same terms of proposed REDD+ projects – as stakeholders, themselves, identified forest areas for protection, identified reasons to implement REDD+ for themselves, and identified deforestation mitigation activities to be supported. Stakeholder feedback was also used to inform viability assessment findings concerning potential program impacts, risks, and mitigation activities to be undertaken – in addition to reviewing existing literature and studies, relevant to the same.

Additionally, BCP’s set of FPIC records for the LCFP shows that, broadly speaking, the following steps took place, in each Chiefdom:

Step 1: Initial “desk review” research to identify potentially viable REDD+ areas (according to liability criteria concerning forest composition, project locations, potential stakeholders, potential project impacts, relevant data/analysis/studies of previous attempts, and potential risks).

Step 2: Initial consultations with key (high level) stakeholders, including district level workshops /meetings, and/or initial consultations with traditional leaders and community institutions. At, or following, these workshops, inputs from key stakeholders were sought with regards to stakeholder identification, initial identification of proposed forest areas for REDD+, high-level stakeholder permission for the viability assessment processes to proceed in each District/Chieftdom, including: viability assessments of proposed forest areas, and FPIC consultations with all potential stakeholders.

Step 3: Participatory forest selection and stakeholder consultations, in line with FPIC, and using PRA/PLA techniques. These included community consultation meetings at Chieftdom, VAG and village levels, as well as individual stakeholder consultations, in some cases, to ensure widespread information dissemination about REDD+

Step 4: Follow-up meetings held at Chieftdom level, to confirm community stakeholder consent. These meetings were held openly and publicly, and members of the wider community were invited to attend and participate, to either provide their feedback, and/or ask questions, before relevant authorities (VAG and CRB representatives, and traditional leaders) were approved by the community, and confirmed by the Chief/Chieftainess as the ultimate authority, to consent to the terms of the proposed project, on behalf of the communities they are legally and culturally/traditionally supposed to represent).

The culmination of these activities resulted in community consent letters for REDD+ implementation, signed by all necessary / relevant authorities, on behalf of the community.

Throughout these processes, community consultations took place in line with a standard “curriculum” that was developed, as outlined in the FPIC SOP. Representatives from BCP and implementing partners – including Government implementing partners – received technical training in REDD+, FPIC and PRA/PLA, prior to undertaking community sensitizations. Locally appropriate informational materials were developed, and refined (based on experience) to communicate key messages about REDD+ to local community stakeholders. These included visual comics, and materials that were translated into local languages, to accommodate the variety of languages spoken by LCFP stakeholders, as well as high levels of illiteracy. Trained facilitators provided information and led meetings/workshops/consultations in local languages. All of these informational materials are available as supporting documents, upon request, and can be reviewed in order to confirm that they conveyed a significant and appropriate amount of information, in order to ensure that stakeholders had been fully informed about the proposed terms and potential impacts of REDD+ project implementation, in line with the principles of FPIC.

As a result of having completed all of these extensive FPIC processes, BCP is confidently able to confirm that the LCFP cannot be considered to be encroaching on any form of land, and that we will not support or undertake any activities that could be considered to be doing so, now or in the future. Where REDD+

activities of the LCFP are being implemented, they are being implemented willingly, and knowingly, by the collective set of community and Government stakeholders to the project. BCP has complied with relevant local and international standards and policies for REDD+ implementation, and the activities of this project are compliant with, and will continue to comply with and be protected by relevant legislation in place (as outlined in Section 2.5).

Moreover, throughout the community consultation and feedback processes, as part of the REDD+ viability assessment process, community stakeholders have consistently confirmed that the REDD+ activities of the LCFP are being implemented on areas of land that are free of encroachment. The REDD+ activities of the LCFP will not cause any evictions or relocations of established communities. In the event that any individual relocations may need to occur – for example, if it is found that an individual or household has recently encroached into an area proposed for REDD+ implementation – then any relocation arrangements will be agreed by all relevant stakeholders, through agreed / appropriate community institutions, and will take place according to *mutual agreement*. The REDD+ activities of the LCFP will not cause any forced relocations or evictions. As the Conservation Fees Guidelines Agreements signed between BCP and local community stakeholders document, an agreed grievance resolution mechanism is in place, and will be supported by BCP through the LCFP, to ensure that any grievances, concerns or questions may be reported and appropriately responded to, in the event that they come up.

2.5.4 Property Rights Protection (G5.3)

The land rights will still vest in the people with Forest user rights being granted to the community by director forestry as in section 31. (1) of the 2015 Forest Act which says “ A community forest management group recognised under section thirty may apply to the Director, in the prescribed manner and form, to enter into a community forestry agreement with the Department in respect of an area or forest for which the community. forest management group is formed. (2) The Director shall, in considering an application under subsection (1), after consultation with the Chief and the local authority in respect of an area or forest for which the community forest management group is formed. this is a guarantee that project activities will not lead to involuntary removal or relocation of property rights holders from their lands or territories, and does not force rights holders to relocate activities important to their culture or livelihood. The targeted forest are arrived at with consensus and the project will not embark on involuntary relocation or removal. the Community Forest Management Agreement provides rights and duties for both parties and gives a guide.

Under LCFP relocation are a preserve of the Community Forest Management Group in consultation with the traditional authority and the local authority.

A similar situation applies to the private properties, there is a contractual agreement in place and no rights are violated.

2.5.5 Illegal Activity Identification (G5.4)

1. selective wood extraction for timber and charcoal production: development of land use plans and developing alternative livelihoods like beekeeping,
2. illegal land allocation by indunas and village headmen: empowering the CFMG and the Community to make their local leaders accountable.

Both illegal activities will also be countered by regular aerial and ground monitoring of the project area.

2.5.6 Ongoing Disputes (G5.5)

No disputes exist in the areas we are implementing the LCFP. What we have as problems are the power vacuum that is created by the death of a Chief, this has succession wrangles which take time to settle but once the chief is put in place then the situation goes back to normal.

2.5.7 National and Local Laws (G5.6)

Environmental Management Act No. 12 of 2011

This Act was passed in the year 2011 and is the Principal Act governing and regulating environmental issues in Zambia. The Act provides for the establishment of the Zambia Environment Management Agency (ZEMA) whose main functions constitutes the protection of the environment and control of pollution in particular so as to provide for the health and welfare of persons, animals, plants and the environment in general.

The Act conforms with the philosophies set out in the Rio and Stockholm Declarations i.e. precautionary principles, polluter-pays principles and transparency and dissemination of information to the general public and public participation (the Act states that the public has the right to participate in decisions concerning the formulation of environmental policies, strategies and plans and to participate in the preparation of laws and regulations on the environment). It states that the Minister shall strive to attain the conservation of biological diversity and fair and equitable sharing of the benefits arising from utilisation of biological diversity. The Government commits itself to cooperating with other governments and local and international organisations in order to protect the regional and global environment. The Agency has various obligations which include *inter alia* conducting surveys on the state of the environment and researching and forecasting environmental changes and undertaking other studies which might contribute to the formulation of policies and preparation of action plans and strategies. In addition, the Agency shall

create and maintain an environmental information registry and an environmental fund. The Act provides that forestry resources will be managed in accordance with the Forests Act.

Note SI No. 28 of 1997 relating to the Environmental Protection and Pollution Control (Environmental Impact Assessment) Regulations, 1997 which sets out in which circumstances an EIA or a Project Brief will be required and how these are to be conducted.

The Zambian Environmental Management Agency has no objection against the implementation of the Lower Zambezi REDD+ Project and has exempted it from any EIAs (Appendix F).

Mines and Minerals Development Act (No. 7 of 2008)

The legal framework for mining and mineral resources development in Zambia is the Mines and Minerals Act of 2008. The Act stipulates regulations in acquiring prospecting licences and mining and mineral rights.

The Act provides for the Director of Mines to give consideration to the need to conserve and protect the air, water, soil, flora, fauna, fish, fisheries and scenic attractions and features of cultural, architectural, archaeological, historical or geological interests when deciding whether or not to grant any mining right. In the event that a mining right is granted it shall be granted under such conditions as may be prescribed by the Minister in relation to the conservation and protection of air, water, soil, flora, fauna, fish, fisheries and scenic attractions and features of cultural, architectural, archaeological, historical or geological interests. The Act provides for the creation of an Environmental Protection Fund. It further provides that a holder of a license shall be strictly liable for any harm or damage caused by mining operations and shall compensate any person to whom the harm or damage is caused. Where damage is caused to the environment or biological diversity, compensation shall include the cost of reinstatement, rehabilitation or clean-up measures which are incurred and where applicable the costs of preventative measures. Note too that any person, group of persons or any private or state organisation may bring a claim and seek redress in respect of a breach or threatened breach of any provision relating to damage to the environment, biological diversity.

Note that a holder of a licence who requires the exclusive or other use of the whole or any portion of the prospecting or mining area for the purpose of the licence, in accordance with the laws relating to such acquisition, may acquire a lease or other right to use the same upon such terms as may be agreed between the holder and owner/ occupier of the land. A holder of a licence shall on demand made by the owner or lawful occupier of the any land subject to a mining right, promptly pay the owner fair and reasonable compensation for any damage done to the surface of the land by the operations. Where compensation is not paid or the owner of the land is dissatisfied with the compensation offered, the dispute shall be determined by arbitration.

It also deals with environmental issues pertaining to prospecting and mining. Reference is made to the Statutory Instrument No. 29 of 1997, a subsidiary legislation of the 2008 Mines and Minerals Act that specifically addresses environmental issues.

The act is relevant to the project due to that activities related to large scale mining will take place within the license area. The act provides for regulations concerning environmental protection during construction, operation and closure of mine sites.

This law is applicable to the project as there are prospecting licenses that overlap part of Rufunsa Conservancy (Appendix G).

Zambia Wildlife Act 2015

The Act provides a legal framework for the Zambia Wildlife Authority to establish, control and manage National Parks, conserve and protect wildlife and objects of interest in National Parks, establish Game Management Areas, licensing of hunting, control of possession of trophies and to control bush fires.

Water Resources Management Act

The Act came into force in 2011 and repealed the Water Act of 1949 and provides for the control, ownership and use of water. The Act defines the functions and powers; provide for the management, development, conservation, protection and prevention of pollution of water resources and its ecosystem. It provides for the issuance of water licences and permits and rules relating to the construction of dams on existing rivers. In addition it provides for a right of easement over land under specified circumstances, which shall include the right of access to any piece of land contiguous to the water which is the subject of a permit. Note there is a difference between public and private water.

Forests Act, 2015

It provides for the establishment and management of National and Local forest conservation and protection of forests and trees, and licensing and sale of forest. It also provides for the establishment of the Forestry Commission and provides for participation of local communities, NGO's and traditional institutions for sustainable management of forest ecosystems and biodiversity. The Act also provides for the implementation of various international conventions relating to the environment. Note the President can declare any land a National Forest and may by compulsory acquisition acquire any land for the purpose of a National Forest. The Act also provides for the declaration of local forests, forest plantations or open areas to be a Joint Forest Management Area.

Zambia Tourism and Hospitality Act No. 24 of 2007

Enacted in 2007, the Act provides for the control of tourism enterprises. The Act, though making no direct reference to environmental protection, does provide for appeals against authorization of tourism projects which are deemed to negatively affect Zambian tourism.

National Heritage Conservation Commission Act No. 23 of 1989

Enacted in 1989, the Act provides for the conservation of ancient, cultural and natural heritage, relics and other objects of aesthetic, historical, pre-historical, archaeological or scientific interest.

The Employment Act (CAP. 268):

Establishes guidelines on employment of an employee in Zambia to be read together with the Minimum Wages and Conditions of Employment (General) Orders. It provides for dismissal of employees and the grievance procedures to be followed as well as basic employment rights relating to leave, working hours, maternity leave, minimum contents to be included in written contracts of service; retirement and redundancy benefits etc.

Minimum Wages and Conditions of Employment Act (CAP. 276) (Statutory Instruments (SI) No. 2 & 3) SI No. 46 of 2012 and the General Order, 2011:

Establishes minimum guidelines for remuneration and conditions of service for employees in Zambia setting out a minimum wage for specific sectors and basic rights and conditions of service.

Industrial and Labour Relations Act (CAP. 269):

Establishes guidelines on employee and employer organisations and on the process of collective bargaining and dispute resolution. It provides for the formation of trade unions and employers representative organisations and recognises the right to strike under specified circumstances. It provides for the settlement of disputes, strikes and lockouts as well as for the use of the Industrial Relations court at which employees can have their grievances and cases heard and determined.

The Land Acquisition Act, Chapter 189 of the Laws of Zambia

The Act makes provision for the compulsory acquisition of land. The President can compulsorily acquire land where he is of the opinion that it is desirable or expedient in the interest of the Republic to do so subject to various obligations which include *inter alia* the requirement to give notice of intention to acquire property and the payment of compensation. Adequate compensations shall be paid and any disputes shall be determined by the courts.

The Lands Act, Chapter 184 of the Laws of Zambia

This Act provides for the continuation of leasehold tenure and the continued vesting of land in the President and the statutory recognition and continuation of customary tenure and it provides for the conversion of customary tenure to leasehold tenure. The Act also provides that a person shall not without lawful authority, continue to occupy vacant land and any person who does shall be evicted. Note that customary land existed in Zambia before the advent of colonialism – under the unwritten indigenous law of tribes. Customary law can be resorted to in the settlement of disputes involving members of tribes and is vital in the settlement of land disputes that may arise in respect of land held under customary tenure.

The project will fully comply with these aforementioned laws.

2.5.8 Approvals (G5.7)

All approvals and signed agreements are listed in Appendix B.

The project has approval from DNPW, Forestry Department and signed Community Forest Management Agreements with traditional authorities. It is in the process of acquiring authorization from Zambia Environmental Management Agency (ZEMA).

2.5.9 Project Ownership (G5.8)

PDR.5

At submission of this document, 10 of the initial 12 CRB project proponents have entered into Community Forestry Management Agreements with the Forestry Department that give the communities the rights to the forest carbon among other things. The documentation for CFMAs, including approvals can be found in Appendix A.

The applications for the remaining 2 CRBs to become community forest management groups (CFMGs), Sandwe and Chitungulu, were submitted to the director of forestry by the Provincial forest officer on the 9th of July, 2019 (Appendix A). A prompt response is expected from the director who the project is in direct contact with. However, should the director not be able to respond quickly, under the 2015 Forest Act, the applications will automatically be approved if they are not acted on within 21 days. After that, the CFMAs can be submitted to the director for approval. The CFMAs have already been received by the provincial forest officer and will be submitted to the director of forestry immediately after the CFMG applications are approved or on the 29th of July when the 21 day limit has passed. The act stipulates that the CFMAs automatically come into effect if they are not acted on within 60 days of being received by the director. This is because there is a substantial consultative process that happens at lower levels first involving district and provincial forest officers before the applications are submitted for final approval.

For the project lands in private ranches, BCP has signed a 30 year agreements giving BCP the rights to emissions reductions and carrying out REDD activities on their lands. This makes BCP the project proponent for these areas. The landowners have 99 year leases for the areas, which will not expire before the conclusion of the project. Documentation of the agreements with BCP and leases can be found in Appendix A.

Finally, in order to represent all of the project proponents in all matters relating to verification, marketing, and sale of VCUs generated by this project, BCP has entered into a communications agreement with the CRB project proponents (Appendix A).

2.5.10 Management of Double Counting Risk (G5.9)

The project is seeking credit under CCB standard v3.1 and VCS standard v3.7. The project is not seeking any other form of social or environmental credit. The project is not aware of any social or environmental credit scheme in which the project would be eligible to participate since such schemes (E.g. Plan Vivo), would consider participation in VCS to be double counting and thus deem the project ineligible. All credits will be issued as VCUs listed in a VCS approved registry and all transactions of credits will be tracked through that registry.

2.5.11 Emissions Trading Programs and Other Binding Limits

The project is not part of any other emissions trading program. Zambia currently does not have any such programs in place. Emission reductions will be sold on the voluntary market.

2.5.12 Other Forms of Environmental Credit

As stated in section 2.5.10, the project has not sought any other form of environmental credit, nor aware of any other form of environmental credit for which the project would be eligible to participate.

2.5.13 Participation under Other GHG Programs

The project is not participating under any other GHG Program. Credits will be sold on the voluntary market.

2.5.14 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

2.5.15 Double Counting (G5.9)

The host country is not participating in any compliance mechanism and credits will be sold on the voluntary market, there is thus no risk of double counting.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology

VCS VM0009 V3.0 Methodology for Avoided Ecosystem Conversion

VCS VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, version 3.0

VCS AFOLU Non-Permanence Risk Tool v3.3

3.1.2 Applicability of Methodology

PDR.1-2

1. The primary driver of deforestation in the baseline scenario is unplanned conversion to cropland, primarily smaller holder farms. Unplanned charcoal harvesting also results in forest loss and degradation. The primary agents of deforestation are small scale farmers and charcoal makers. These drivers and agents are consistent with those described in section 6 of the VM0009 v3.0. The end land use in the baseline scenario is non-forest (cropland) and the project activity is AUDD.
2. According to historical satellite imagery used to define the project accounting areas (see section 3.2.1), the project accounting areas were all forest as per the Zambian definition¹³ of forest at least 10 years prior to the beginning of the project accounting period.
3. According to results from analysis of conversion within 120 meters of threatened boundaries (Table 16), some parts of the project accounting area were within 2 kilometers of areas that had been converted to cropland, the end land-use in the baseline scenario.
4. N/A for F-U3

¹³ **PDR.3** The Forest Act (Commencement) Order, 2015 definition: “forest” means any land with a tree canopy cover of more than ten percent and area of more than zero point five hectares and includes young stands that have not yet reached, but are expected to reach, a crown density of ten percent and tree height of five metres that are temporarily under stocked areas;

5. N/A for F-U3
6. N/A for F-U3
7. The project accounting areas do not contain peat soil.
8. A reference area has been identified (see Figure 28) for the F-U3 baseline type that is similar to the project area as per the methodology requirements and had at least as much forest area as the project accounting areas at the start of the historical analysis in 1986.
9. There was sufficient historical satellite imagery (see Figure 28) to cover more than 90% of the reference region at least twice during the historical reference period.
10. The project activities (see section 2.1.11) are designed to mitigate ecosystem conversion by addressing agents and drivers of conversion.
11. The activity-shifting leakage areas and proxy areas are accessible to the project for every monitoring period.
12. N/A
13. The project is located in a tropical ecosystem.
14. There will be little or no livestock grazing in the project area and absolutely no manure management in the project area.
15. N/A

1.1.1 Project Boundary

PDR.11-13

Source		Gas	Included?	Justification/Explanation
Baseline	Flux in carbon pools	CO ₂	Yes	As required per VM0009, the project will evaluate fluxes in CO ₂ from AGOT. Fluxes from BGOT and SOC will also be tracked as a substantial portion of the total system carbon is stored in these pools. AGMT, DW, and WP will not be included as neither baseline nor project scenario include significant timber harvesting. AGNT will not be included as baseline scenario does not include perennial tree

Source		Gas	Included?	Justification/Explanation
				crops.
	Biomass Burning	CO ₂	No	Accounted for under carbon pool flux
		CH ₄	No	Conservatively excluded as per VM0009 v3.0
		N ₂ O	No	Conservatively excluded as per VM0009 v3.0
	Livestock	CH ₄	No	Baseline emissions from livestock grazing are conservatively assumed to be zero as per section 6.5.1 of VM0009 v3.0
		N ₂ O	No	Excluded as per VM0009 v3.0 applicability requirement 14
Project	Flux in carbon pools	CO ₂	Yes	Most prominent GHG sources in baseline and thus will also be evaluated in the with project condition including fluxes in CO ₂ in AGOT, BGOT, and SOC.
	Biomass Burning	CO ₂	No	Accounted for under carbon pool flux
		CH ₄	No	Conservatively excluded as per VM0009 v3.0
		N ₂ O	No	Conservatively excluded as per VM0009 v3.0
	Livestock	CH ₄	No	Monitoring flights over the project areas have only detected livestock in Mnkhanya, Msoro, and Nsefu project areas. The peoples of the project areas are not traditionally pastoral, but do sometimes keep oxen for plowing farms. The livestock observed in the project area were likely for that purpose as they were only observed in areas that had been encroached by farmers. The pilot observing the cattle estimated that the total number of cows in those areas is not more than 200 ¹⁴ . Therefore, there are certainly fewer than 1000 cattle, 1000 goats, and 1000 sheep in the project area. The annual methane emissions from that number of livestock would be equivalent to 1,386 tCO ₂ and therefore is far less than 5% of the GHG emission reductions anticipated from this project (see Appendix G for calculations). This remains true even if the number of livestock increased 10 fold. The project is not

¹⁴Personal communication from Dr. Hassan Sachedina

Source		Gas	Included?	Justification/Explanation
				promoting livestock grazing in the project area and will be preventing exactly the type of encroachment that has been associated with livestock in the project areas. Thus, there is no expected increase in grazing in the project scenario and therefore GHG from livestock are de minimis and can be excluded.
		N ₂ O	No	Excluded as per VM0009 v3.0 applicability requirement 14
	Synthetic Fertilizer	N ₂ O	No	The project is not promoting the use of synthetic fertilizers and no increase in use is expected in the project scenario relative to the baseline scenario. Therefore, project emissions from synthetic emissions are zero and therefore de minimis and excluded.

A map of the project area and the project zone with reference and leakage areas is shown in Figure 16. The project area and project zone are where activities described in Section 2.1.11 will be located.

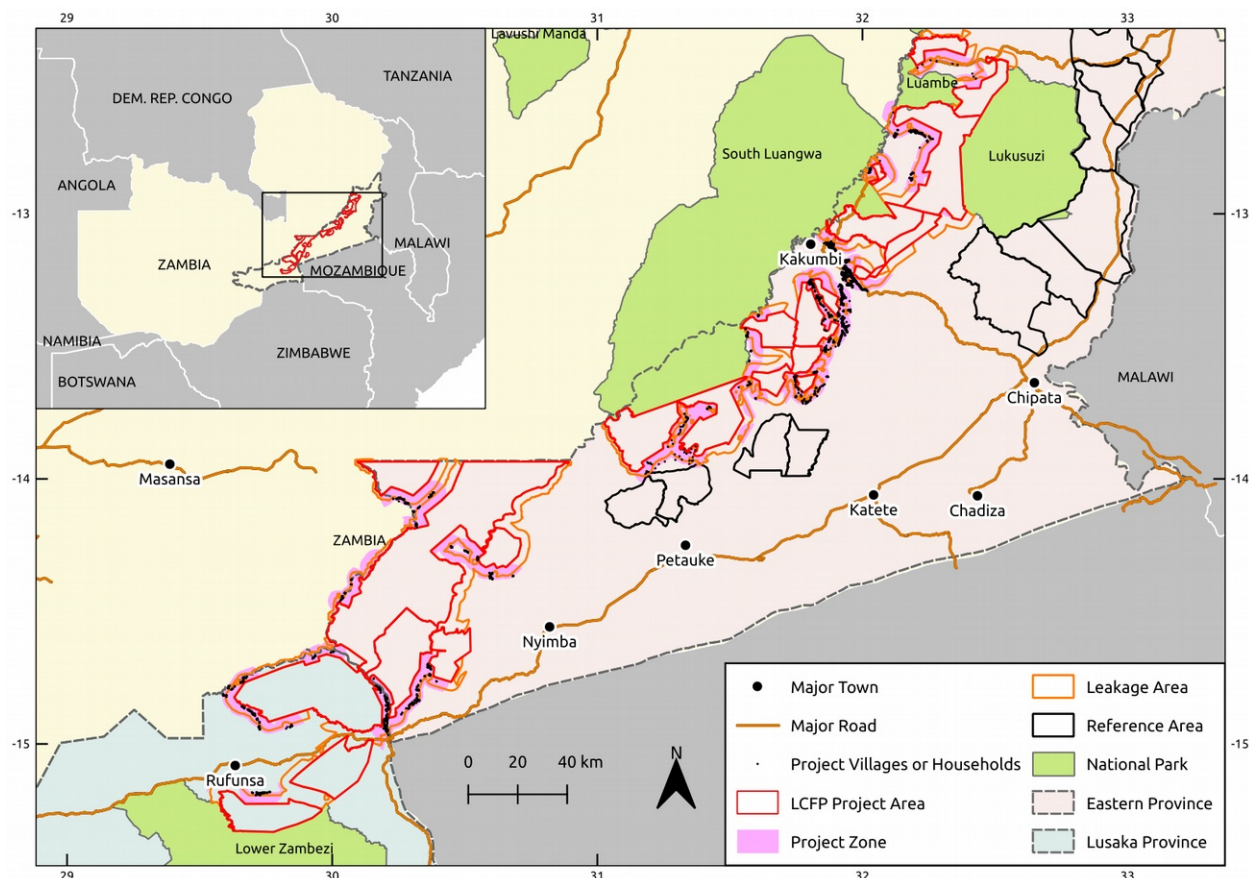


Figure 16: Project area, project zone, leakage areas and reference areas

1.1.2 Baseline Scenario

The baseline scenario was determined from a 2017 study of agents and drivers of deforestation in the cheifdoms where the project areas are located (Appendix K). The study was conducted by an international team from Unique forestry and land use GmbH on behalf of BCP. The study was participatory and aimed to build a common understanding of the baseline scenario with input from a wide variety of stakeholders. In addition to focus group discussions with community members from different chiefdoms and a literature review, the researchers also consulted a wide range of experts from relevant Zambian government ministries and BCP.

The results of the Unique study were supplemented with a GIS analysis of deforestation for Lusaka and Eastern provinces conducted by BCP. The analysis used freely available PALSAR radar data mosaics

products¹⁵ to map forest in 2007 and deforestation annually from 2007-2010, from 2010 to 2015, and annually from 2015 to 2017. These date ranges correspond to the all of available PALSAR mosaic products from the PALSAR I and PALSAR II sensors. The PALSAR sensors are cloud penetrating L band radar which are ideally suited for mapping woodlands in the tropics where cloud cover and dry season leaf fall frequently make mapping with optical sensors challenging. The analysis used HV backscatter data, which previous research has shown is particularly sensitive to variations in biomass in Miombo woodlands¹⁶. Using the HV value for 582 training points collected from high resolution imagery in Google Earth for the year 2016, a HV threshold was identified that generated an forest / non-forest map following the Zambian forest definition with 97% accuracy. Then, a python script (available for the VVB to review) was written to generate the gross deforestation analysis using the changes in HV data from one year to the next.

Then, to analyze agent mobility and constraints to deforestation, the areas of deforestation were overlaid on maps of other factors likely to relate to deforestation including slope, elevation, soil type, streams, precipitation, vegetation type, and distance to roads. Based on a visual analysis, the most prominent factors were found to be distance to roads and slope. To identify specific cutoff points beyond which deforestation is unlikely, a histogram analysis was performed for the two most prominent factors, using the areas of deforestation from 2007 to 2017 as a mask.

PDR.17

As required by VM0009 v3.0, the project used the VCS Tool VT0001 (version 3) for the assessment of additionality. This is described in section 1.1.3. This tool shows that the identified baseline type, deforestation for small holder agriculture expansion and some charcoal production, is the most plausible baseline scenario.

PDR.18

The study identified the following main agents of deforestation and their mobilities:

1. Smallholder farmers. They either expand from their own farms, or look for forested land where they can develop a new farm. A GIS analysis of the PALSAR deforestation analysis found the median distance for new farms from old farms is about 250 meters with a range from about 50 meters to just over 5km. Over an 8 year period from 2009 to 2017, 99% of deforestation fell within 3.58 km of deforestation that occurred between 2007 and 2009. Farmers are constrained by steep slopes with 99.9% of deforestation from 2007 to 2017 occurring in areas with less than 45% slope. Finally, it was also observed that 99% of deforestation occurred within 6.5 km of a road.

¹⁵ Available from https://www.eorc.jaxa.jp/ALOS/en/palsar_fnf/fnf_index.htm

¹⁶ Mitchard, E. T. A., et al. "Using satellite radar backscatter to predict above ground woody biomass: A consistent relationship across four different African landscapes." *Geophysical Research Letters* 36.23 (2009)

2. Charcoal producer – their mobility depends on the type of producer. Most charcoal making in the project chiefdoms is opportunistic and occurs when farmers need extra cash while they are clearing land for cultivation. Dedicated charcoal makers have greater mobility, but generally, then will not go further than 4 km from a road to produce charcoal.
3. Firewood collectors – Most firewood collectors near the project area are doing so for person use, but there may be a few commercial collectors collecting on behalf of institutions. Their mobility is similar to dedicated charcoal makers.

The study identified the following main drivers of deforestation:

1. Agricultural expansion;
2. Charcoal productions; and
3. Firewood extraction

PDR.19-20

Detailed narratives, descriptions and useful statistics for each agent:

1. Smallholder farmers are a predominant deforestation agent in the project reference region. Large-scale farming is not common here. Smallholder farmers are characterised by having small farm sizes, an average of 2 hectares or less, and the practice of subsistence farming. They are highly dependent on agriculture, and have little to no alternative means of livelihood. They clear forestland to practise low-input cultivation, resulting in low yields and rapid exhaustion of the soils. Maize is the most common crop grown, in addition to other crops such as sunflower, beans, cotton, and tobacco. When the soils have become less- or unproductive, new lands are cleared. According to consultations and interviews, all rural households are engaged in small-scale farming. Even the few people having alternative incomes such as from charcoal production or salaried jobs (e.g. school teachers) are involved in small-scale farming. In the reference region, the percent of rural population ranges from 60 to 97% (see section 4.4.1 for details). With annual average human population growth rate of about 2 to 4% in the region, this agent group is exerting enormous pressure on forest loss, and expected to continue doing so for the foreseeable future.
2. Charcoal producers can be categorised into two groups: 1) occasional producers – those who produce charcoal infrequently since they do not depend on it as their primary means of livelihood; and 2) regular producers – those who are involved in charcoal production as their main source of income and livelihood. The first group produces charcoal in order to take care of occasional cash needs or as an opportunistic additional income during land clearing for cultivation, or as a safety net against poor crop yields experienced in a particular year or season. The second group produces charcoal regularly for sale either independently or in collaboration with charcoal traders.

Dedicated charcoal producers often move from place to place in search of abundant wood resources, and sometimes in response to increased enforcement of production and trade regulations. Though the population of charcoal producers is unknown and hard to determine due to lack of monitoring coupled with frequent producers' mobility, according to interviews, in areas where productions are concentrated, more than half of the households are reportedly engaged in charcoal production – either as occasional or regular producers. With increased urbanisation rate¹⁷, the number of charcoal consumers is on the increase and can be realistically expected to grow in the foreseeable future, driving demand and production upwards.

3. Firewood collectors: Every household in rural areas collects and/or uses firewood as the primary source of energy for cooking and heating. Besides households, other main collectors and consumers of firewood are institutions – including educational/training institutions; cottage and small-scale businesses such as burnt-brick makers, and restaurants and lodges; and small to medium-sized industries such as beer breweries. Firewood is preferred by all these groups of users because it is a cheaper and readily available energy source. As rural population continues to grow, so will the collection and consumption of firewood– because alternative energy sources for cooking and heating that are affordable and readily accessible by the rural population and institutional consumers are non-existent. According to consultation and interviews, rural households traditionally collected mainly dead or dry wood for firewood, hence, did not cut standing live trees in significant numbers. However, with increasing users' population, tree cutting for firewood is now common since deadwood alone is insufficient to cater for all firewood needs. The average firewood consumption of a household is estimated at about 1.6 ton/year in Zambia (Kalinda et al., 2008). Institutional consumers on the other hand mainly cut standing live trees, in addition to harvesting wood from forestland that is being cleared for agriculture. They are an important group because their activity is endorsed by the government – since they provide a revenue source to the FD through payment of harvesting/royalty fees for firewood collected legally. It was acknowledged during consultations with FD officials that many of them are, however, also collecting firewood without payment or following legal procedures.

Detailed narratives, descriptions and useful statistics for each driver:

1. Expansion of small-scale agriculture is by far the most important driver of deforestation in the reference region. The region contributes a substantial part of crop production in the country – in some cases producing a quarter to a half of the total production of a particular crop in the country in a particular year – for example: maize, sunflower, cotton and tobacco (Tembo and Sitko, 2013).

Historically, crop cultivation in the region involved practicing shifting cultivation with long fallow periods to allow the soils to regain its fertility as a result of re-growth of vegetation. The practice

¹⁷ Urbanisation rate of Zambia is estimated at about 4.4% p.a. (The World Factbook, 2017).

was as follows: after clearing forested land and planting on it repeatedly for 4 to 5 years, the land was abandoned and left to fallow for a long time – sometimes 20-30 years. During this period, forests would have re-grown on the land. However, with increasing sedentary population, this practice has all but disappeared. According to FGDs and published literature, a few farmers who still practise fallow do so in areas with abundant land, and for short periods of about 1-3 years (Kwesiga et al., 2005). Hence, permanent conversion of forests to agricultural land is now the norm, which is driving deforestation in the region. Figure 3 below shows examples of new forest conversion to agricultural land in the project region. Fires are commonly used in the process.



Figure 17: Deforestation due to agricultural expansion

According to MTNER (2002) quoted in Vinya et al., 2012, land clearing for agriculture accounts for an estimated 90% of deforestation in the country. In addition, from both our consultations and a previous survey by COMACO in the region, the evidence shows that the majority but not all deforested lands have been converted to agriculture. Hence, we concluded that the majority of deforestation – some 70-90% can be attributed to agricultural expansion, with the rest being attributable to other drivers, which are subsequently discussed.

2. Charcoal production results in deforestation because of the nature of the production system: Charcoal producers practise mainly clear-cutting as opposed to selective cutting of trees, which is recommended – as it allows smaller diameter and other trees to remain growing on the site. Selective cutting is likely to be confined to places where preferred species are abundant. However, all charcoal producers met in this study indicated that selective tree cutting is not a common practice. In addition to clear-cutting, the construction of earth kilns used in the conversion of the wood to charcoal results in further tree clearing at the production sites; sometimes the kilns can measure up to 40 m long by 40 m wide. Besides not practising selective tree cutting, other cutting techniques, which are recommended for sustainable tree utilisation, such as making a clean and slanting cut close to the tree base, are also not followed by charcoal

producers. These poor harvesting practices greatly diminish the ability of cut trees to re-grow after charcoal production.



Figure 18: Charcoal along the Great East Road in Rufunsa district

How much charcoal production contributes to deforestation is hard to exactly quantify because of the intricate link between it and agriculture: Forests may be cleared for charcoal production and later taken over for cultivation; or the forest is cleared with the objective of cultivation and the harvested wood opportunistically used for charcoal production. Therefore, the primary reason for clearing is not possible to exactly ascertain and attribute deforestation to it. Nonetheless, to put it into perspective, considering household consumption alone, and using population data¹⁸ and consumption figures¹⁹ of 1.4 tonnes/household/year, we estimated charcoal consumption at the urban household level in the districts comprising the reference region to result in extraction of about 797,400 tonnes of wood per year in 2016. Taking 81% of the extracted wood as non-renewable (thus leading to deforestation), this consumption would equal to about 8,100 ha deforested due to charcoal production in the reference region districts per year. With average annual deforestation in Lusaka and Eastern Provinces estimated at around 50,000 ha/year in the 10 years before 2017, the above estimate of charcoal-induced deforestation translates to 16% of total deforestation in the historical reference period. Therefore, in addition to expert consultations and review of available literature, we concluded that charcoal production is a major driver of deforestation – contributing some 15-30% of all deforestation.

¹⁸ CSO, 2012. 2010 Census of population and housing, Volume 11: National descriptive tables.

¹⁹ Kalinda, T., Bwalya, S., Mulolwa, A., & Hantuba, H. (2008). Use of Integrated Land Use Assessment (ILUA) data for forest and agricultural policy review and analysis in Zambia. Lusaka: FAO & MTENR.

3. Household generally harvest both dead/dry wood and live trees for use as firewood. However, when dead/dry wood supply is insufficient they cut trees and dry them before use¹⁹. Institutional, cottage and business users, however, primarily obtain firewood through cutting of live trees.

Based on the process of fuelwood extraction, and discussions with various stakeholders, we concluded that firewood extraction would contribute primarily to forest degradation – as long as the harvested area is left alone to regenerate. The time required for such tree regeneration is estimated at between 8 to over 20 years²⁰. However, the harvested sites are usually not left to recover for this period of time; hence, permanent loss of the forest cover occurs – either as a result of agriculture eventually expanding to the harvested sites or persistent exposure to other disturbances such as grazing, fires, and combinations of these. This makes estimating the direct contribution of firewood collection to deforestation hard to exactly determine. Nonetheless, considering household consumption alone – using population data¹⁸ and average consumption figures of 1.6 tonnes/household/year, we estimated firewood consumption at the household level alone in the districts comprising the reference region to be 588,700 tonnes in 2016. To put it into perspective, taking 81% of the extracted wood as non-renewable²¹ (thus leading to deforestation), this consumption would equal to about 6,000 hectares deforested due to firewood extraction annually. The average annual deforestation in the region during the historical reference period was estimated at 51,913 ha/year; hence, the above estimate of firewood-induced deforestation is equivalent to 12% of total deforestation in the historical reference period. From this breakdown and expert consultations, firewood extraction was thus estimated to contribute considerably to deforestation – at about 10-15% of total deforestation.

Therefore, in summary, expansion for small holder farming is the primary driver of deforestation. Charcoal production and firewood collecting may also play a role, but probably mostly only contribute to deforestation when they are followed by conversion to farming and otherwise cause degradation. All of these drivers result in unplanned conversion.

PDR.21

No external drivers were quantified in a way that could be used as a covariate in a deforestation model.

²⁰ Chidumayo E. N., Gumbo D. J., (eds) 2010. The dry forests and woodlands of Africa: managing for products and services. London, UK: Earthscan.

²¹ UNFCCC CDM, 2012. Thirty-fifth meeting Report, Annex 20.

1.1.3 Additionality

As required by VM0009 v3.0, The VCS Tool VT0001 (Version 3) Tool for the Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities” was used to demonstrate project additionality.

Applicability Conditions

- a) Project activities do not violate any applicable laws (Section 2.3.16)
- b) The baseline methodology (VM0009) allows for a stepwise approach justifying the most appropriate baseline scenario.

The Steps of VT0001 have been applied as follows:

PDR.99 A list of alternative land use scenarios to the project.

Step 1. Identification of alternative land use scenarios to the proposed VCS AFOLU project activity

Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity:

- i. Based on historical land use change assessments ten years prior to the project start date and taking into consideration the state of Game Management Areas in Zambia (Lindsey *et al.* 2014) it is highly likely that the without project land use scenario will continue namely spreading subsistence agriculture leading to deforestation and other degradation activities such as charcoal production and harvesting of non-timber forest products. Without project land use scenarios are described in section 2.2.1 and 3.1.4 based on the prescribed analyses of Drivers and Agents of Deforestation (Appendix K).
- ii. The proposed Avoided Unplanned Deforestation (AUD) project activity without being registered as a VCS project needs to be considered based on the tool (project activity without carbon financing. This is not a credible scenario as the project area with its various REDD Zones is not likely to be conserved without ongoing outside funding sources. Current income generation from tourism activities cannot sustain forest conservation efforts (let alone the wildlife populations they are mandated to protect) in GMAs nor Private Ranches in GMAs (see Lindsey *et al.* 2014 & 2013 for a review). GMAs are focused on protecting wildlife populations and human settlements are not illegal in a GMA.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations:

- a) Human settlements are allowed within GMAs and with permission from traditional authorities. The subsequent clearing of land and burning of biomass that accompanies subsistence agriculture, is however, controlled by the Forest Act of 2015 and not allowed without a permit, but this is rarely enforced. As per the requirements of the tool, since the project occupies two provinces, the smallest administrative unit encompassing the project area is the whole country. A representative study of all of GMAs in the country found that by 2010, 39.5% of the total GMA area in the country had been encouraged by humans, show that definitely, enforcement does not happen on more than 30% of GMA area²². The reality is though that the Forestry Department is unable to enforce this law across Zambia and even more so in remote GMAs where the deforestation rate has been found to be higher than in open areas. The prevalence of non-compliance far exceeds 30% and it is estimated that only 5% of Zambia's forest produce is officially permitted²³.
- b) Outcome of Step 1b: Due to systematic non-enforcement of legal requirements all the scenarios listed in Step1a are plausible and are an alternative to the proposed project activity.

PDR.100 Justification for the selected baseline scenario.

Sub-step 1c. Selection of the baseline scenario:

The most likely baseline scenario is Option i) Step 1a. The step wise approach followed to determine the baseline is outlined in figure 2 on page 44 of VM0009 V3.0. The methodology requires the identification of primary agents and drivers of deforestation, secondary agents and drivers, and identifying external drivers of conversion. There are several barriers to the implementation of the proposed activity (Lindsey *et al.* 2013 & 2014), which is why Option i is the most likely baseline scenario in the absence of the project.

PDR.101 An investment or barriers analysis proving that the project is not the most economical option.

Step 2. Investment analysis:

Sub-step 2a. Determine appropriate analysis method:

²²Lindsey, P.A., Nyirenda, V.R., Barnes, J.I., Becker, M.S., McRobb, R., Tambling, C.J., Taylor, W.A., Watson, F.G. and t'Sas-Rolfes, M., 2014. Underperformance of African protected area networks and the case for new conservation models: insights from Zambia. *PLoS one*, 9(5), p.e94109.

²³Gumbo, D. J., Moombe, K. B., Kandulu, M. M., Kabwe, G., Ojanen, M., Ndhlovu, E. and Sunderland, T.C.H., 2013. Dynamics of the charcoal and indigenous timber trade in Zambia: A scoping study in Eastern, Northern and Northwestern provinces. Occasional Paper 86. CIFOR, Bogor, Indonesia.

The proposed project will not generate any other income other than VCS related income. Therefore, the project can apply the simple cost analysis (Option 1).

Sub-step 2b. - Option 1. Apply simple cost analysis

The costs associated with this project are described in the financial analysis submitted with the non-permanence risk report (Appendix H). This financial analysis also shows that the only income from the proposed project activities is from GHG credit sales. Therefore, the project satisfies the criteria for the simple cost analysis.

PDR.102 A common practice analysis including a list of project activities and the drivers of conversion that they address.

Step 4. Common practice analysis:

There are currently two AUD projects in the same geographical area as the proposed activity. There are also other conserved forests in the geographic area, but these are all National Parks or National Forest Reserves. National Parks are managed by the DNPW and are subsidized by government and receive income from tourism. National Forest Reserves fall under the mandate of the Forestry Department and are poorly subsidized and many are encroached and non-functioning. The GMAs in which the AUD activity fall are mandated to protect wildlife and are in a continuous state of decline with deforestation rates higher than in open areas (Lindsey *et al.* 2014). There are other private ranches in the geographical region that aim to conserve wildlife and forests, but again they are at much reduced scale and all rely on private funding and supplementary income from trophy hunting.

There are numerous differences between the LCFP and the two other AUD projects in the geographical area. One project is only a fraction of the size of the LCFP and has been implemented on private, titled land. The other project falls within open land and is also only 1/3d of the size of LCFP. The other project is solely community governed. LCFP is distinct here as well as it is governed in partnership with community structures (CRBs and chiefs) as well as DNPW and Forestry Department.

The LCFP project is one of the first of its kind within Zambia and was conceived in the early 2010's as a pilot project to demonstrate the use of carbon finance for the protection of wildlife corridors between Zambia and Malawi and the development of the surrounding rural communities. Prior to LCFP, no other project had undertaken such a diverse range of forest protection and development activities on the same scale.

PDR.103 Evident compliance with the minimum requirements of VT0001 v3.0.

The LCFP is thus conclusively distinct from other similar activities and thus satisfies the additionality criteria.

1.1.4 Methodology Deviations

Demonstration of Forest Age

The methodology requires (in PDR.22) that a digital GIS-based map of the project accounting areas, including aerial or satellite imagery showing that they are comprised of forests or native grasslands as of the project start date and 10 years prior to the project start date. The project did not meet this requirement because it chose to map forests in the project area with PALSAR radar data and the earliest available data for that sensor is from 2007. However, this is justified on several grounds. The project areas are virgin forests that were previously remote and are inaccessible. Thus, there is no history of large disturbances in the area that would have been followed by regeneration between 2005 and 2007. An accurate 2007 forest map of these areas should be virtually indistinguishable from a 2005 map. Additionally, using the 2007 PALSAR imagery results in a much more accurate map of the forests that existed 10 years prior to the start date than would be possible with optical imagery. The optical imagery best suited to mapping the project area is generally from April-June. It is at this time of year, that relatively cloud free imagery can be obtained when the forests still have leaves and are not dotted with burn scars. Even so, the passing of just a couple of weeks during this period, which is immediately after the rains, results in significant changes in the appearance of the forest in optical imagery. The project initially tried mapping forests with optical imagery, but found significant changes between years that were the result of seasonality rather than real changes in forest areas. These effects are much reduced when using PALSAR radar data. This is because, in addition to being cloud penetrating, L band radar has a relatively long band-width and thus does not generally interact with the leaves of trees, but does interact with large branches and stems²⁴. Finally, VM0009 does not require a remote sensing analysis of an area, but rather to visual show that it is unconverted as of the project start date and 10 years prior. Such an approach is highly subjective since unconverted grasslands and farmlands may at times appear similar. With the PALSAR based analysis, all non-forest areas, whether natural or not were excluded. Thus, using the PALSAR data improved the accuracy of the analysis and did not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.

Leakage Plot Protocol

There was a minor deviation of the visual approach for leakage monitoring (Appendix R, VM0009). Instead of Forest Inventory Team members estimating degradation, they were required to geo-reference every felled tree in the permanent plots and the total number of felled trees were compared against the average density of trees/ha in the project accounting area, again avoiding biased estimations/erroneous

²⁴Mitchard, E. T. A., et al. "Using satellite radar backscatter to predict above ground woody biomass: A consistent relationship across four different African landscapes." *Geophysical Research Letters* 36.23 (2009)

counting of trees in 1ha plots. This value was then placed in a degradation category as per VM0009. This approach removed all potential bias and greatly improves the robustness and repeatability of degradation monitoring. This deviation does not affect the conservativeness of any emissions reductions as it is in fact an improvement of the prescribed procedure and removes observer bias that could result in degradation in the leakage area being underestimated.

Neither of the relate to any other part of the methodology except for monitoring and measuring.

The project plans to use two different allometric equation to estimate above ground biomass. One was developed from data provided by Prof. Chudimayo from Zambian Miombo (see Appendix G) and was tested on trees up to 55 cm DBH. It will be used to estimate the biomass of trees up 63 cm dbh, with the biomass of trees between 55 cm and 63 cm dbh assumed to be equal to a 55 cm dbh tree. Then, a second equation developed by Mugasha et al. in Tanzanian Miombo (see Appendix G) will be used to estimate the biomass of trees between 63 cm dbh and 110 cm dbh with trees with a greater than 110 cm dbh assumed to have a biomass equal to a 110 cm dbh tree. This parsing is necessary because the methodology requires that the directly measured biomass of the largest tree in the representative sample of destructively harvested trees be greater than the biomass predicted by the equation, and that the value of the first order derivative of the largest tree in the representative sample and inventory not differ by more than 10%. The first selected allometric equation does under predict the biomass of the largest tree in the representative sample, but the difference of the first order derivative values was greater than 10%. The second allometric equation doesn't pass the first test (it slightly over predicts the biomass of the largest tree in the sample). However, compared to the Zambian equation, it predicts lower biomass for every size tree. Therefore, the equations were applied as described above to enable to project to make estimate of the biomass of larger trees that fell outside the range of the size of trees used to develop the equations. Only 33 trees out of 3589 trees in the inventory had a dbh above 55 cm and only 3 had a dbh above 110 cm. Applying the equations in this way is conservative and erases the possibility of the equation over predicting biomass of trees outside the size classes for which it was tested. This deviation only relates to how biomass is measured for monitoring purposes and doesn't affect any other part of the methodology. It results in a more conservative estimate of biomass and emissions savings. Therefore, it meets the criteria for an acceptable methodological deviation.

3.2 Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

Project Accounting Areas

The first part of determining the baseline emissions with VM0009 is to define the project accounting area, which is the area that will be subject to the baseline model of deforestation and for which a appropriate proxy, reference, and leakage areas will also be created. This is done based on analyzing agent mobility and the suitability of the project area for conversion under the baseline scenario.

PDR.23

As described in detail in section 1.1.2, a GIS analysis of the PALSAR deforestation map from 2007 to 2017 was used to identify key constraints to conversion. Two factors stood out most prominently- distance from roads and slope. These factors make sense since roads provide access to forested areas for farmers and charcoal makers and farmers, who are responsible for most deforestation, prefer to cultivate on relatively flat ground which will better hold soil moisture and is easier to work with oxen (which some farmers have). The analysis found constraints to conversion at greater than 45% slope and greater than 6.5 km distance from a road. Therefore, to calculate the project accounting area, a polygon buffer covering 6.5 km from roads was clipped from the project area. Then, areas of high slope and areas that were not forest according to the PALSAR analysis in both 2007²⁵ and 2015, were also removed from the accounting area polygon. The results (**PDR.22**), are shown in figures Figure 19, Figure 20, Figure 21, Figure 22, Figure 23, Figure 24, Figure 25, and Figure 23.

Baseline Type

The next part of the methodology is to determine the baseline type following the diagram on page 50 of VM0009 v3.0. Under VM0009 v3.0, there are baselines for forest and grassland. While there are natural grasslands within the project area and they are occasionally under threat of conversion to cultivation, the project has chosen to only consider forest conversion. As stated in section 1.1.2, the drivers of deforestation result in unplanned conversion. Finally, the rate of conversion along the project area perimeter has been low. Thus, as under VM0009 v3.0, the baseline type is considered F-U3.

PDR.32

Next, in order to use an F-U3 baseline, the methodology requires a spatial analysis to show that the project area is within 120 meters of deforestation that occurred within 10 years prior to the project start date. This analysis was conducted as part of the work to determine r_U (described in more detail at X). The results show that 7.6% of the threatened project boundary is within 120 meters of conversion that happened between 2007 and 2015²⁶.

²⁵ This deviation from the methodology is explained and justified in section 1.1.4.

²⁶ See section 1.1.4 for explanation and justification of this deviation

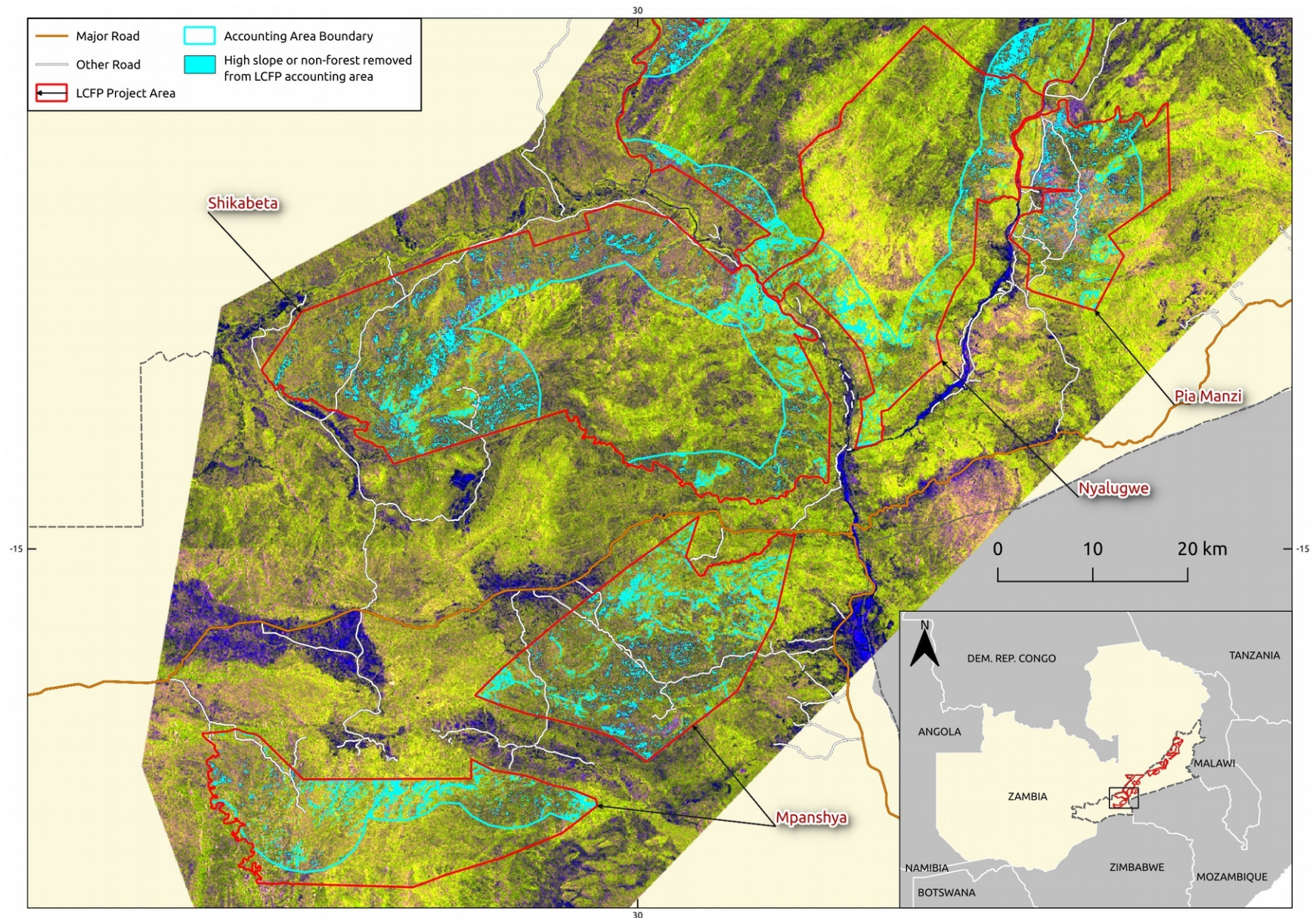


Figure 19: Mpanshya, Shikabeta, Nyalugwe, and Pia Manzi Accounting Areas with 2015 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

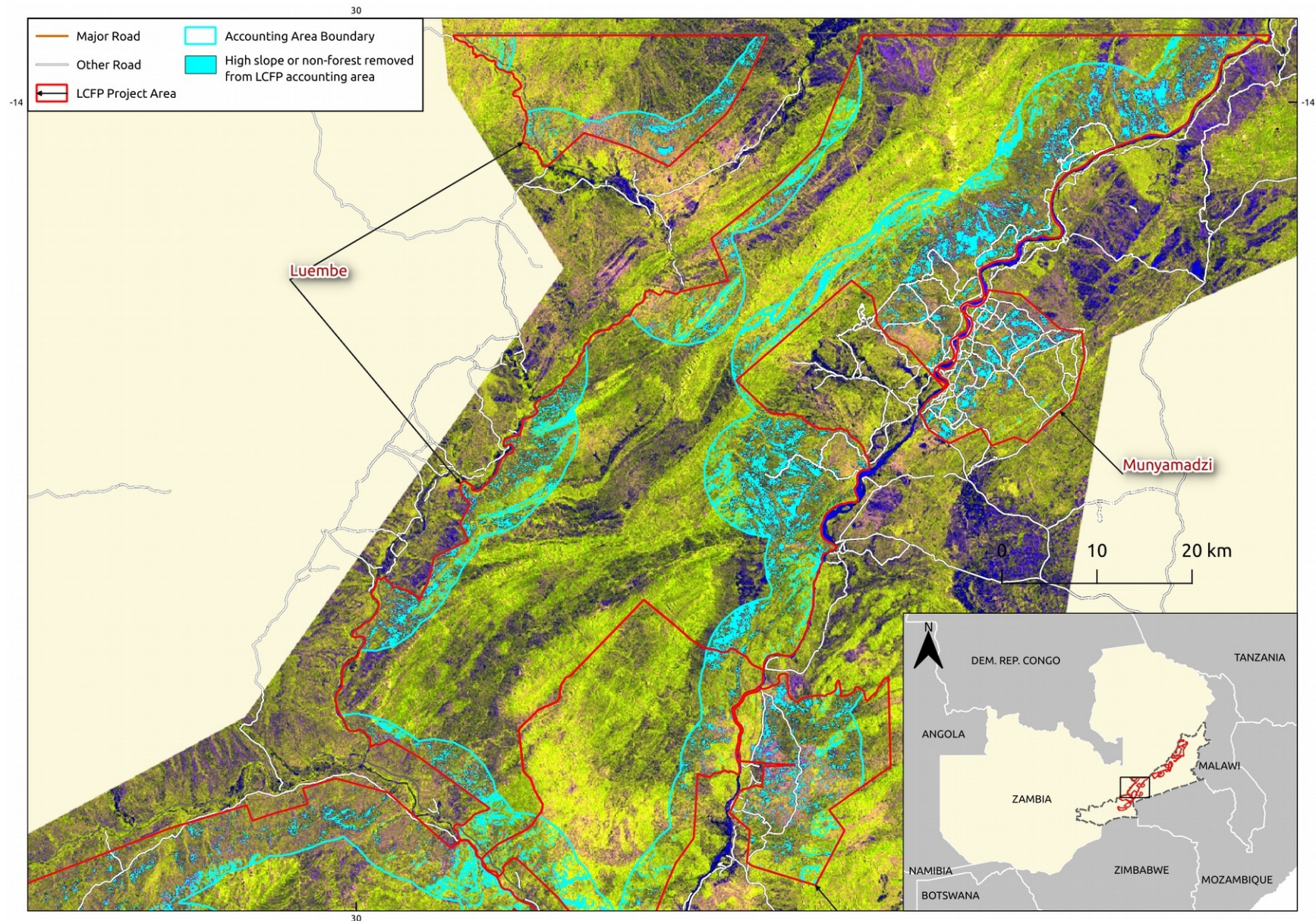


Figure 20: Luembe and Munyamadzi Accounting Areas with 2015 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

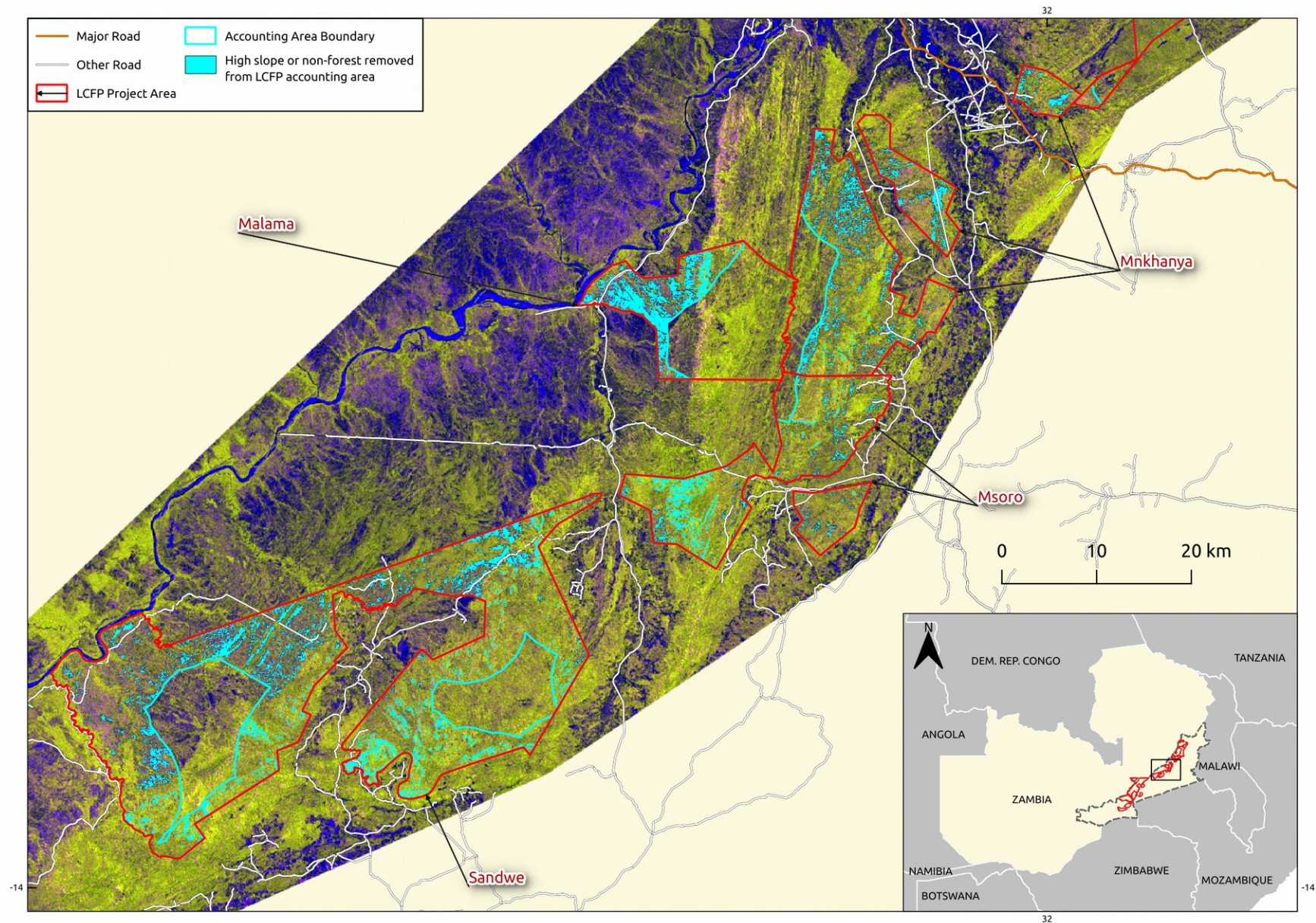


Figure 21: Sandwe, Malama, Msoro, and Mnkhanya Accounting Areas with 2015 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

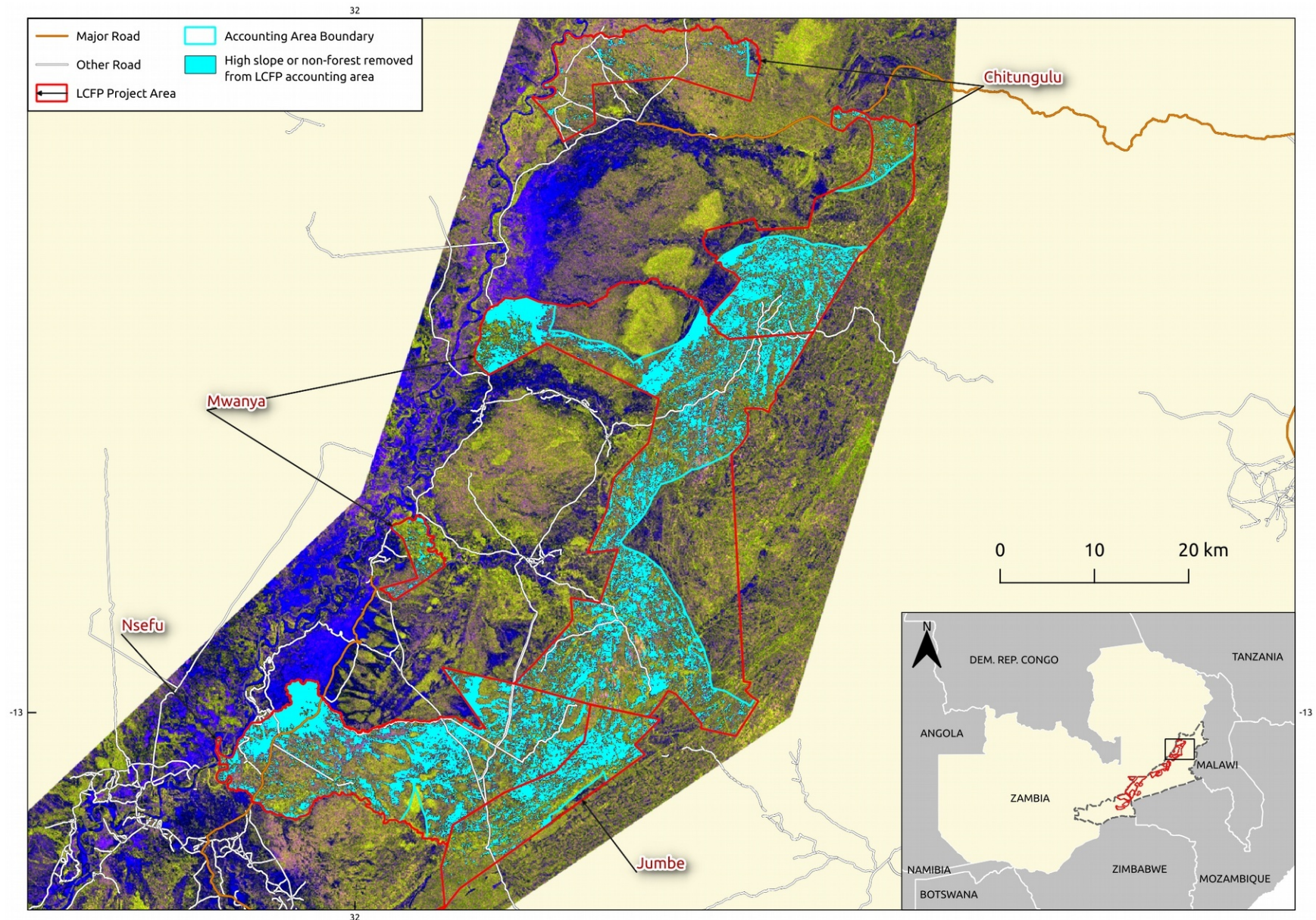


Figure 22: Jumbe, Nsefu, Myanya, and Chitulungu Accounting Areas with 2015 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

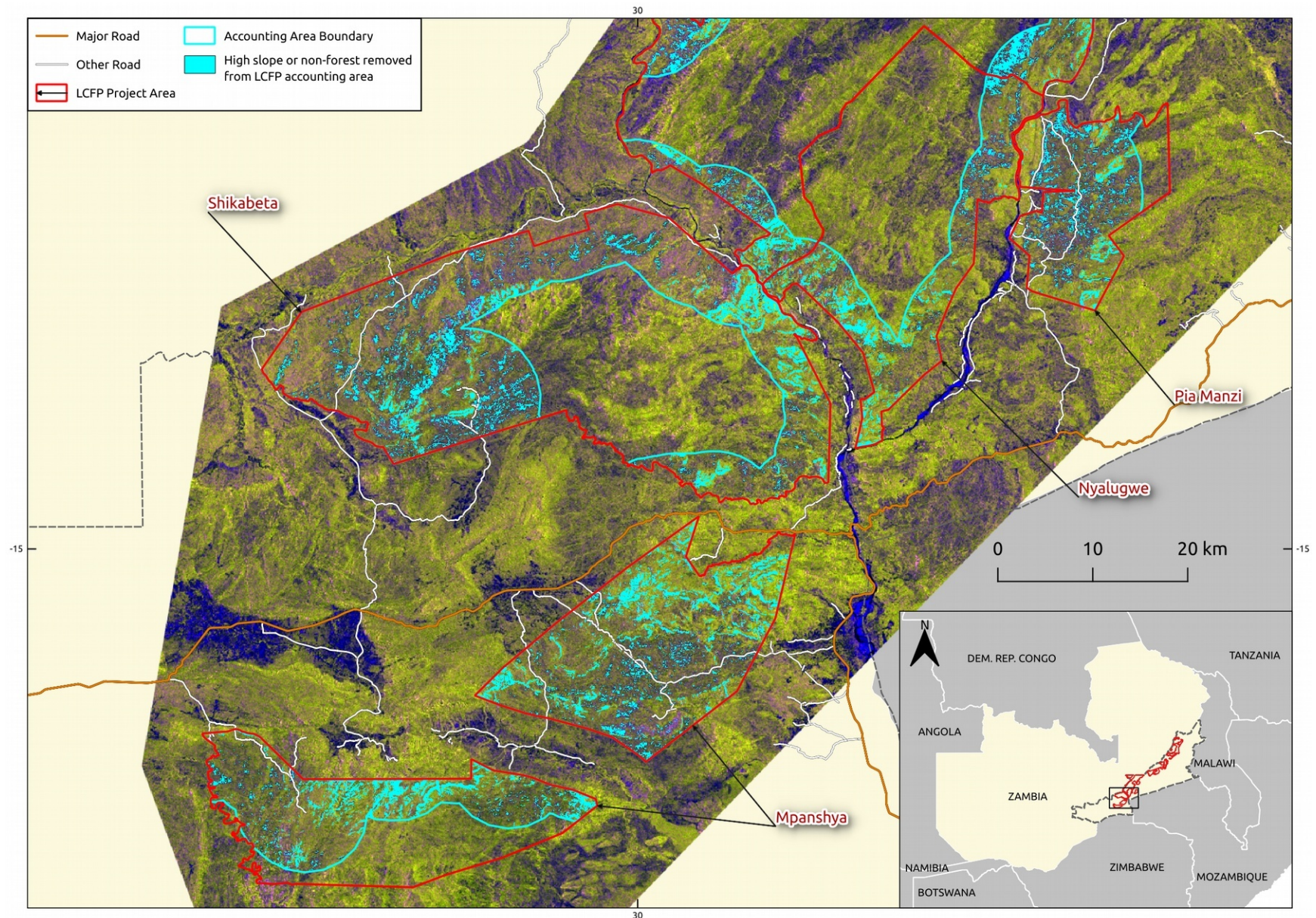


Figure 23: Mpanshya, Shikabeta, Nyalugwe, and Pia Manzi Accounting Areas with 2007 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

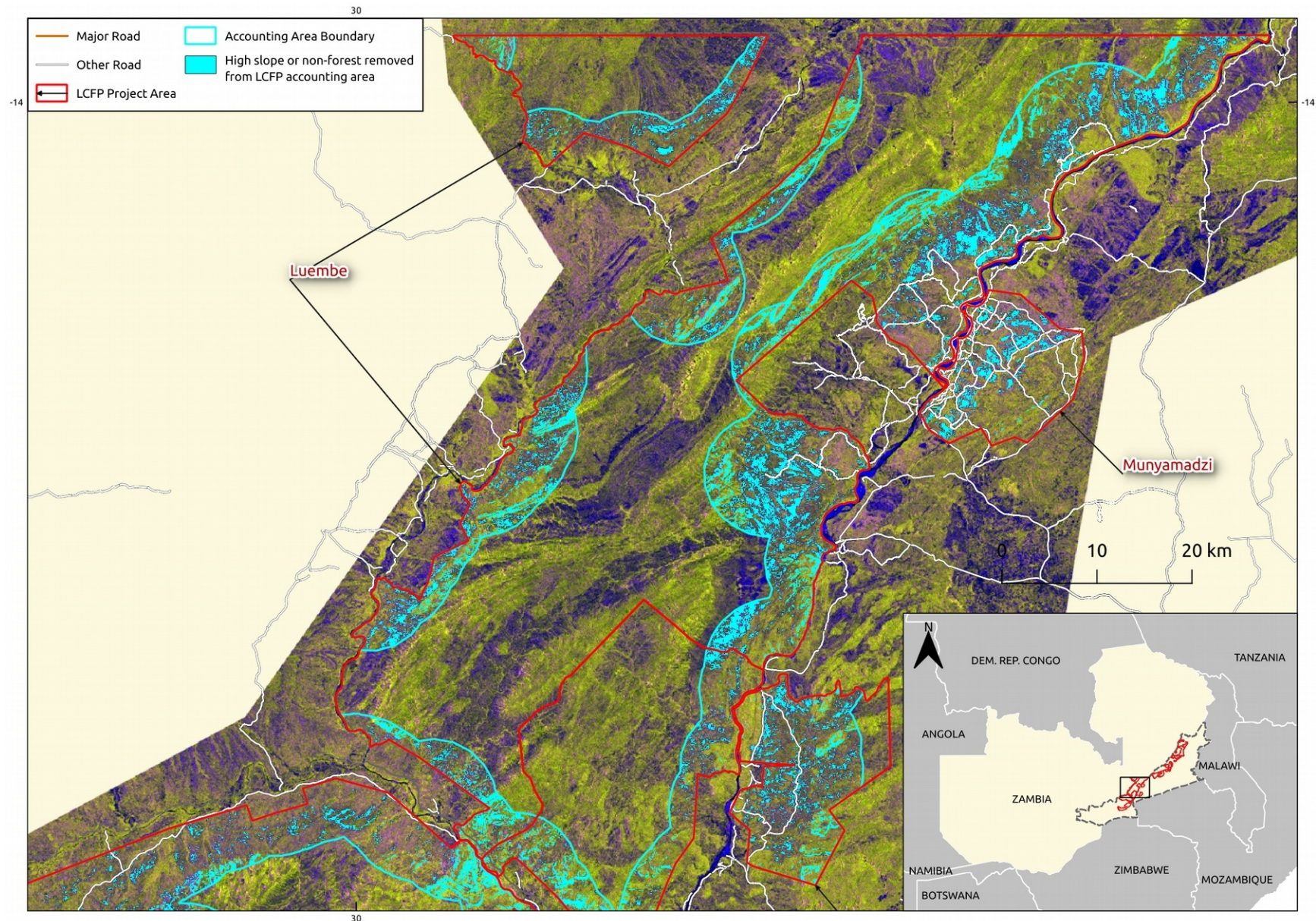


Figure 24: Luembe and Munyamadzi Accounting Areas with 2007 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

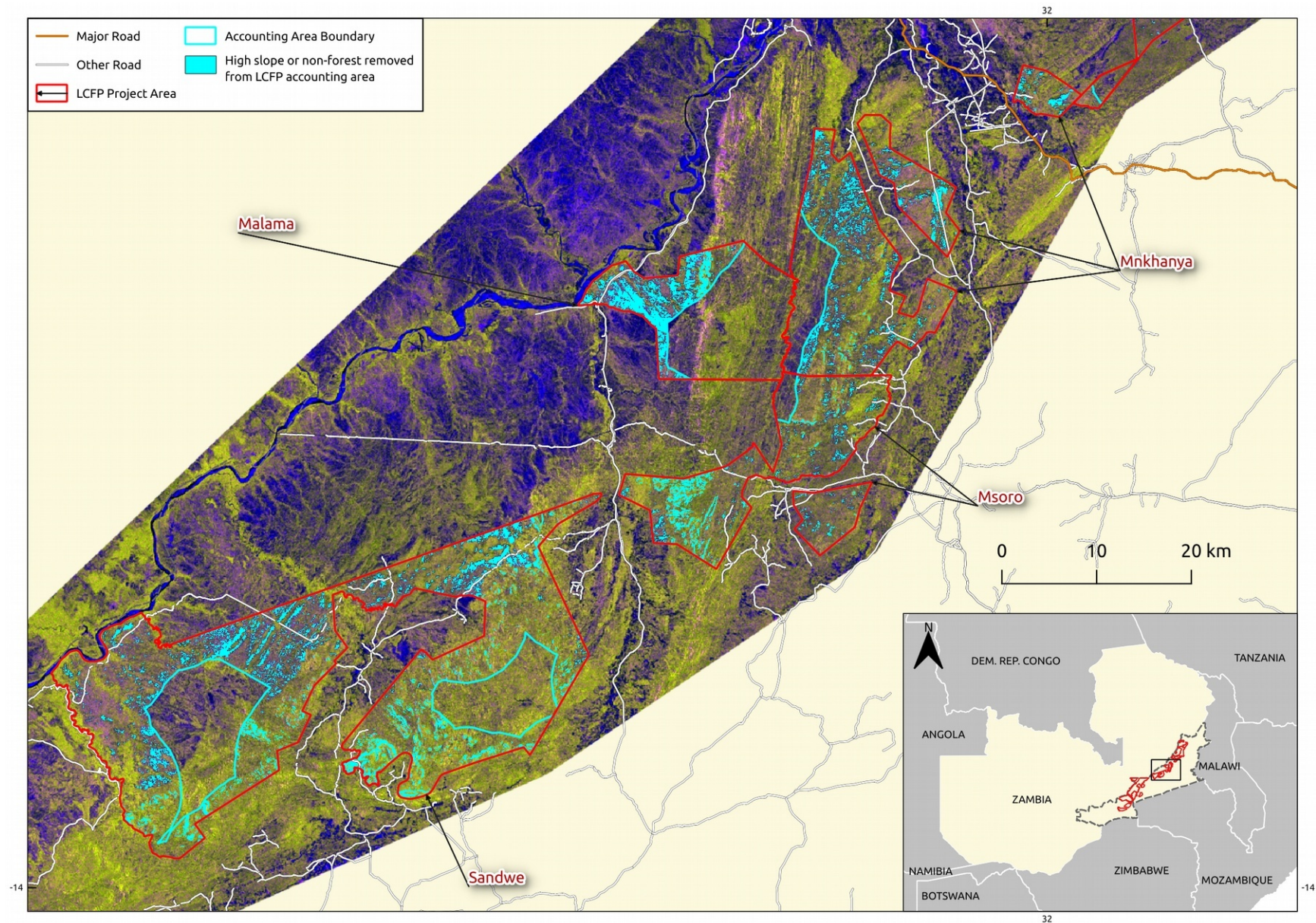


Figure 25: Sandwe, Malama, Msoro, and Mnkhangya Accounting Areas with 2007 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

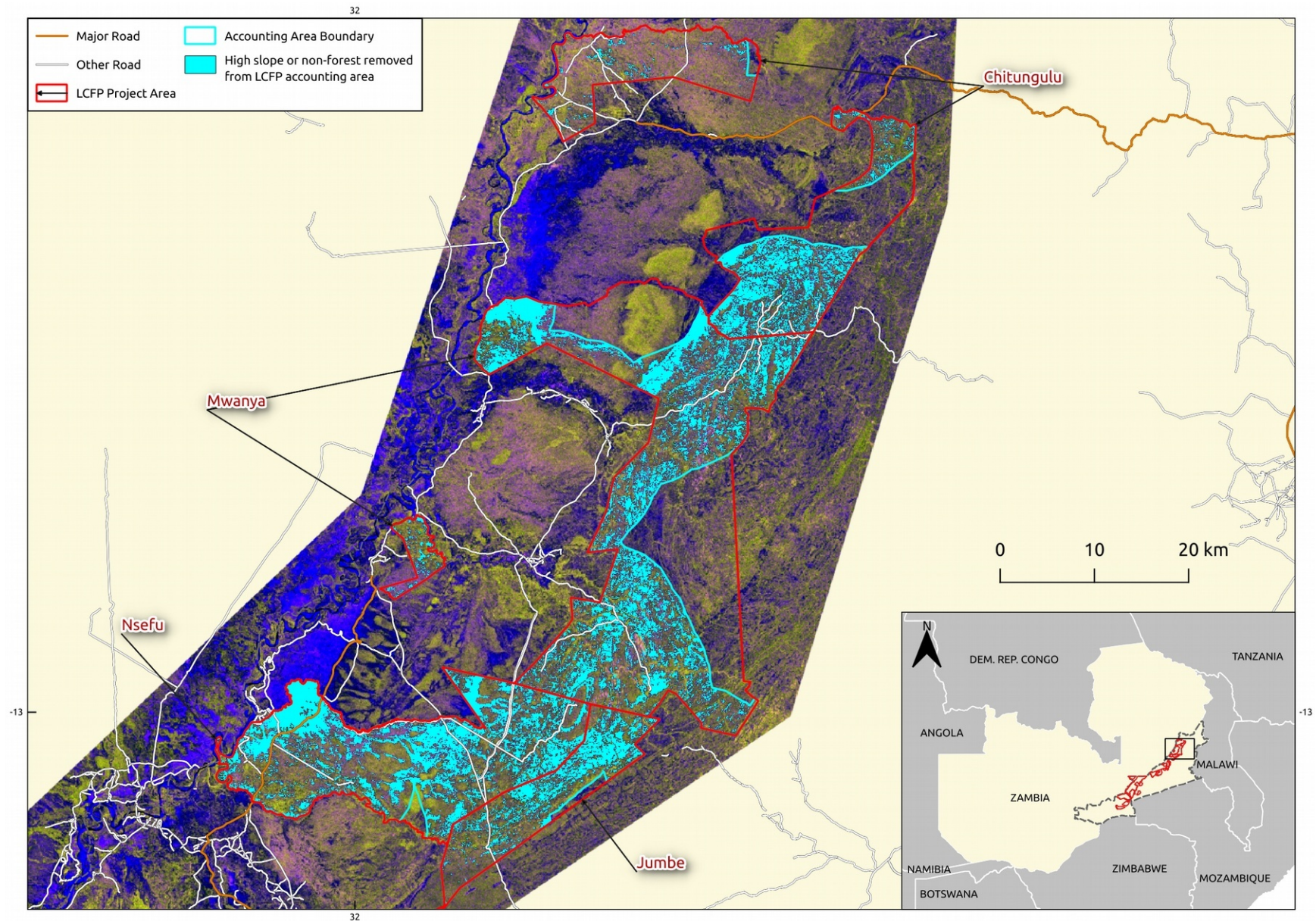


Figure 26: Jumbe, Nsefu, Myanya, and Chitungulu Accounting Areas with 2007 PALSAR Mosaic (R: HH backscatter, G: HV backscatter, B: HH / HV). Dark blue colors indicate non-forest.

Proxy Area Selection

VM0009 v3.0 uses proxy areas to estimate the residual carbon stocks after conversion over time. This is accomplished with fixed plots that are remeasured at least once every 5 years.

PDR.35

A general map of the proxy area boundaries can be found in Figure 11.

PDR.36

The following figures provide the maps of the proxy areas:

- Figure 2 (Elevation)
- Figure 3 (Slope)
- Figure 7 (Land-Cover)
- Figure 4 (Soil Class)
- Figure 5 (Precipitation)
- Figure 6 (Historical Natural Vegetation)
- Figure 30 (Access points (roads))
- Figure 30 (Markets (major towns))
- Figure 30 (Chieftdom land, national, or local forest reserve (Not in a National Park))

PDR.37

Since the biomass proxy plots must be revisited frequently, the project restricted the selection of biomass proxy areas to communities that the project knew would be agreeable to having proxy plots on their lands. Additionally, to make it easy to visit the areas with community representatives, the proxy areas were limited to a 2 km buffer around roads passing through the favorable communities. However, in all other ways, the proxy areas were similar to the project accounting areas:

- The proxy areas cover the same historical forest types as what as in the project accounting areas (Miombo, Mopane, and Munga Woodlands).

- The proxy areas cover the same soil types as the project accounting area (chromic luvisols, and lithosols)
- The proxy areas cover the same annual precipitation levels as the project accounting area (roughly 800 to 950 mm)

The soil proxy area is the proxy area for BCP's other REDD+ project known as Lower Zambezi REDD Project, which is also in Lusaka province. It has the same type of conversion in the baseline scenario as LCFP and the main soil type is

PDR.38

After defining the general proxy areas as described above, the final proxy area was reduced to areas that were mapped as deforested between 2010 and 2017 in the PASLAR deforestation analysis described in section 1.1.2. Thus all of the proxy area was converted prior to any plot work. The final proxy area was 3840 ha.

Carbon Pool Baseline Scenarios

PDR.39

As justified in section 1.1.1, there are 3 carbon pools considered in this project. Table 11 provides a qualitative description of the baseline scenario for each pool. The project accounting area SOC value was not yet available at the submission of this document.

To estimate above-ground biomass, two peer-reviewed allometric equations for Miombo woodland were used. One was from Zambia and the other was from Tanzania. This was necessary because the Zambia equation was only evaluated for trees up to 55 cm dbh. The Tanzanian equation covers trees up to 110 cm dbh. Both equations were developed from trees found in Miombo woodland and there was substantial overlap in species between the equations.

Table 11: Carbon Pool Baseline Scenarios

Carbon Pool	Baseline Scenario	Carbon Stock Pre-Conversion (tCO ₂ e/ha)	Carbon Stock Post-Conversion (tCO ₂ e/ha)
Above Ground	Immediate release of carbon from burning of above-ground tree biomass. The burning can either happen	155.5	10.13

Other Tree (AGOT)	on site, or offsite if the biomass is removed for charcoal or firewood. Carbon stocks estimated from fixed sample plots in project accounting area as described in section 3.3.3, Appendix P, and Appendix G.		
Below Ground Other Tree (BGOT)	Decay or partial removal of below ground tree biomass (roots) as the result of land-use conversion. Carbon stocks for this pool were determined using a local peer reviewed root-shoot ratio of 0.54.	83.98	5.47
SOC	Decays over time as a result of land-use conversion up to the post-conversion carbon stock. Carbon stocks are estimated from fix plots as described in section 3.3.3, Appendix P, and Appendix G.	?	131.7

Reference Area Selection

VM0009 uses a cumulative deforestation model to estimate baseline emissions. This model is calibrated based on the cumulative proportion of conversion over time. This is determined by visual tracking changes in points (pixels) across a reference area during a reference period that is representative of the future for the project area in the baseline scenario.

PDR.40

The following figures provide the maps of the reference region:

- Figure 2 (Elevation)
- Figure 3 (Slope)
- Figure 7 (Land-Cover)
- Figure 4 (Soil Class)
- Figure 30 (Access points (roads))
- Figure 30 (Markets (major towns))
- Figure 30 (Chieftdom land, national, or local forest reserve (Not in a National Park))

The reference region consists of chiefdom land, national, and local forest reserves. Forest reserves were included because the PALSAR based deforestation analysis (section 1.1.2) suggests that they are not protected are are open access lands similar to any chiefdom land.

PDR.41

The reference area is 556,309 ha. Of the 2600 image interpretation points (Figure 28), 2248 points were interpreted as forest in 1986. This is equivalent to 86.4% of the points, which would mean that approximately 480,993 ha of the reference region in 1986 was forested. The project accounting area is 485,495, which is less than 1% different in size.

PDR.42

The Drivers and Agents study commissioned for this project (Appendix K), found the main agents of deforestation across Lusaka and Eastern Provinces to be small holder farmers, charcoal producers and consumers and firewood collectors and consumers. The underlying causes of deforestation included growing rural populations with a dependence on agriculture, poor farming practices and weak regulatory systems for natural resource management and charcoal production. The study goes on to describe how at both traditional leadership and government levels there is a lack of land and resource use planning, even in Chiefdom's near to National Parks and Game Management Areas where Community Resource Boards have been established. The study found the "the CRBs do not have the technical and financial capacity to protect, manage, and monitor forests independently."

In focus group discussions for the same study, community members reported that there is a steady migration of people from higher elevation areas, where the reference area is situated, to lower elevation areas, where the project areas are. In short, the baseline agents of deforestation are often from the very same communities as those currently placing resource use pressure on the project accounting areas. Hence it is considered that the institutional or local management practices of the baseline agents in the reference area are similar to those that would have been applied to the project accounting area in the baseline scenario.

PDR.43

The reference area consists of 10 administrative wards within Eastern Province, which have been mostly cleared of forest and represent the expected baseline scenario within the project accounting area. As with the project accounting area, areas of with a slope greater than 45% were excluded from the wards. The rationale for this exclusion was that 99.9% of the forest loss seen over Eastern and Lusaka Provinces, between 2007 and 2017, occurred on slopes of less than 45%.

PDR.44

The reference area is 556,309 hectares, which is larger than the 485,495 hectares that comprise the project accounting area. Eight out of ten of the reference area wards are adjacent to the project area wards, and two are adjacent to other reference area wards. This adjacency in location is considered to satisfy most of the similarity requirements between reference and project accounting areas, and in particular those relating to the location of the agents of conversion.

The key drivers of deforestation in the reference area are described in the Drivers and Agents study commissioned for this project (Appendix K). This study looks at the drivers of deforestation across the entire Eastern and Lusaka Provinces, and so includes both the project area and the reference area. In this study population density was “identified as the driver variable that possesses a clear relationship with historical deforestation, and thus, can explain the quantities of deforestation in the historical reference period in [Lusaka and Eastern Provinces].” Hence predicted population density for the wards containing the project accounting area was examined in relationship to current population density in the reference area. The methodology used is described below, and similarity is considered to be sufficient in that the average 2045 population density for project accounting area wards is 19.3 people per km², and for the reference area wards in 2010 it was 29.6 people per km². Although the average population density is higher in the reference area, it should be noted that a historical growth rate was used to predict the future population density in the project accounting area wards, and this is likely lower and less realistic than a modeled growth rate would be.

The Zambian Central Statistics Office provides ward level population data from the 2010 census²⁷. This data was accessed and joined to a spatial data representing wards digitized from a paper map provided to the author, Nicholas Eubank, by the Electoral Commission of Zambia and provided by the Stanford University Library²⁸. As habitation of National Parks is not permitted, the areas of National Park were removed from the wards spatial data to calculate new population density estimates in the remaining extents of each ward. Lastly population growth rates per district, also published by the Central Statistics Office but accessed from the table in the drivers and agents study (Appendix K), were used to predict 2045 population density. The resultant ward data for the wards containing the project accounting area are shown in table 1. Note that the population figures for Mambwe are for the entire constituency as individual ward populations were not available. Also note that the growth rate for Rufunsa district was used for wards in Chongwe district, as Chongwe was only recently separated from Rufunsa to become its own district. Table 2 shows the 2010 population density for the reference area wards, and it should be noted that the population figures for Chamtowa and Kachama were combined as these wards were not distinguished in the spatial data.

²⁷Zambia 2010 census descriptive tables, CSO

²⁸Zambia Constituencies and Wards shapefile, Stanford University

Table 12: Population density figures for the project accounting area wards

Province	District	Ward	Total area	Area minus NPs	Population	2010 pop density	Annual growth rate	2045 pop density
Eastern	Lundazi	Lumimba	2228	1266	8679	6.9	3.2	20.7
Eastern	Mambwe	Mambwe	5959	5726	68918	12.0	3.8	44.4
Eastern	Petauke	Lusangazi	1288	839	2216	2.6	2.7	6.7
Eastern	Nyimba	Chinambi	1356	1351	9213	6.8	1.9	13.2
Eastern	Petauke	Mawanda	1822	1266	11809	9.3	2.7	23.7
Lusaka	Chongwe	Shikabeta	1978	1975	1954	1.0	3.4	3.2
Lusaka	Chongwe	Mankanda	1436	1433	4042	2.8	3.4	9.1
Lusaka	Chongwe	Rufunsa	2739	1456	15724	10.8	3.4	34.8
Eastern	Nyimba	Luangwa	3945	3935	2292	0.6	1.9	1.1
Eastern	Nyimba	Chamilala	1325	1321	3970	3.0	1.9	5.8
Eastern	Nyimba	Mombe	692	690	4889	7.1	1.9	13.7
Eastern	Nyimba	Kaliwe	313	312	8651	27.7	1.9	53.6
Eastern	Lundazi	Chibande	1577	908	9385	10.3	3.2	31.1
Eastern	Lundazi	Lukusuzi	2095	700	2155	3.1	3.2	9.3

Table 13: Population density figures for the reference area wards

Province	District	Ward	Area	Population 2010	Population density 2010
Eastern	Chipata	Nthope	1089	32993	30.3
Eastern	Chipata	Chipangali	687	20558	29.9
Eastern	Petauke	Singozi	331	12802	38.7
Eastern	Petauke	Ukwimi	359	9945	27.7
Eastern	Petauke	Chisangu	253	8610	34.0
Eastern	Petauke	Mateyo Mzeka	340	10111	29.8
Eastern	Lundazi	Diwa	1057	23410	22.1
Eastern	Lundazi	Kamimba	488	11306	23.2
Eastern	Lundazi	Chamtowa and Kachama	489	22042	45.1
Eastern	Lundazi	Wachitangachi	508	7960	15.7

In terms of the mobility of the agents of conversion, the project accounting areas have been limited to those areas within 6.5km of roads. The entire of the reference area wards are also within 6.5km of roads.

In terms of landscape similarity, the only major difference between the reference area and project accounting areas wards is one of elevation. The reference area is mostly on the plateau and has a median elevation of 986 meters above sea level. The project accounting area is mostly in the Luangwa Valley and has a median elevation of 611 meters above sea level. However, as supported by the focus group discussions held during the drivers and agents study (Appendix K), the baseline agents of deforestation are migrating from the plateau and evidently finding the landscape suitable for the agricultural, charcoal production and firewood collection practices that are driving deforestation across the region.

Reference Period and Historic Imagery to Parameterize Cumulative Deforestation Model

PDR.48

The start of the reference period is May, 1986 and the end is May, 2017.

PDR.49

Table 14: List of available historical imagery for the reference region

Sensor	Path/Row or Tile Name	Date
Landsat 5	169/069	1986-05-04
Landsat 5	170/070	1986-05-11
Landsat 5	170/070	1991-05-09
Landsat 5	169/069	1993-06-08
Landsat 5	169/069	1999-05-24
Landsat 5	170/070	2001-05-04
Landsat 5	169/069	2007-04-28
Landsat 5	170/070	2009-05-26
Sentinel-2	T36LTK	2016-05-07
Sentinel-2	T36LUK	2016-05-07
Sentinel-2	T36LVL	2016-05-07

Sentinel-2	T36LVM	2016-05-07
Sentinel-2	T36LWM	2016-05-07
Sentinel-2	T36LTK	2017-05-22
Sentinel-2	T36LUK	2017-05-22
Sentinel-2	T36LVL	2017-05-22
Sentinel-2	T36LVM	2017-05-22
Sentinel-2	T36LWM	2017-05-22

PDR.50

No specific timeline of events is known for the agents of conversion.

PDR.51

Deforestation in Eastern Province started in the southern side of the province near settlements along the Great East Road. As population has expanded, deforestation advanced towards the Luangwa valley, passing through the reference areas on the way towards the project area. No specific events regarding this movement were identified, so the project simply selected a reference period based on the best²⁹ historical imagery that covered the reference areas. Landsat 5 imagery was used for older periods, while Sentinel-2 imagery was used for the last two years of the period. The May, 1986 Landsat scenes provided the earliest suitable imagery. Two years past the start date were included so that Sentinel-2 imagery, which only started to cover the reference areas in 2016, could be used to aid the land-cover interpretation. With a resolution of 10 meters, it is much easier to interpret than Landsat imagery, so this helped to improve the accuracy of the analysis.

PDR.52-58

The selected imagery was chosen specifically to avoid clouds and was thus almost entirely cloud free, so there was 100% double coverage, meaning that all parts of the reference region were covered by the selected imagery at least twice during the historical reference period. This is visualized in Figure 28. Figure 27 shows that the time series is stationary (evenly distributed through the timeline).

To develop the point interpretation sampling frame, 2600 random points were generated using this QGIS random points in polygons function. These are also shown in Figure 28.

²⁹ Best imagery was identified as being almost completely cloud free and from a time of year when trees still have leaves.

The analysis of conversion in the interpolation points was conducted in Earth Engine (script is available to share with VVB). This made it very easy to examine each point systematically and rapidly flip from one image date to the next. As shown in Table 14, all of the imagery used was 30 meter resolution or less. The registration for the imagery was visually checked before starting the analysis. All of the images were well registered with less than 10% RMSE with the exception of the the Landsat 5 image from 1999. The shift in that image was taken into account while interpreting the images.



Figure 27: Line Plot do Demonstrate Approximate Stationarity of Historical Imagery

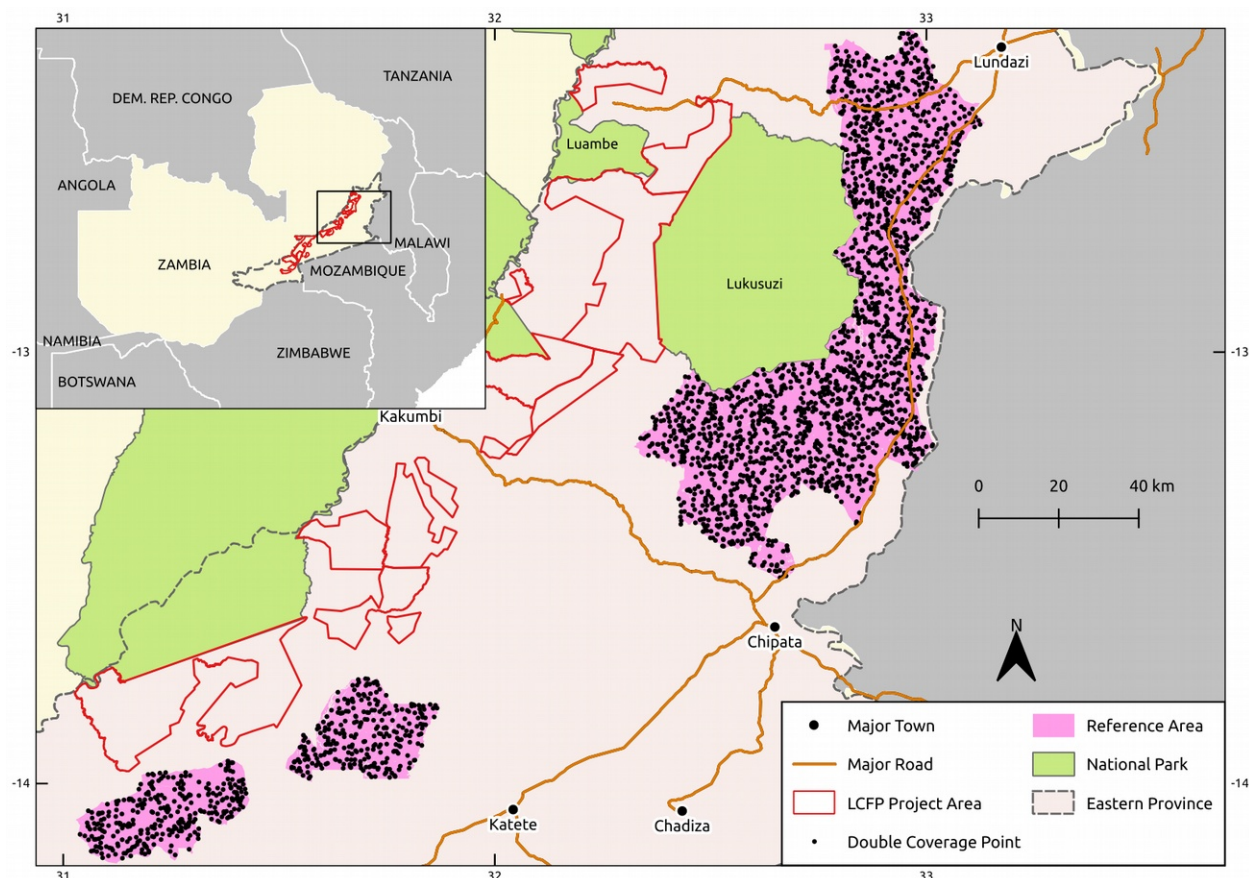


Figure 28: Map of reference area and double coverage points (100%)

Point Interpretation and fitting the Logistic Function

PDR.59-62

VM0009 allows for using parameters (θ) other than time to improve the fit of the logistic function. However, the project only tested a single model with time and found that it provided a good fit. Therefore, no other parameters were considered.

PDR.63

The point interpretation analysis was conducted using a script in Earth Engine to display the historical imagery and make it easy for the user to navigate to each of the 2600 points (script available for VVB review). A protocol was developed to guide the 3 interpreters who conducted the work. The protocol was

developed by a BCP employee who was an expert in remote sensing in Miombo forests in Zambia and very familiar with interpreting a variety of imagery types. The protocol describes, with illustrations:

- How to discern conversion based on changes in pixels, the shapes of changes, surrounding context, and using supplementary higher resolution imagery.
- Being aware of seasonal changes that occur in vegetation, which was most prominent in June imagery (though the historical imagery was specifically selected to avoid this issue as much as possible)
- Defined deforestation as a clearing of 0.5 ha in order to reduce errors associated with image speckle or degradation that did not lead to deforestation. A 5 x 5 grid of Landsat pixel boundaries was overlayed on the imagery and the protocol required that the interpretation pixel (the center pixel) be have an orthogonal connection to at least 5 other pixels to be considered deforested.

PDR.64-65

A sample of 100 points was checked by one of the other interpreters for each of the 3 interpreters. On average, there was 95% agreement. No systematic errors were detected, but following guidance from VM0009 v3.0, it was decided to set all values after observed conversion to conversion, so the data was recoded with a formula in LibreOffice.

PDR.66-67

The logistic function (Figure 29) was fit using an R script. The results and standard errors are shown in table 15. Additionally, the standard deviation of observed conversion $\hat{\sigma}_{EM}$ was estimated using equation F.13 from VM0009 v3.0. $\hat{\sigma}_{EM}$ was found to be 0.4273.

$$\hat{\sigma}_{EM} = \sqrt{\left[\sum_{i \in J} w_i o_i \right] \left[1 - \sum_{i \in J} w_i o_i \right]}$$

Equation F.13

Table 15: Logistic function fitting results from R

Coefficient	Estimate	Standard Error
α	0.2188	0.02932
β	0.0003271	0.000006976

PDR.74-75 No covariates to conversion were used.

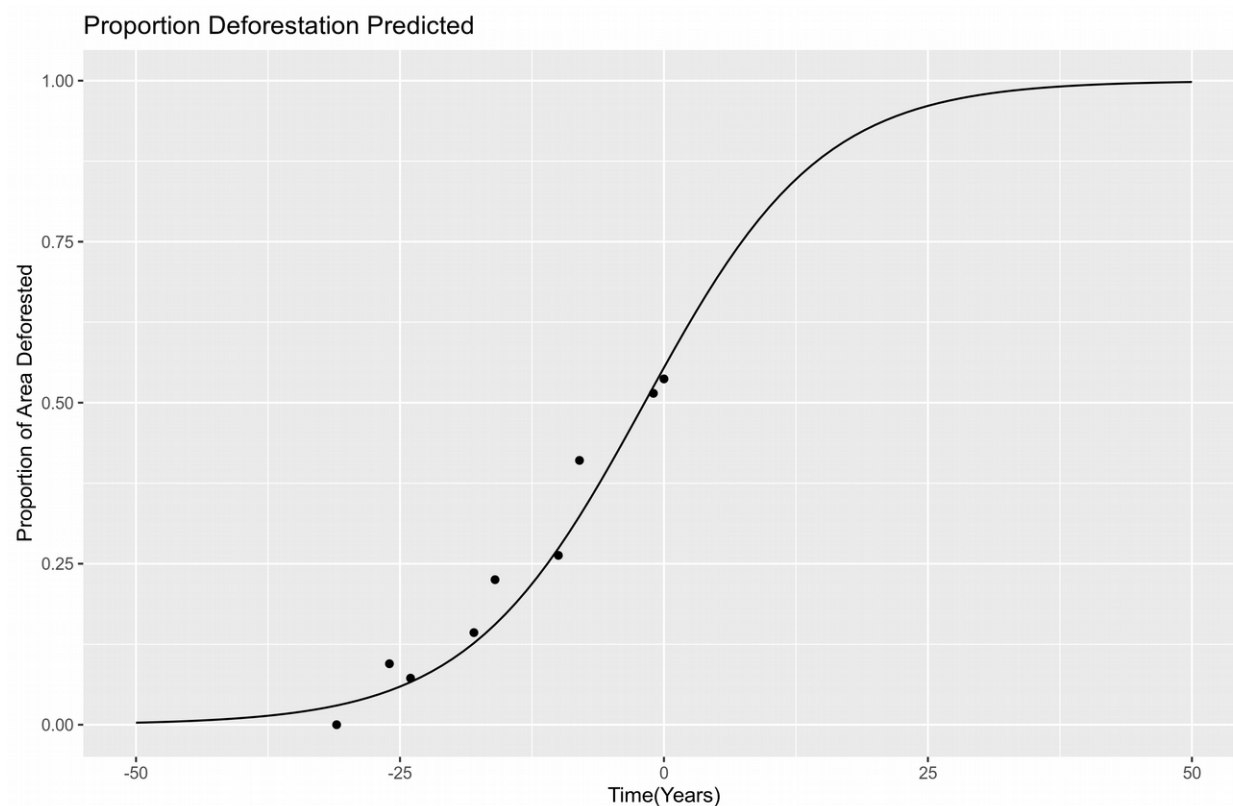


Figure 29: Logistic function used in CDM

Lag between the onset of degradation and the beginning of conversion

PDR.79-80

VM0009 v3.0 optionally allows projects to calculate the parameter q , which is the number of days between the onset of degradation and the beginning of conversion was previously estimated to be 620 days. This came from a participatory rural appraisal with 28 farmers in Lusaka province near BCP's other REDD project, in 2013.

Proximity of project area to recent change

PDR.81-83

VM0009 v3.0 requires that project calculate the parameter $\hat{r}_U$ as the ratio of converted perimeter to total threatened perimeter. Converted perimeter is defined as the perimeter of accounting areas that are within 120 meters of conversion that happened in the previous 10 years. The project did a GIS analysis to determine $\hat{r}_U$ using the PALSAR deforestation analysis described in section 1.1.2 as the source of information about recent deforestation, though due to the dates of available PALSAR imagery, the analysis only went back 8 years to 2007.

The threatened perimeter is perimeter neighbored by forest that is under threat similar to the project accounting area. Identifying these areas required a three step process. First, all of the interior boundaries for the accounting areas were removed because they are more than 6.5 km from a road and thus considered inaccessible to agents of deforestation (as per the accounting area definition). Then the remaining boundaries were checked against high resolution imagery in Google Earth to see if they were bordered by natural non-forest areas such as wide rivers and grasslands. All such areas were also removed. Finally, the remaining perimeter was rasterized into pixels matching the size (approximately 24 x 24 m) and spacing of pixels in the PALSAR deforestation analysis, and raster math was used to remove the parts of the perimeter that were non-forest in 2007 or fell on slopes greater than 45%.

Then, to map the converted perimeter, a raster buffer of 5 pixels (approximately 120 meters since pixels are approximately 24 m accross) was made for areas mapped as deforested in the PALSAR deforestation analysis between 2007 and 2015. Then band math was used to identify the threatened boundary pixels that overlapped with the converted areas or within the 5 pixel buffer around converted areas. $\hat{r}_U$ was then calculated as the ratio of the threatened boundary pixels that overlapped with conversion or the buffer compared to the total threatened boundary pixels. The results are shown in table 16.

Table 16: Results from analysis of conversion within 120 meters of threatened boundaries

Threatened boundary pixels	Threatened boundary pixels near conversion from 2007-2015	$\hat{r}_U$
41091	3449	0.084

Soil Carbon Decay

The project used the default value of 0.2 for the soil decay parameter λ_{SOC} . The project is in a tropical region and thus qualifies under VM0009 to use the default value.

PDR.84-92 Not applicable because λ_{SOC} was not estimated from data or literature.

PDR.93-98 Not applicable because this is the first validation.

3.2.2 Project Emissions

There are no planned project emissions. As described in section 1.1.1, emissions from livestock and fertilizer are considered de minimus for this project. Unplanned project emissions will be calculated as per equation F.41 in VM0009 v3.0. Additionally, changes in project carbon stocks will be tracked with 122 fixed carbon inventory plots as described in section 3.3.3.

3.2.3 Leakage

Activity-shifting leakage is estimated by directly observing conversion and degradation in 1 ha square plots in an activity-shifting leakage area and comparing it to the baseline emissions estimated by the Leakage Emissions Model (equation F.45 of VM0009 v3.0).

PDR.105-109 A map of the delineated boundaries.

A general map of the activity-shifting leakage area boundaries for the initial project accounting areas, access points (roads), and locations of important markets (major towns) is shown in X. The deforestation agent mobility analysis showed that agents have limited mobility and generally don't move far from existing farms over a short time period (the 10 baseline validity). As described in section 1.1.2, 99% of deforestation from 2009 to 2017 occurred within 3.58 km of deforestation that occurred between 2007 and 2009. Therefore, the project concluded it was likely any activity shifting leakage would also mostly likely occur within 3.58 km of the project area boundary. Therefore, the project defined the leakage area as all forested areas within 3.58 km of the project area boundary and outside of a national park, but also added the accounting area constraints of being no more than 6.5 km of a road, and on slopes less than 45%. Forest reserves were included in the leakage area, just as they were included in the reference areas because the deforestation mapping suggested that they are accessible to the same deforestation agents that would also deforest the project accounting area. The same PALSAR based map of forest persistence between 2007 and 2015 used to define the project accounting areas was used to identify forested areas in the leakage area. Additionally, to avoid having leakage plots fall into areas that are already highly fragmented (unlike the project accounting area) a 81x81 moving window was used to calculate the number of neighboring deforested pixels for project accounting area forested pixels. The 99.5% threshold (756 deforested neighbors) was identified and the potential leakage area pixels were filtered to remove those that were above the threshold. Finally, parts of the leakage area that fell in chiefdom that are not working with the LCFP project were removed. This was the only way to guarantee that the project would have access to the leakage areas. This resulted in a leakage area of 319,182 ha, which, while is somewhat smaller than the project accounting area, is sufficiently large to accommodate any potential leakage. In summary, the leakage area is as accessible or more accessible than the project accounting area and given the close proximity to the project accounting area, has similar elevation, slope, land-cover, soil, and proximity to markets.

The following figures provide the maps required by the methodology:

- Figure 2 (Elevation)
- Figure 3 (Slope)
- Figure 7 (Land-Cover)
- Figure 4 (Soil Class)
- Figure 30 (Access points (roads))
- Figure 30 (Markets (major towns))
- Figure 30 (Chieftdom land, national, or local forest reserve (Not in a National Park))

PDR.110-117 The project did not consider market leakage because the project will not significantly affect the supply of wood products or commodities. This is because few wood products are taken from the project area in the baseline scenario and most farming in the baseline scenario is subsistence.

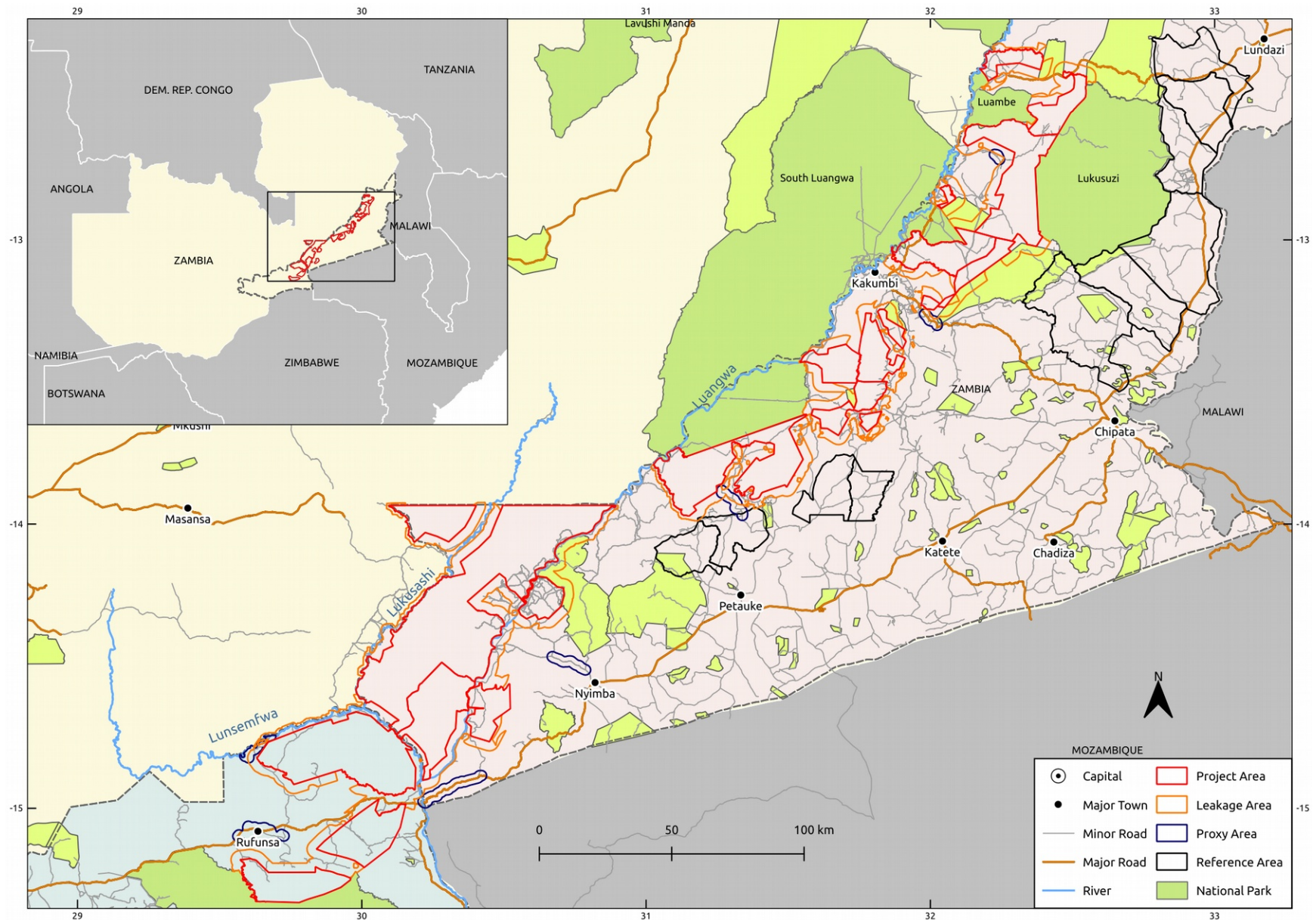


Figure 30: Location of roads, major towns (markets) in relation to leakage areas

3.2.4 Net GHG Emission Reductions and Removals

PDR.118-119

Net GHG emission reductions and removals are calculated using equation F.56 from VM0009 v3.0 and Appendix G. All of the required inputs for the calculations are listed in sections 3.3.1 and 3.3.2.

The project had not yet completed its SOC estimates from the project accounting area, so for the purposes of calculating ex-ante emissions reductions, The average project accounting area SOC value from BCP's other REDD project, the Lower Zambezi REDD Project, was used.

Leakage estimates were also based on experience with its other REDD project, where there has been zero leakage after 10 year.

Project performance compared to the baseline was estimated as starting at 90% in year 1, rising to 95% in 2020, and then 99% in year 2022. This is also based on experience from BCP's other REDD+ project where encroachment was reduced to near zero after a few years of project implementation.

PDR.120 Not applicable because project activities do not include biomass burning, charcoal production, or logging.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	VCS Risk Buffer Allocation (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2015-2018	4,835,545	0	531,910	483,555	3,820,081
2019	1,695,591	0	186,515	169,559	1,339,517
2020	1,840,621	0	202,468	92,031	1,546,122
2021	2,030,036	0	223,304	101,502	1,705,230
2022	2,214,811	22,148	243,629	22,148	1,949,034
2023	2,367,900	23,679	260,469	23,679	2,083,752
2024	2,579,723	25,797	283,770	25,797	2,270,156
2025	2,763,587	27,636	303,995	27,636	2,431,957
2026	2,980,175	29,802	327,819	29,802	2,622,554

2027	3,154,321	31,543	346,975	31,543	2,775,802
2028	3,379,351	33,794	371,729	33,794	2,973,829
2029	3,430,791	34,308	377,387	34,308	3,019,096
2030	3,828,359	38,284	421,119	38,284	3,368,956
2031	3,711,551	37,116	408,271	37,116	3,266,165
2032	4,039,265	40,393	444,319	40,393	3,554,553
2033	3,886,754	38,868	427,543	38,868	3,420,344
2034	4,099,805	40,998	450,979	40,998	3,607,828
2035	4,293,415	42,934	472,276	42,934	3,778,205
2036	4,108,524	41,085	451,938	41,085	3,615,501
2037	4,144,999	41,450	455,950	41,450	3,647,599
2038	4,120,498	41,205	453,255	41,205	3,626,038
2039	4,489,426	44,894	493,837	44,894	3,950,695
2040	3,864,744	38,647	425,122	38,647	3,400,975
2041	4,135,310	41,353	454,884	41,353	3,639,073
2042	3,824,867	38,249	420,735	38,249	3,365,883
2043	3,411,357	34,114	375,249	34,114	3,001,994
2044	3,587,097	35,871	394,581	35,871	3,156,645
2045	3,023,432	30,234	332,578	30,234	2,660,620
Total					

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation

PDR.121

Data / Parameter	α
Data unit	Unitless
Description	Combined effects of β and θ at the start of the historic reference period
Equations	[F.3], [F.5], [F.6], [F.8]
Source of data	Reference area and historic reference period

Value applied	0.2188
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	β
Data unit	Unitless
Description	Effect of time on the cumulative proportion of conversion over time
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8]
Source of data	Reference area and historic reference period
Value applied	0.0003271
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	γ
Data unit	Days
Description	Time shift from beginning of historic reference period to project start date
Equations	[F.3], [F.6]
Source of data	Historic reference period
Value applied	10,589

Justification of choice of data or description of measurement methods and procedures applied	Time in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	Provide any additional comments

Data / Parameter	λ_{soc}
Data unit	Proportion (unitless)
Description	Exponential soil carbon decay parameter
Equations	[F.9], [F.33]
Source of data	VCS AFOLU default for tropical regions
Value applied	0.2
Justification of choice of data or description of measurement methods and procedures applied	Conservative default based on IPCC value.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	$\hat{\sigma}_{EM}$
Data unit	Standard deviation (unitless)
Description	The estimated standard deviation of the state observations used to fit the logistic function
Equations	[F.12], [F.14], [B.31]
Source of data	Remote sensing image interpretation. Analysis undertaken as described in section 3.2.1 of PD.
Value applied	0.4273

Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>B</i>
Data unit	Set
Description	The set of all selected carbon pools in biomass. Is a subset of <i>C</i>
Equations	[F.17], [F.18], [F.23], [F.50]
Source of data	PDD
Value applied	2
Justification of choice of data or description of measurement methods and procedures applied	AGOT and BGOT, See section 1.1.1 for justification
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>C</i>
Data unit	Set
Description	The set of all selected carbon pools
Equations	
Source of data	Monitoring records
Value applied	3
Justification of choice of data or description of measurement methods and procedures applied	AGOT, BGOT, SOC, See section 1.1.1 for justification

Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>J</i>
Data unit	Set
Description	The set of all observations of conversion. When superscripted with a monitoring period, the conversion observations are taken for leakage analysis
Equations	[F.13]
Source of data	Remote sensing image interpretation or field observations in the leakage area
Value applied	13,194
Justification of choice of data or description of measurement methods and procedures applied	Analysis undertaken as described in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	<i>M</i>
Data unit	Set
Description	The set of all monitoring periods
Equations	[F.32], [F.33], [F.36], [F.54], [F.56]
Source of data	Monitoring records
Value applied	27
Justification of choice of data or description of measurement methods and procedures applied	The project crediting period is 30 years and after this first verification, the project will use annual verification. Thus there will be 27 verification.
Purpose of data	Calculation of baseline emissions

Comments	
----------	--

Data / Parameter	A_{PAA}
Data unit	ha
Description	Area of project accounting area
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8], [F.41], [F.52], [F.57]
Source of data	GIS analysis prior to sampling
Value applied	485,495
Justification of choice of data or description of measurement methods and procedures applied	Area within the project area that is on slopes of less than 45%, less than 6.5 km from a road, and was forest at least 10 years before the project start date. Described in more detail in in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	A_{PX}
Data unit	ha
Description	Area of proxy area
Equations	[F.57]
Source of data	GIS analysis prior to sampling
Value applied	3840
Justification of choice of data or description of measurement methods and procedures applied	Areas converted from forest to non-forest between 2010 and 2017 falling within 2 km of buffer around roads passing through communities favorable to having proxy plots on their lands. Described in more detail in in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	n_d
------------------	-------

Data unit	
Description	Number of spatial points in the reference area
Equations	[F.14]
Source of data	Remote sensing image interpretation
Value applied	2600
Justification of choice of data or description of measurement methods and procedures applied	Value deemed sufficient as per VM0009 to adequately quantify forest state in the reference area. Described in more detail in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	o_i
Data unit	Binary
Description	State observation for the sample point in the reference area
Equations	[F.13]
Source of data	Remote sensing image interpretation
Value applied	0 or 1
Justification of choice of data or description of measurement methods and procedures applied	Data collected as per the methodology. Described in more detail in section 3.2.1 of PD.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	P_{LME}
Data unit	Proportion (unitless)
Description	Portion of leakage related to market
Equations	[F.51]

Source of data	Analysis of drivers of deforestation (Appendix K)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Few wood products are taken from the project area in the baseline scenario and most farming in the baseline scenario is subsistence.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	q
Data unit	Days
Description	Lag between start of degradation and conversion
Equations	[F.3], [F.4], [F.5]
Source of data	Consultation with agents of deforestation in near project area in Lusaka province – 28 farmers were questioned to determine how long they usually wait after clearing an area before ploughing it.
Value applied	620
Justification of choice of data or description of measurement methods and procedures applied	Based on observations in the reference area, hence applicable to the project area.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	r_{RS}
Data unit	Unitless
Description	Expansion factor for above-ground biomass to below-ground biomass (root/shoot ratio)
Equations	[F.30]

Source of data	Chidumayo, E. N. 2013. Forest degradation and recovery in a miombo woodland landscape in Zambia: 22 years of observations on permanent sample plots. Forest Ecology and Management 291:154-161.
Value applied	0.54
Justification of choice of data or description of measurement methods and procedures applied	The root-shoot ratio was determined within a study site that overlaps one of the same provinces the project is in.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	r_u
Data unit	Proportion (unitless)
Description	Onset proportion of conversion immediately adjacent to project area
Equations	[F.4]
Source of data	GIS analysis and image interpretation
Value applied	0.084
Justification of choice of data or description of measurement methods and procedures applied	Result of remote sensing analysis described in more detail in section 3.2.1.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t
Data unit	Days
Description	Time since project start date
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8], [F.10]
Source of data	Calendar

Value applied	1340
Justification of choice of data or description of measurement methods and procedures applied	Project start date is May 1 st , 2015 and first monitoring period ended December 31 st , 2018.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t_i
Data unit	Days
Description	The point in time of the observation made at point i
Equations	[F.11], [A.6]
Source of data	Remote sensing image interpretation
Value applied	1-11341
Justification of choice of data or description of measurement methods and procedures applied	The historical reference period stretches from May 4 th , 1986 to May 22 nd 2017, which is 11341 days in total.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t_{PL}
Data unit	Days
Description	Length of project crediting period
Equations	[F.5]
Source of data	PD
Value applied	10957
Justification of choice of data or description of	The project crediting period is from May 1 st , 2015 to April 30 th , 2045 which is equivalent to 10957 days.

measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	w_i
Data unit	Unitless
Description	Weight applied to the i^{th} sample point in the reference area
Equations	[A.6], [F.13]
Source of data	Remote sensing image interpretation
Value applied	Varies (see Appendix O)
Justification of choice of data or description of measurement methods and procedures applied	Calculated as specified in VM0009
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	x_i, y_i
Data unit	Geographic coordinates
Description	Latitude and Longitude of the i^{th} sample point
Equations	[A.6], [F.11]
Source of data	Remote sensing image interpretation
Value applied	Varies (see appendix Appendix O)
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions

Comments	
----------	--

3.3.2 Data and Parameters Monitored

Data / Parameter	$A_{P1}^{[m=0]}$
Data unit	ha
Description	Area of project accounting area stratum 1 prior to first verification event
Equations	[F.24]
Source of data	GIS analysis prior to sampling
Description of measurement methods and procedures to be applied	GIS analysis removing areas in the project area that were non-forest prior to the project start date, on slopes greater than 45% or in areas farther than 6.5 km from a road.
Frequency of monitoring/recording	First monitoring period / 10 years
Value applied	485,495
Monitoring equipment	N/A
QA/QC procedures to be applied	Double checking GIS analysis
Purpose of data	Calculation of baseline emissions
Calculation method	GIS area calculation
Comments	

Data / Parameter	$C_B^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline carbon stocks at the end of the current monitoring period

Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.57]
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value applied	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.
Comments	

Data / Parameter	$C_{B\ BGB}^{[m]}$
Data unit	tCO ₂ e
Description	Carbon not decayed in BGB at the end of the current monitoring period
Equations	[F.16]
Source of data	Equation F.32 from VM0009
Description of measurement methods and procedures to be applied	Ongoing forest monitoring as per Appendix P
Frequency of monitoring/recording	Based on peer reviewed literature and prescribed equation in VM0009
Value applied	84,940

Monitoring equipment	See Appendix P
QA/QC procedures to be applied	See Appendix P
Purpose of data	Calculation of baseline emissions
Calculation method	Applied equation F.32 in VM0009 3.0
Comments	

Data / Parameter	$C_{B\text{ SOC}}^{[m]}$
Data unit	tCO ₂ e
Description	Carbon not decayed in SOC at the end of the current monitoring period
Equations	[F.16]
Source of data	Project accounting area sampling and application of equation F.33 from VM0009 3.0
Description of measurement methods and procedures to be applied	Accounting area sampling as per the protocol in Appendix P and application of equation F.33, VM0009 V3.0
Frequency of monitoring/recording	Fixed once off sampling, not repeated as per VM0009
Value applied	33515
Monitoring equipment	See Appendix P standard soil sampling procedures and equipment.
QA/QC procedures to be applied	See Appendix P.
Purpose of data	Calculation of baseline emissions
Calculation method	Application of equation F.33 VM0009 V3.0 using SOC values from the project accounting area.
Comments	SOC data at validation comes from LZRP REDD project as LCFP SOC data not ready at document submission date.

Data / Parameter	$C_{Bb}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline scenario average carbon stock in selected carbon pools
Equations	[F.18] [F.24]
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value applied	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.
Comments	

Data / Parameter	$C_{B\ BM}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline carbon stocks in biomass at the end of the current monitoring period
Equations	[F.19], [F.20], [F.21], [F.24], [F.52]
Source of data	Proxy area sampling

Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	At first verification and baseline revision
Value applied	15.6
Monitoring equipment	See Appendix P
QA/QC procedures to be applied	Review of monitoring records as described in Appendix P and re-sampling of 20% of plots
Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations. Averages calculated as per equation B.9 VM0009 V3.0
Comments	

Data / Parameter	$C_{B\text{ SOC}}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Baseline soil carbon stocks at the end of the current monitoring period
Equations	[F.25], [F.27], [F.28]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	Once at first verification
Value applied	131.7

Monitoring equipment	See Appendix P, standard soil sampling equipment
QA/QC procedures to be applied	Review of monitoring records at next baseline re-evaluation
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0
Comments	

Data / Parameter	$C_P^{[m]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks at the end of the current monitoring period
Equations	[F.41], [F.57]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Plot monitoring as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value applied	239.5
Monitoring equipment	See Appendix P, standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_P^{[m-1]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks at the beginning of the current monitoring period
Equations	[F.41]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value applied	239.5
Monitoring equipment	Review of monitoring records, see Appendix G.
QA/QC procedures to be applied	Review of monitoring records, re-sampling of 20% of plots.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_P^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks prior to first verification event
Equations	[F.7]
Source of data	Project accounting area sampling

Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	239.5
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{P1BM}^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks in stratum 1 prior to first verification event
Equations	[F.24]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	239.5

Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{P1BM}^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project carbon stocks in stratum 1 prior to first verification event
Equations	[F.24]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	239.5
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.

Comments	
----------	--

Data / Parameter	$C_{PBM}^{[m=0]}$
Data unit	tCO ₂ e
Description	Project carbon stocks in biomass prior to first verification event
Equations	[F.19], [F.20], [F.21], [F.52]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	116,291,351
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{Pb}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Average carbon in biomass in the project accounting area
Equations	[F.17]
Source of data	Project accounting area sampling

Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of biomass plots annually
Value applied	239.5
Monitoring equipment	Appendix P describes the plot based monitoring equipment.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{P1b}^{[m]}$
Data unit	tCO ₂ e / ha
Description	Average carbon in biomass for project accounting area stratum 1
Equations	[F.23]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of biomass plots annually
Value applied	239.5

Monitoring equipment	Appendix P describes the plot based monitoring equipment.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$C_{P\text{SOC}}^{[m=0]}$
Data unit	tCO ₂ e / ha
Description	Project soil carbon stocks prior to first verification event
Equations	[F.25], [F.27], [F.28]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	192.6
Monitoring equipment	Appendix P, standard soil sampling equipment
QA/QC procedures to be applied	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Comments	This soil data is borrowed from LZRP project since LCFP

	accounting area soil data was not yet available at submission.
--	--

Data / Parameter	$E_{\Delta GER}^{[m]}$
Data unit	tCO ₂ e
Description	GERs for the current monitoring period
Equations	[F.55], [F.57]
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	

Data / Parameter	$E_{\Delta GER}^{[i]}$
Data unit	tCO ₂ e
Description	GERs for monitoring period i
Equations	[F.54]
Source of data	Area measurements

Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Prior monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	

Data / Parameter	$E_{\Delta NER}^{[i]}$
Data unit	tCO ₂ e
Description	NERs for monitoring period i
Equations	[F.56]
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Prior monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment

QA/QC procedures to be applied	Review of NER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with VCS AFOLU Non-Permanence Risk Tool and equation F.55 as per section 8.4.3 of VM0009 V3.0
Comments	

Data / Parameter	$E_B^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions at the end of the current monitoring period
Equations	[F.15]
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.16.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	

Data / Parameter	$E_B^{[m-1]}$
------------------	---------------

Data unit	tCO ₂ e
Description	Cumulative baseline emissions at the beginning of the current monitoring period
Equations	[F.15]
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.16.
Frequency of monitoring/recording	Prior monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	

Data / Parameter	$E_{B\Delta}^{[m]}$
Data unit	tCO ₂ e
Description	Change in baseline emissions
Equations	[F.9], [F.10], [F.14], [F.53], [F.57]
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.15.

Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.15 from VM0009 V3.0
Comments	

Data / Parameter	$E_{B \Delta BGB}^{[i]}$
Data unit	tCO ₂ e
Description	Change in baseline emissions from below-ground biomass during monitoring period i
Equations	[F.32]
Source of data	Monitoring the proxy area
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions

Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{BASOC}^{[m]}$
Data unit	tCO ₂ e
Description	Baseline change in emissions from soil carbon
Equations	[F.15],[F.33]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P
Frequency of monitoring/recording	At first verification
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{BASOC}^{[i]}$
Data unit	tCO ₂ e

Description	Baseline emissions from soil carbon in monitoring period i
Equations	[F.33]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P
Frequency of monitoring/recording	At first verification
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{BBGB}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from below-ground biomass at the end of the current monitoring period
Equations	[F.31]
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of	Every monitoring period

monitoring/recording	
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{BBGB}^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from below-ground biomass at the beginning of the current monitoring period
Equations	[F.31]
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.

Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	0 for the first monitoring period

Data / Parameter	$E_{BBM}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from biomass at the end of the current monitoring period
Equations	[F.16], [F.30], [F.52]
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	See Appendix E
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard vegetation sampling techniques, quantification of deforestation in the reference area and application of equation F.4 and F.24, VM0009 V3.0
Comments	

Data / Parameter	$E_{BSOC}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from soil carbon at the end of the current monitoring period
Equations	[F.16], [F.26]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	

Data / Parameter	$E_{BSOC}^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative baseline emissions from soil carbon at the beginning of the current monitoring period
Equations	[F.26]
Source of data	Measurement in the project area

Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification
Value applied	
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	

Data / Parameter	$E_{BA}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions allocated to the buffer account at the end of the current monitoring period
Equations	[F.55]
Source of data	VCS Non-Permanence Risk Report, See Appendix H
Description of measurement methods and procedures to be applied	See Appendix H – standard VCS procedure used.
Frequency of monitoring/recording	Non-permanence risk revised at every baseline re-assessment.
Value applied	

Monitoring equipment	None
QA/QC procedures to be applied	Annual review of risk criteria and disturbance monitoring
Purpose of data	Calculation of emissions submitted to VCS buffer account
Calculation method	Procedure followed in VCS non-permanence risk tool and emissions allocated to buffer account subtracted from GERs.
Comments	

Data / Parameter	$E_L^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions from leakage at the end of the current monitoring period
Equations	[F.44]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	

Data / Parameter	$E_L^{[m-1]}$
Data unit	tCO ₂ e
Description	Cumulative emissions from leakage at the beginning of the current monitoring period
Equations	[F.44]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	

Data / Parameter	$E_{L\Delta}^{[m]}$
Data unit	tCO ₂ e
Description	Change in emissions due to leakage
Equations	[F.53]
Source of data	Measurements in the leakage area(s)

Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	

Data / Parameter	$E_{L\ ASF}^{(m)}$
Data unit	tCO ₂ e
Description	Cumulative emissions from activity-shifting leakage in forested strata at the end of the current monitoring period
Equations	[F.45]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	

Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	

Data / Parameter	$E_{LME}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative emissions from market leakage at the end of the current monitoring period
Equations	[F.45]
Source of data	N/A
Description of measurement methods and procedures to be applied	Land-use by deforestation agents is subsistence and therefore all market leakage is captured by monitoring in activity shifting leakage areas.
Frequency of monitoring/recording	N/A
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	N/A
Purpose of data	N/A
Calculation method	N/A
Comments	

Data / Parameter	$E_{P\Delta}^{[m]}$
Data unit	tCO ₂ e
Description	Change in project emissions
Equations	[F.53]
Source of data	Monitoring records for Forest Fire, Burning, logging, wood products, and natural disturbance events
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques and remote sensing to determine the extent of fires and other disturbances.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	Standard vegetation sampling equipment and satellite imagery.
QA/QC procedures to be applied	Remote sensing, patrols and field measurements
Purpose of data	Calculation of project emissions
Calculation method	Monitoring data from areas of Forest Fire, Burning, logging, wood products, and natural disturbance events and equation F.41 VM0009 V3.0.
Comments	

Data / Parameter	$E_{P\Delta BRN}^{[m]}$
Data unit	tCO ₂ e
Description	Cumulative project emissions due to burning at the end of the current monitoring period
Equations	[F.41]
Source of data	Monitoring plots in the project accounting area

Description of measurement methods and procedures to be applied	Application of equation F.42 using burning monitoring data as input.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	Standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records
Purpose of data	Calculation of project emissions
Calculation method	Monitoring data from areas of Burning and equation F.42 VM0009 V3.0.
Comments	

Data / Parameter	$E_U^{(m)}$
Data unit	tCO ₂ e
Description	Cumulative confidence deduction at the end of the current monitoring period
Equations	[F.55]
Source of data	N/A
Description of measurement methods and procedures to be applied	Application of equation F.57 which is the a linear combination of weighted standard errors of estimates from baseline emission models and carbon stock measurements.
Frequency of monitoring/recording	Every monitoring period
Value applied	

Monitoring equipment	N/A
QA/QC procedures to be applied	Review of calculations
Purpose of data	Calculation of project emissions
Calculation method	Equation F.57 from VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{[m]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation at the end of the current monitoring period
Equations	[F.46], [F.47], [F.48], [F.49]
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{[m]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation at the end of the current monitoring period
Equations	[F.46], [F.47], [F.48], [F.49]
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{[m=0]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation prior to first verification event
Equations	[F.48]
Source of data	Monitoring in the leakage area

Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	First monitoring period
Value applied	
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{(m=0)}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation prior to first verification event
Equations	[F.48]
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	First monitoring period
Value applied	

Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

3.3.3 Monitoring Plan

Table 17. Summary of monitoring activities.

Monitoring Activity	Timing	Methods
Trespassing and boundary transgressions	Monthly patrols backed up with aerial surveys at least once per dry season.	Scouts patrol the project are on an ongoing basis, it is their core duty
Leakage monitoring	Annually	Forest Inventory team following VM0009 and remeasuring 100% of leakage plots each year as per protocol in Appendix R.
Plot measurements	Annually	Forest Inventory Team following VM0009, remeasuring 20% of plots in project accounting and project proxy areas as per protocol in Appendix P.
Disturbance monitoring	One a year using satellite imagery from April-May	Remote sensing using Sentinel-2 or Landsat imagery. Followed up with aerial surveys and, or scout patrols.

PDR.122

The principle means of monitoring project emissions is with inventory plots in project accounting areas where above ground biomass is measured by recording tree dbh. Allometric equations are used to convert the measures to above and below ground biomass. Soil samples are also collected during the first monitoring period to establish the soil carbon stocks for the project accounting area. VM0009 requires that all such plots must be measured during the first monitoring period and then 20% measured during each subsequent monitoring period. After mapping the project accounting area described in section 3.2.1, the project used gis software to generate 122 plots randomly (Figure 31) within the project accounting area polygon. The sample size was estimated based on the variance in biomass inventory plot data from BCP's other REDD project, the Lower Zambezi REDD project.

Plot monitoring is conducted according to the BCP field sampling manual (Appendix P). Optimum plot size and minimum tree diameter were determined under BCP's other project, the Lower Zambezi REDD Project. Circular plots (12m radius) are used due to their simplicity, ease of layout and reduced measurement error. A randomly generated plot location is navigated-to using a handheld GPS or Android Smart Phone. When the plot location is reached, the centre is permanently marked with a steel rebar driven into the ground.

Trees are identified to species level as far as possible. The use of a generic allometric equation to calculate AGOT mass fortunately negated mistakes or uncertainties with tree species identification.

As explained above, plots are re-measured on a rotating basis, with 20% re-measured per year, however stock estimates are only updated every 5 years after all plots have been re-measured.

In addition to the inventory plots, human and natural disturbances in the project accounting area will be mapped using visual interpretation of Sentinel-2 and if needed Landsat data. This will be done once a year using the most cloud free imagery from April-May. This represents the end of the raining season and beginning of the dry season. This is generally the best time of year for mapping woodlands in Zambia as there is less cloud cover, trees generally still have leaves, and the under story grass has started to dry out providing better contrast between grassy and wooded areas. The woodlands in the project area are fire adapted and fires generally don't cause massive disturbance. April-May imagery helps avoid mistakenly mapping fire scars as disturbance, when the fire likely only burned the under story grass. With April-May imagery, only permanently damaging fires will be mapped as disturbance.

When significant disturbances are detected in an area not covered by an inventory plot, a polygon will be drawn around the area and the per hectare emissions factors determined from the plot inventory work will be applied to the area to calculate project emissions.

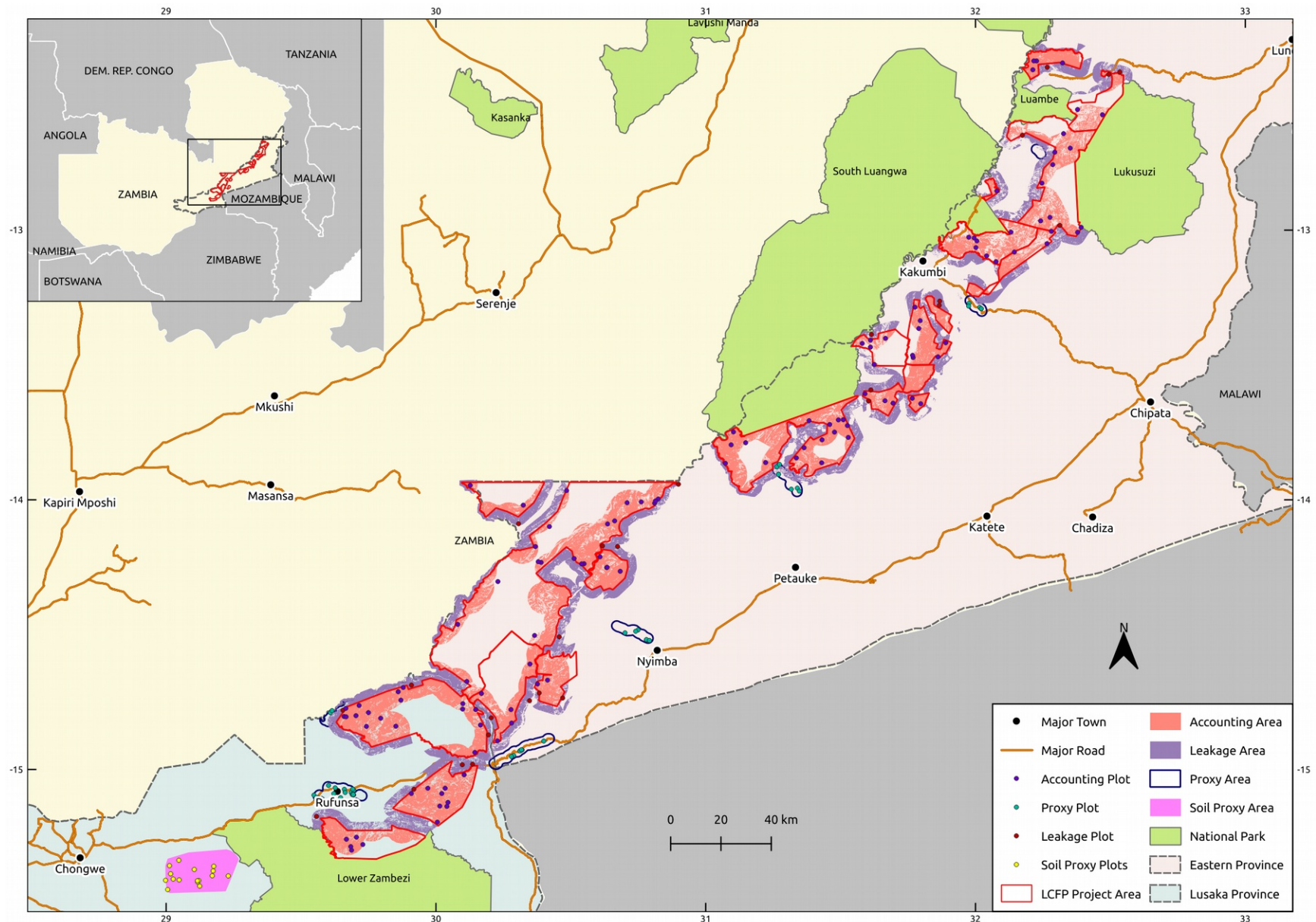


Figure 31: Accounting, Proxy, and Leakage Plots

PDR.123

Proxy plots are used to estimate residual biomass after conversion and the end point for soil carbon decay. For estimating residual biomass, after mapping the proxy areas as described in section 3.2.1, the project generated a random sample of 45 points within the proxy areas. Variability in proxy plots is high since some farmers leave large isolated trees in their farms and invariably, some proxy plots will include these trees. Going based on experience from the previous LZRP REDD project, BCP settled on a sample size of just 45 points. This is because while the variability with proxy plots is high relative to the proxy plot average, that variability is low relative to the accounting area carbon stocks and thus represents a relatively small portion of the overall uncertainty in baseline emissions.

Since proxy soil carbon is only measured once, it is important to select samples from areas that represent the end point of conversion. This data was already available from the previous LZRP REDD project for 16 soil plots from older farms in that project's proxy area.

The same biomass and soil sampling protocols used for accounting areas (Appendix P) were used for proxy areas as well.

PDR.124

Leakage plots in the leakage area are used to estimate activity-shifting leakage from the project accounting areas. The project generated a random sample of 25 points within the leakage area where 1 ha leakage plots could be established. Equation B.31 as per VM0009, suggested the minimum number of plots required was 22.

The leakage plot protocol is described in Appendix R.

Organizational Structure, responsibilities and competencies

BioCarbon Partners has a team of experts responsible for the carbon stock monitoring and standards compliance. As per VM0009, carbon stock monitoring is done using a combination of permanent plots in the project accounting area, proxy area, and leakage area, and remote sensing. Scout teams, are used to confirm encroachment and deforestation in the project area as part of the community engagement process to resolve encroachment (Figure 32). The carbon accounting department oversees all forest related data collection, processing, storage, GIS and remote sensing and electronic data capture systems with a central database (Figure 33).

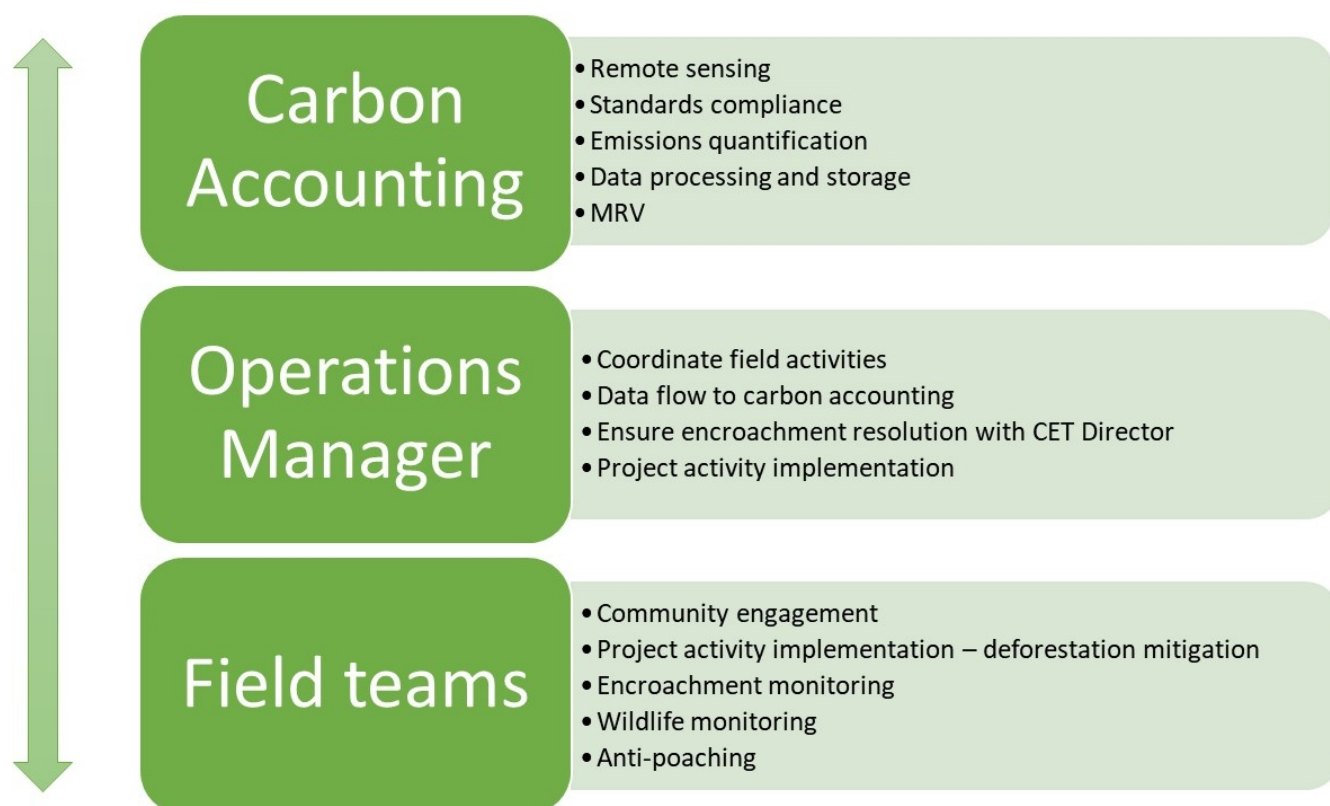


Figure 32. BCP Carbon Monitoring Process.

Both the carbon accounting manager and remote sensing and GIS specialist have post graduate degrees with a strong focus on African ecology and collectively have more than 20 years' experience in biological inventories, remote sensing and GIS.

Data Management

The process followed for field data collection is outlined in Figure 33. It is a feedback system that relies heavily on training, review and good communication. All training is followed by trial runs of the sampling procedure and then actual data collection under supervision. Data is continuously submitted from the field to the carbon accounting department, and feedback is provided if any corrective actions are required.

Remote Sensing data does not have a field component, but all data is also stored on Gdrive.

Once data has been error checked and cleaned it is safely stored online using Google Drive and then transferred to an online relational database.

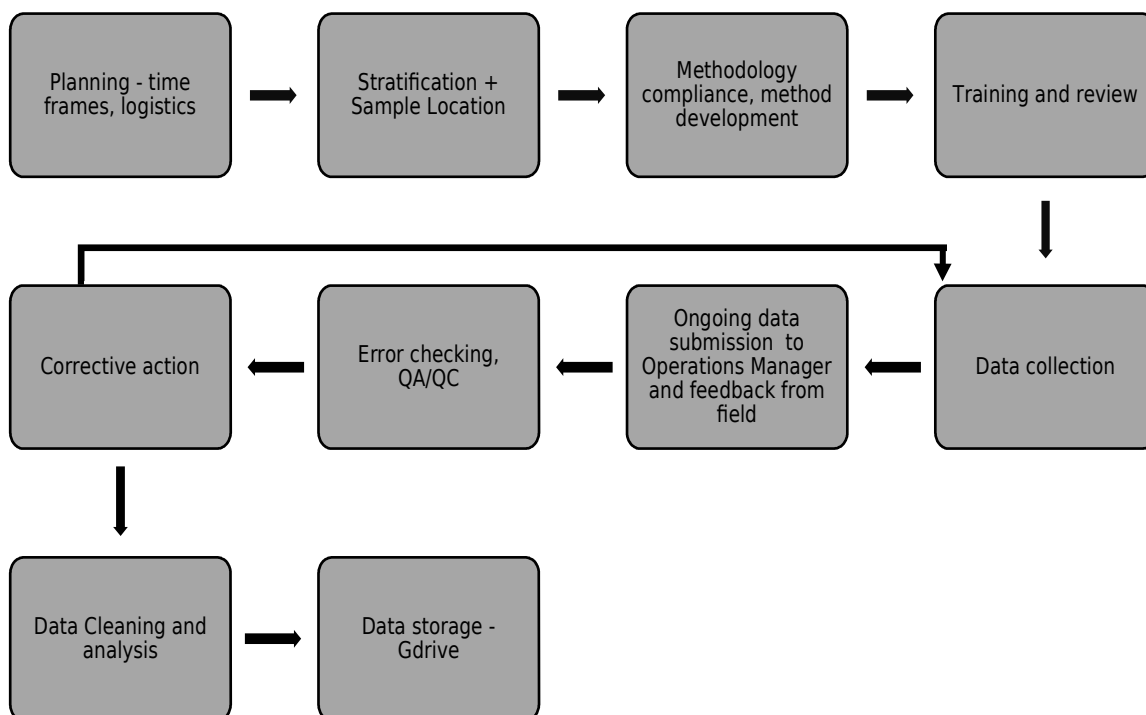


Figure 33. BioCarbon Partners data collection, QA/QC and storage process.

Quality Assurance and Quality Control

All data submitted to the central database is subjected to quality control by appropriate staff. The majority of field data is collected using digital tools on smart phones and are either directly loaded to the database or saved on Gdrive where it is checked and then transferred to the database.

For remote sensing data, accuracy assessments are undertaken for all LULC analysis as described in the Baseline Scenario component of the PD.

3.3.4 Dissemination of Monitoring Plan and Results (CL4.2)

All verification documentation is made available to communities and other stakeholders. Project activities are discussed at community consultation meetings, reported in newsletters and updates posted on the BCP website.

The monitoring plans for this project will be shared as a publicly available appendix to this project document on the VCS / CCB project databases (<https://www.vcsprojectdatabase.org/>). Verified monitoring reports will also be made publicly available through the same online database.

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (GL1.1)

The impacts of climate change are primarily reported using the outputs from General Circulation Models (Grotch and MacCracken, 1991, Boko et al., 2007). These models are used to understand how climate is likely to change in the future and thus contribute to informed decision making regarding adaptation and potential mitigation. These models, however, are operated at a coarse resolution and are often inappropriate for local scale projects such as the Luangwa Community Forest Project discussed in this document. As an alternative, this project makes use of the UNDP Climate Change Country Profiles published by McSweeney et al. (2010). The document outlining projected changes for Zambia is available from the following url: <http://www.geog.ox.ac.uk/research/>.

Projections of future climate change are primarily reported using temperature and rainfall variables with the regional models outputting statistics related to how both temperature and rainfall are likely to change in the next 30 to 90 years (Figures 21 & 22). Primary findings for Zambia are as follows (McSweeney et al. 2010):

Temperature: Mean annual temperature is projected to increase by 1.2 to 3.4°C by 2060 and 1.6 to 5.5°C by 2090. The range of change predicted under any one emissions scenario is between 1.5 and 2.5°C. Regionally, the rate of warming is slightly more rapid in the southern and western regions of Zambia. All projections indicate a substantial increase in the number of both days and nights considered 'hot' in the current climate. All projections indicate that the number of days and nights considered 'cold' will decrease. In addition, 'cold' nights will decrease in frequency more rapidly than 'cold' days.

Rainfall: Projections of mean rainfall do not indicate large changes in annual rainfall, however, seasonally the range of projections from different models is large. Ensemble projections indicate a decrease in September-October-November (SON) rainfall by 2090 while December-January-February (DJF) is set to increase by 2090, particularly in the north-east of the country. The proportion of total rainfall that falls in heavy events is projected to increase annually, but mainly in DJF and March-April- May (MAM).

Projections indicate that maximum 1 and 5 day rainfall events may increase in magnitude in DJF and MAM.

Projections reported above indicate that the LCFP area is likely to become warmer over the next 30 years with an increase in the frequency of extreme rainfall events particularly in the months of DJF. The Zambian climate is also especially sensitive to the El Niño Southern Oscillation (ENSO) and La Niña (Anti-ENSO) phenomenon. However, climate projections of El Niño events show wide disagreements between models adding additional uncertainty. What is however understood without any uncertainty is that the people living in the project zone are highly vulnerable to any changes in the area brought about by climate or any other anthropogenic driver. As such, the project activities listed earlier seek to improve resilience and adaptation of local peoples, while taking cognisance of the projected climate changes with special emphasis on what appears to be a change in the seasonality of the current rainfall regime. Should temperatures increase and rainfall become more erratic—as is projected, and reported to be already happening by local communities in the project zone—deforestation pressure is likely to increase under the baseline scenario as people's dependence on charcoal and poaching could increase in the case of failed or marginal crops. Projections suggest that the area will become hotter and more importantly there appears to be a decrease in the number of cold days, which could have a negative impact on both the fauna and flora within the project area. More frequent drought years and more intense flooding are other likely impacts of climate change in the Luangwa valley.

Temperature:

- Mean annual temperature is projected to increase by 1.2 to 3.4°C by 2060 and 1.6 to 5.5°C by 2090. The range of change predicted under any one emissions scenario is between 1.5 and 2.5°C.
- Regionally, the rate of warming is slightly more rapid in the southern and western regions of Zambia.
- All projections indicate a substantial increase in the number of both days and nights considered 'hot' in the current climate.
- All projections indicate that the number of days and nights considered 'cold' will decrease. In addition, 'cold' nights will decrease in frequency more rapidly than 'cold' days.

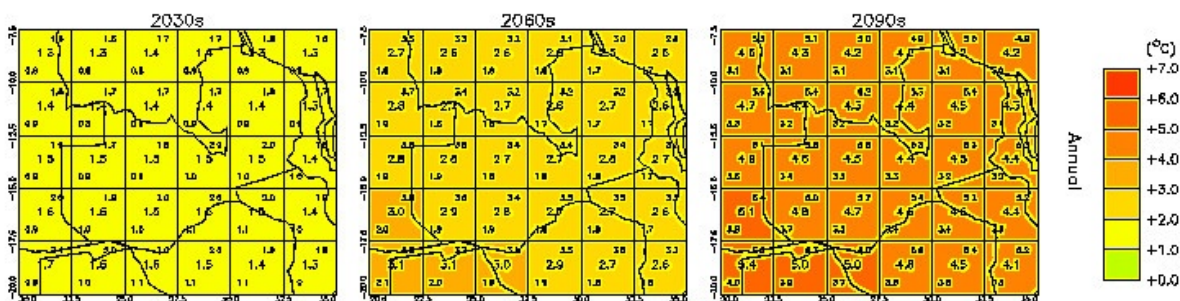


Figure 34. Spatial patterns of projected change in mean annual temperature for 10-year periods (SRES A2 scenario). All values are anomalies relative to the mean climate of 1970-1999. In each grid box, the central value gives the ensemble median and the values in the upper and lower corners give the ensemble maximum and minimum.

Rainfall:

- Projections of mean rainfall do not indicate large changes in annual rainfall, however, seasonally the range of projections from different models is large. Ensemble projections indicate a decrease in September-October-November (SON) rainfall by 2090 while December-January-February (DJF) is set to increase by 2090, particularly in the north-east of the country.
- The proportion of total rainfall that falls in heavy events is projected to increase annually, but mainly in DJF and March-April- May (MAM).
- Projections indicate that maximum 1 and 5 day rainfall events may increase in magnitude in DJF and MAM.

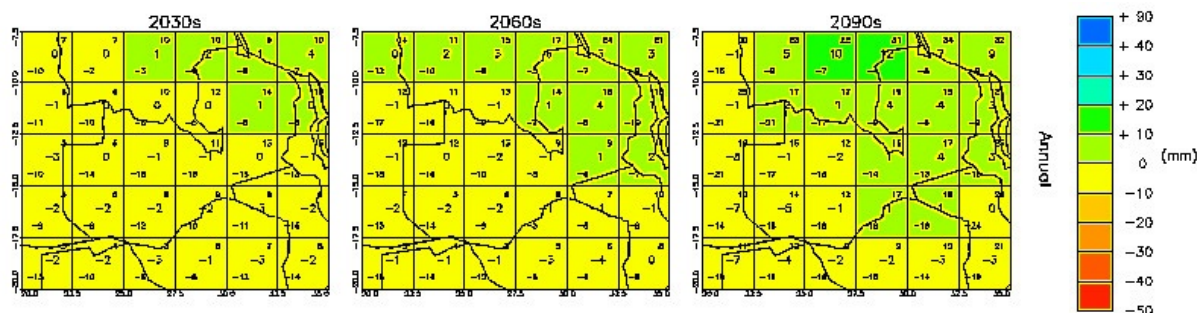


Figure 35. Spatial patterns of projected change in monthly precipitation for 10-year periods in the future under the SRES A2 scenario. All values are anomalies relative to the mean climate of 1970-1999 given in mm. In each grid box, the central value gives the ensemble median and the values in the upper and lower corners give the ensemble maximum and minimum.

3.4.2 Climate Change Impacts (GL1.2)

Community Well being

Climate change impacts will have a negative effect on the livelihoods of these people, as their resilience to shocks associated with climate change is very low. An example of this is the dry land farming commonly practiced. This form of subsistence agriculture is particularly dependent on seasonal rainfall, which is predicted to change over the next 30 years.

Biodiversity Conservation Status

Anticipated climate changes are likely to have an impact on the well-being of biodiversity conservation by reducing the amount of dry season water available in the ecosystem—in the case of a drought—reducing the amount of available habitat to species. This could result in habitat compression and degradation around remaining water sources. More erratic rainfall could also compromise biodiversity by affecting fire regimes, if more fuel is created in heavier rainfall periods, but longer hotter dry seasons might exacerbate late season hot fires.

3.4.3 Measures Needed and Designed for Adaptation (GL1.3)

Community Adaptation

Project benefits seek to reduce vulnerability of local communities to climate change through income diversification and improved agriculture practices. Vulnerability will be reduced through diversification and improved livelihoods with the introduction of a number of project-based activities designed to increase the resilience of local communities. Projected changes to the climate suggest that the dry season may become longer with reduced rainfall in the SON period, exacerbated by an increase in extreme rainfall events in the DJF period. Mean annual rainfall is projected to remain largely unchanged, however, the increase in the extreme rainfall events and the increase in mean annual temperature suggest that risks associated with droughts and or floods should be planned for using flexible, reactive and anticipatory management strategies. Additionally, while projections may be uncertain, in a semi-arid region such as the LCFP any social development project should account for the natural variability inherent in these systems.

Crop Failure Prevention

Due to the projected increases in temperature and the variable nature of precipitation in the future, crop failure threatens both food security as well as biological diversity. Agents within the project zone are more likely to exploit natural resources through poaching and charcoaling activities if their crops fail. In order to mitigate this, in the short term, agents will be encouraged to practice conservation farming methods such

as minimum tillage and moisture conservation practices. In addition, beekeeping will bring additional income streams and so will benefit sharing in the form of conservation fees.. Improved farming practices will thus help to buffer local agents against projected environmental change through diversification of livelihoods.

Social services

Education is vitally important for the uplifting of the local communities. Education will provide opportunities for climate change adaptation by providing opportunities to community members that go beyond subsistence agriculture. As such, the project proponents will use carbon financing to refurbish at least one local school that has fallen into disrepair. In addition to the school refurbishment, the project proponent will also refurbish a disused borehole that will provide much-needed water to and local community and a tree nursery.

Honey Production

The project proponent will support the establishment of a honey project by providing support to these communities. In addition, proponents will also provide access to both local and regional markets using transport networks established to support the eco-charcoal business. This activity will diversify household incomes and provide a buffer in case of climate induced agricultural shocks.

Biodiversity Adaptation

The main biodiversity adaptation of the project is securing the long term conservation of a vast wildlife corridor along a north-south rainfall gradient and ensuring access of wildlife to critical water sources.

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CM1.1)

Although the LCFP areas fall under the administrative structures of the local authority, traditional customary governance under chiefdoms plays a very important role. The traditional system involves the Chiefs, Indunas and the headmen. The system of Indunas and headmen is the main way of organizing local communities and involvement of the Indunas and headmen is a prerequisite for obtaining local consensus regarding developmental projects. The traditional system manages access to land and undertakes conflict resolution of civil cases. The traditional authority is complimented by formal political authority, which is vested in the area councilor and the area Member of Parliament.

The main ethnic groups found within the LCFP accounting for 80% of the population are Soli found in Rufunsa, Nsenga in Nyimba and Petauke, Kunda in Mambwe and Tumbuka and Bisa in Lundazi. The Chewa and Ngoni speaking people, though considered a minority and accounting for 20% of the population have quite a considerable amount of influence in the traditional and cultural practices in Nyimba, Mambwe and Lundazi. Households primarily identify as Christian.

In Eastern Province in general, more than 75% of households are poor (less than \$1.25 a day) and around 60% of the population is living in extreme poverty (Tembo & Sitko, 2013). Approximately 1.5 million people live in Eastern Province, and 87.4% reside in rural areas. The majority of households live without electricity or public water or sanitation services and are primarily reliant on subsistence agricultural activities. The average household landholding is 2.54 ha, with 1.86 ha under cultivation. The main subsistence crops include maize, cotton, and groundnuts, where harvested crops make up 64% of total household income. Other sources of income include off-farm income (21.9%), fruit and vegetable production (10.5%), and livestock sales (2.5%) (Tembo & Sitko, 2013).

Poverty and lack of alternative livelihoods in rural areas cause these rural populations to depend principally on smallholder agriculture for food and income, and on firewood for cooking and heating. The agriculture practiced involves poor farming practices such as burning of cropland and crop residues, and continual cultivation of fertility-mining crops. In urban areas, poverty and low incomes result in high dependence on charcoal for cooking and heating. With increasing population both in rural and urban areas, these situations translate into increased demand for agricultural land, firewood, and charcoal – resulting in forest exploitation and ultimately deforestation. These chains of events are illustrated in Figure 8 above. They are relevant now and for the foreseeable future.

Dependence on forests for livelihoods varies across households in the project area. Around 15% are not dependent on the forest for consumption or income benefits. At the same time, 84% depend on the forest for consumption, and 19% depend on the forest for income. The most common forest products collected for income, as reported by head of households, are charcoal, mushrooms, and fuel wood. The most common forest products collected for consumption are fuel wood, wood for poles, and fibers, followed closely by mushrooms (Huntington et al, 2016).

Average household size in the chiefdoms stands at 5.7 persons. Female-headed households account for 36%; Elderly households, defined as households where head of household is aged 65 years and older, accounts for 6.6%; While child-headed, defined as households where head of household is aged below 18 years accounts for 0.1 %. (Source: LCFP household registry)

Counting the most educated person in a household, the average number of years of education in the area is around 6.9. In around 1/5th of households, adults have no formal schooling. Another fifth of households have at least one member that has completed Grade 7. Wives have an average of 5 years of education, where 1/5th have no schooling, and 27% have completed at least Grade 7 (Huntington et al, 2016).

Zambian villages tend to be built around a few family groups. In the project zone 74% (3199) of households are related to the headman, and 69% (2205) of these households are in the same immediate family as the headman. A smaller percent of households are related to the chief. 12% (535) of households report being related to chief, and 37% (196) of those households are in the chief's immediate family. 64% (2790) of the sample is monogamously married and 6% (262) is polygamous. A slight majority of the sample (52%) has household land that is passed down from the husband's family, whereas 32% of respondents report that land the household farms is passed down through the wife's family (Huntington et al, 2016).

Community Resource Boards (CRB), made up from constituent Village Action Groups (VAG) have been registered by the Forest Department as Community Forest Management Groups (CFMG) and are the primary governance institutions managing REDD+ forests and ensuring the distribution of benefits. CRBs and VAG leaders are elected through special elections overseen by DNPW and stakeholders (like BCP or hunting concessionaires) every 5 years. CRBs fall under the authority of the Chief but also require input from District government, DNPW and other stakeholders.

1.1.5 Interactions between Communities and Community Groups (CM1.1)

Female heads of household are older, less likely to be married, and have smaller households. They also tend to have fewer assets and are poorer. Female heads of household also worry less about the lack of jobs and bad roads, and more about food scarcity. They report far lower levels of income from all sources (Huntington et al, 2016). It is important the LCFP supports this group.

About 50% of the population is under the age of 35 and qualify as youths. Demographically, young respondents are more likely to be married, have fewer children, and smaller households. Youth respondents are also more likely to live in the village in which they were born. As for assets, while younger household heads have fewer farming tools and livestock, they are more likely to have electronics (generally cell phones). With respect to development challenges, youth respondents are mostly concerned about a lack of jobs and the travel time to nearest market (Huntington et al, 2016). This grouping needs to be supported as well by the LCFP.

The CRBs and VAGs are the key community groups involved in REDD+ in the LCFP. There is considerable risk of elite capture or outright theft of benefits from REDD+ by these groups. This has been reported historically in conjunction with CRBs. The LCFP is taking precautions to guard against this kind of corruption by ensuring transparency in benefit distribution and project selection and implementation as well as ensuring the CRB members (i.e. the entire chiefdom) understands their role as watchdogs of the CRB.

1.1.6 High Conservation Values (CM1.2)

High Conservation Value	Nyani falls
Qualifying Attribute	Historical sites
Focal Area	Mwanya Chiefdom.

1.1.7 Without-Project Scenario: Community (CM1.3)

The without project community scenario is described in section 2.1.5, 2.1.11 and 2.2.3. Without the project communities will continue to rely on subsistence agriculture and other unsustainable practices unable to escape from the cycle of poverty. Deforestation will continue with subsequent loss of biodiversity further eroding the ability of the ecosystem to provide benefits to communities.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CM2.1)

Community Group	Smallholder farmers
Impact(s)	Conservation agriculture
Type of Benefit/Cost/Risk	Actual resilience to droughts and pests, diversification of crops
Change in Well-being	Diversity of crops, drought resilience, diversification of diet, improved health

Community Group	Smallholder farmers
Impact(s)	Beekeeping
Type of Benefit/Cost/Risk	Actual Improved income and diversification of income
Change in Well-being	Reduction in tree cutting from traditional methods of honey harvesting, support the propagation of bees otherwise threatened by use of chemicals and pesticides from crops, diversification of household source of income.

Community Group	Smallholder farmers
Impact(s)	Infrastructure projects
Type of Benefit/Cost/Risk	Benefits related to improved water access, access to education, access to health and support for agricultural businesses.
Change in Well-being	Reduced cost of living. Improved livelihoods.

4.2.2 Negative Community Impact Mitigation (CM2.2)

Potential negative impacts revolve around the possibility of conflicts arising out of perceptions of unfair benefit sharing. It is critical that benefit distribution is carried out in a transparent, participatory fashion that ensures the participation of all VAGs. Financial expenditures by the CRB and VAGs must be prominently shared with constituents. Decisions on where to invest benefits must be carried out in a fair manner.

4.2.3 Net Positive Community Well-Being (CM2.3, GL1.4)

Beekeeping, Infrastructure development, conservation fees and conservation agriculture have been implemented to assist communities with climate change adaptation as well as more general community development. Please see section 3.4 for further details.

4.2.4 High Conservation Values Protected (CM2.4)

The project will not have negative impacts on HCVs important to community wellbeing as these areas are all protected by the REDD+ designation.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (CM3.1)

Stakeholders who depend on illegal activities to support their livelihoods (i.e. poaching, illegal charcoal) will experience short term negative impacts from project activities that seek to curb these activities. At the same time however the project will provide alternatives to these activities that are capable of replacing this income. The positive impacts to stakeholders are increases in quality of life from livelihood and social services projects. For example, in the Lower Zambezi REDD+ Project, a five year repeat-round survey suggested that average household level income increased 450% over a 5 year period (Mint, 2017). This illustrates the types of gains which can be targeted through BCP's social impact delivery program.

The measures to mitigate negative impacts to poachers are driven by a combination of job creation, and livelihood alternatives. The project has created approximately 60 community jobs, mainly for Community Scouts (CS). These CS are hired through BCP support via CRB contracts and their role is to protect community resources, including wildlife habitat. The income from these jobs goes directly into the Chiefdom, and it is fair to say that poachers have an equal, if not advantage in applying for CS jobs. The CS role requires physical endurance, knowledge of bushcraft, and a willingness to spend one's work time in the outdoors. People with poaching experience are naturally inclined to be high performing CS with a change in mindset towards legally earned income and with the passing of training.

BCP's regular investment in in service training of CS serves as a deterrent to would be poachers. The information regarding recurrent training is shared with communities via BCP's community engagement team. These deterrents help would be poachers with decision-making tools to not invest effort into illegal activities with a higher risk of CS intervention.

A key part of the strategy employed is a heavy investment in projects which improve well-being. For example, investments in water, education and health services, as well as community transport (such as boats) provide households with indirect improvements in quality of life. Direct improvements in quality of life are provided through BCP's investment in scalable food security safety net and income diversification schemes such as honey and conservation farming.

A further measure to reduce negative impacts to poachers is BCP's use of 'Community Covenants' for community projects BCP invests in. These provide a mechanism by which the community can engage with and help would be poachers with social support as an incentive to stop poaching and risk undermining the Covenant or the CFMA which includes a definition of non-performance as poaching.

These measures above are also applicable to illegal charcoal producers. In addition, BCP has initiated a pilot Eco-Charcoal project which supplies legal, sustainable charcoal the market. The model is being implemented in Rufunsa District under LZRP, however this project model is scalable to Open Areas and operates under the principle of "Poachers to Gamekeepers". This principle is to provide illegal charcoal producers with a pathway to becoming legal charcoal producers. Currently, there are no restrictions on people joining the Eco-Charcoal project, and the vision in time is to offer it for expansion to community areas where there is demand, illegal charcoal, and a desire for alternative livelihoods.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

Stakeholders who depend on illegal activities to support their livelihoods (i.e. poaching, illegal charcoal) will experience short term negative impacts from project activities that seek to curb these activities. At the same time however the project will provide alternatives to these activities that are capable of replacing this income.

As described above, there are no other stakeholder groups in or outside of the project zone that are anticipated to be negatively impacted by the project (apart from groups engaged in illegal activities). In the event of negative impacts, these should be reported and resolved in accordance with BCP's grievance mechanism procedures.

Activities that have been undertaken to avoid / mitigate negative impacts on stakeholders include:

- Careful assessment of existing local conditions and stakeholder dynamics, prior to project implementation, as part of the initial project viability assessment and design process (which was thoroughly documented and shared at the time of our original validation);
- Significant stakeholder consultation prior to project implementation, as part of the Free, Prior and Informed Consent (FPIC) procedures followed, and as outlined in BCP's Standard Operating Procedure (SOP) for FPIC. This included consultation with relevant government authorities and

community leaders, to identify any potential project impacts and to ensure that the project would be designed to avoid negative impacts, and to ensure that adequate feedback and reporting mechanisms would be in place between BCP and local stakeholders, to ensure that any problems, concerns, questions or updates could be communicated effectively between the implementing parties of the project, to ensure their resolution in an effective and timely manner;

- BCP maintains regular community presence and undertakes regular assessments of program activities and impacts (as described in section 4.3, below, in greater detail), to ensure that any potential negative impacts would be identified, reported, documented, and addressed/resolved within in a timely manner;

BCP additionally works closely with local stakeholder governance institutions and local government partners, in order to promote a truly community-driven project implementation model, with the support and knowledge of local government and local traditional leaders. This model is intended to ensure that the project is implemented in ways that are locally appropriate, to minimize any potential local risks, and ensure that any potential problems are identified, reported and responded to appropriately.

4.3.3 Net Impacts on Other Stakeholders (CM3.3)

As detailed in section 4.3.2, there will be no net negative impacts on other stakeholders.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

BCP commits to monitoring key community impacts from the LCFP, in the short and long term. Table 18 lists key indicators that BCP will track to assess the community impacts of the LCFP, as well as the mechanisms and timelines for these assessments to take place.

Table 18. Community indicators being monitored.

Indicators	Mechanism	Frequency
Number of community stakeholders + beneficiaries Including a gender disaggregation, to track the number of women impacted by the project	Stakeholder / Beneficiary Registry	5 years (baseline = 2017)
Key socio-economic indicators of community dynamics and conditions, including:	Social Monitoring Program – Impact Evaluation Surveys	5 years (baseline = 2020)

- Household incomes - Household wealth indicators, including assets and self-reported wealth rankings, going beyond “income” - Livelihood dependencies - Specific dependencies on forest resources - Agriculture indicators tracking dependencies on subsistence agriculture (household consumption) as well as commercial agriculture / agriculture for income - Ethnicity (including to track minority ethnic status) - Self-reporting of participation in LCFP activities, and perceptions of “benefits” from the project	(qualitative and quantitative data collected to assess key socio-economic indicators among a representative sample of project beneficiaries)	
Project-specific impact assessments, tracking impacts of specific livelihoods / deforestation mitigation activities on specific (targeted) well-being indicators for community participants (different for each livelihoods project)	Social Monitoring Program – Project-Specific Impact Evaluation Surveys (qualitative and quantitative data collected to assess key socio-economic indicators among a representative sample of project beneficiaries)	TBD / Different for different projects. Goal = Minimum every 2 years, beginning from project implementation
Annual Reports, highlighting: - Number of project stakeholders and beneficiaries, including breakdowns by gender - Value of funds distributed to communities, through the program - Community infrastructure supported by the program (including: number of infrastructure, type of infrastructure, locations, and who are / how many beneficiaries) - Direct beneficiaries / impacts from Deforestation Mitigation and Livelihoods projects: - What projects are supported, where (including	Ongoing Monitoring and Evaluation (M&E) data-collection and reporting - Quarterly reports (until 2019) - Annual reports (present – ongoing)	Ongoing / quarterly and annual data collection – with reports compiled on a minimum annual basis (quarterly from present to 2019, under the USAID-funded CFP)

maps); ◦ How many beneficiaries (including gender breakdowns, indicating specific impacts on women); ◦ Qualitative and quantitative descriptions of impacts on community beneficiaries; ◦ Maps / descriptions of links to conservation activities (e.g.: where are the activities taking place, in relation to protected forest areas? Does conservation data (patrol data, # of arrests / apprehensions, GIS/RS forest cover data) show any link between community and conservation impacts? • Local employment • “Success stories” highlighting specific program impacts on specific individuals’ lives • Narratives and qualitative data illustrating project impacts on “poorest of the poor” and that impacts are being shared with beneficiaries from a diversity of ethnicities, gender, age and socio-economic status (to show that project impacts are reaching intended beneficiaries).		
Monthly Reports from Community Institutions, tracking key conservation data (number / location of forest monitoring patrols, and any instances of deforestation), and community engagement data (number of stakeholders sensitized)	CRB Operational Reports	Monthly

It should be noted that under the USAID-funded CFP, scheduled to run from 2017-2019, BCP has developed a Monitoring and Evaluation (M&E) plan that meets USAID specifications (which are generally considered to be a high standard) to track LCFP impacts on local communities and biodiversity, on a regular basis. BCP already therefore has data collection, tracking, monitoring and reporting systems in place, to ensure our compliance with USAID standards for M&E. This systems are expected to continue even after the USAID-funding under the CFP ends, to allow us to continue tracking key indicators of program impacts, on a regular (minimum annual) basis.

BCP’s deforestation mitigation projects implemented under the LCFP are all collectively designed and intended to achieve meaningful and measurable conservation outcomes (forest protection within identified REDD+ zones) and community impacts. In order to achieve our goals, sustainably, in the long term, BCP aims to achieve the following:

- Widespread impacts, to as many community stakeholders / beneficiaries as possible;
- Meaningful and measurable impacts, as outlined in the table, above;
- Targeted benefits for “those who need it most” / “poorest of the poor” – as these are the people our program most needs to engage with, in order to meaningfully address poverty, marginalization and lack of access to sustainable alternatives, as contributing drivers of deforestation;
- Non-discriminated impacts, targeted to reach those who need it most, including members of minorities and/or marginalized groups, to ensure that local development takes place in a sustainable and meaningful manner.

The collective sets of data, studies and information available to describe current stakeholder community dynamics, as listed in Section 2.1.6 and 4.1.1, illustrate that the vast majority of LCFP stakeholders can be considered to be among “the poorest of the poor,” by virtue of living in poor rural communities in Zambia, a developing country with high, documented levels of rural poverty, high rates of population growth, strong dependency on natural resources and subsistence agriculture for livelihoods, and lack of access to social services and infrastructure – all basic indicators of well-being. The baseline survey results additionally illustrate that the vast majority of project stakeholders can be considered to be poor, rural households, heavily dependent on natural resources and subsistence agriculture for their livelihoods.

The LCFP is designed to positively impact community stakeholders, and to bring significant positive impacts to “those who need it most,” in meaningful, measurable, and documented ways. It is the function of our M&E systems, therefore, to ensure that BCP is collecting data that will allow us to track our impacts and achievements, in a credible, defensible and documented way. This data will be useful for tracking and documenting impacts, as well as identifying “lessons learned” and informing management decisions, in line with BCP’s “adaptive management” approach to REDD+ implementation.

It is worth noting that BCP has a credible track record for being able to deliver significant, meaningful, positive impacts to local community stakeholders from REDD+: the latest data from a 5-year impact evaluation / social monitoring follow-up survey from BCP’s other verified REDD+ project – the pilot LZRP taking place in Rufunsa – documents that BCP *has* the ability to channel significant, positive impacts to local community stakeholders through REDD+. We intend to replicate this model for success, to similarly benefit community stakeholders of the LCFP.

4.4.2 Monitoring Plan Dissemination (CM4.3)

All verification documentation is made available to communities and other stakeholders. Project activities are discussed at community consultation meetings, reported in newsletters and updates posted on the BCP website.

4.5 Optional Criterion: Exceptional Community Benefits

4.5.1 Exceptional Community Criteria (GL2.1)

The LCFP intends to achieve “exceptional community” impacts according to two / both criteria. First, it is true that for the LCFP, communities are recognized as having management rights to land in the project area, and also rights to claim that their activities will generate / cause the project’s climate, community and biodiversity benefits. This is confirmed in Sections 2.5 and 2.1.8, above, which, document that community stakeholders are recognized -- in the Laws of Zambia, as well as according to local customs / culture / traditions -- as having the rights to manage land in the project area, and to undertake activities to generate the projects’ impacts and benefits.

Second, in addition to this, the project zone for the LCFP is taking place in Zambia, which is widely recognized as a “developing” country. According to the UNDP Human Development Report 2016 (http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/ZMB.pdf), Zambia is a “medium human development” country, with a Human Development Index (HDI) value of 0.579, ranked at 139 of 188 countries/territories -- a rank shared with Bangladesh and Ghana. Furthermore, as sections 2.1.6, 4.1.1 and 4.1.4 describe in further detail, numerous studies and data sets -- including the recent Baseline Survey among project stakeholders within the project zone, undertaken in 2015 -- document that the LCFP is impacting stakeholders living in areas of the country that are characterized by primarily rural and poor populations, with high levels of dependency on diminishing natural resources, and other additional indicators of relatively low well-being; who stand to be significantly and meaningfully impacted by this project, especially when compared to the potential without-project scenario that is described in Section 4.1.4.

4.5.2 Short-term and Long-term Community Benefits (GL2.2)

The project activities in section 4.2.3 outline all the positive community impacts. Short term impacts have already become available to communities in the form of payment of conservation fees and investment in new infrastructure projects. Long term benefits will come from long term expansion of beekeeping and conservation farming activities. In addition, conservation fee / Revenue share payments will contribute annually to the long term development of infrastructure in these communities.

4.5.3 Community Participation Risks (GL2.3)

Section 2.5.3 (FPIC) describes how all members of the community have been engaged and are allowed to participate and benefit from the project. The vast majority of community members are smallholders and all benefit through community participation and through a participation process are not at risk of not benefitting. Households are registered with the project to ensure their participation and wellbeing are tracked. As part of the CFMA, communities are also obliged to report on projects rolled out with conservation fees to ensure that there is no “elite capture” or misappropriation of funds.

4.5.4 Marginalized and/or Vulnerable Community Groups (GL2.4)

Community Group 1	Small holder farmers
Net positive impacts	Improved incomes from alternative livelihoods. Improved access to critical infrastructure (water, health, education). Improved farm/crop management and linkages to markets.
Benefit access	Ensure transparency in benefit distribution and planning for long term development.
Negative impacts	Ensure participation of woman and elderly headed households in project activities. Ensure their participation in project selection and siting.

4.5.5 Net Impacts on Women (GL2.5)

The LCFP recognizes the importance of mainstreaming gender in the promotion of sustainable use and management of natural resources under a REDD+ project. The project will work to mainstream gender in its activities with the overall goal to ensure that women and men equally access, participate, and benefit from natural resource management projects, and that gender inequalities in activities and outcomes of its works are reduced or eliminated.

To do this we will implement targeted activities to challenge gender stereotypes, support women to take up leadership roles in natural resources, and providing economic opportunities for women, through

livelihood empowerment support and linkages to stable markets. In addition our M&E system is gender responsive and helps the project to collect gender disaggregated data, track gender results, assess impacts as well as monitor access, participation, and benefits among males and females and incorporate remedial action that redresses any inequalities in project implementation.

4.5.6 Benefit Sharing Mechanisms (GL2.6)

The project's benefit sharing model is based on the principle that benefits from community-based natural resource management utilized on communal projects have more advantages over systems that only allow revenue to be spent on individual community members. Using community-based natural resource management as a tool for local area development gives community members the flexibility and decision-making power over the use of their natural resources. This approach supports the building of strong local governance institutions that facilitate the delivery of communal benefits and encourage wide participation in decision-making.

BCP is aware that the potential revenue from REDD could be substantial and in some instances exceed the revenue from hunting concessions that the communities are accustomed to managing. In this regard, BCP has worked with participating communities and the Department of National Parks and Wildlife to strengthen financial systems for managing community revenue from natural resource management.

BCP developed guidelines for participating communities to consider, these guidelines apportioned the revenue from REDD, ensuring that communal projects are allocated the largest portion of revenue. Community consultations to review and revised these guidelines were held starting at the lowest governance structure; the village action group, up to the community resources board. Most important, the benefit sharing guidelines recommend that communal projects selected by the Village Action Groups are presented to the Community Resources Board where they are consolidated and the final decision on communal projects to implement from revenue generated from REDD is made.

The basic steps of the process are as follows:

1. Community General Meeting:

Community meeting for all community members over the age of 18 is convened by BCP and the Community Resources Board where a cheque for the revenue from REDD is presented to the community.

2. Village Action Group Meetings:

The Village Action Group committee convenes a meeting with all members of the community over the age of 18 to review community needs and compile a list of communal projects to implement. Budgets are developed at Village Action level and these along with records of meetings held are submitted to the Community Resources Board.

3. Community Resources Board Meeting:

A board meeting for the Community Resources Board is convened at which representatives from the individual village action groups present to the board their list of selected projects and budgets. Using PLA tools, the board considers each selected project on its merits and perceived benefits to the community. The board makes a final decision on which communal projects to implement after which the final decision made is communicated to the Village Action Groups.

BCP has also worked with Community Resources Boards to develop bylaws concerning the use of funds from REDD, these bylaws require Community Resources Boards to stick to approved budgets and account properly for the expenditures related to all approved projects. With the successful implementation of some community projects, Community members are showing more confidence in the system to deliver development in their villages.

4.5.7 Benefits, Costs, and Risks Communication (GL2.7)

Throughout the design, establishment and implementation of LCFP, the community has been integral part of all processes and activities. BCP has implemented a continuous, community-oriented communication strategy that has included information about predicted and actual benefits, costs and risks associated with implementing a REDD+ project.

4.5.8 Governance and Implementation Structures (GL2.8)

Planning and decision making process starts at Village Action Group level where meetings which include all community members of 18 years and above are consulted. Whilst key decisions relating to LCFP are made at Community Resources Board, extensive consultation is done at village level through the Village Action Group. Decisions made at this level with consultation form the village action groups and traditional leaders include the approval of the project zones and community forest reserve boundaries, management and by-laws and REDD+ revenue distribution.

These local governance structures; Community Resources Boards are in line with the Wildlife Act No. 14 of 2015 and have been duly recognized as Community Forest Management Groups in line with the Forest No 4 of 2015 of the Laws of Zambia. BCP has signed MoUs, and agreements with each participating community through the Community Resources Board, which obliges BCP to generate and trade in carbon offsets on behalf of the community for 30 years.

4.5.9 Smallholders/Community Members Capacity Development (GL2.9)**Capacity building for local Governance Structures.**

Representatives from all participating Community Resources Boards were involved in the project design process. In addition, the representatives from these Community Resources Boards participated in a four-day training course on Governance and Accountability as well as a three-day course in Financial Management led by the Department of National Parks and Wildlife and Forest Department.

Capacity building to smallholders / community members on project implementation.

Additional trainings have been provided to small-scale farmers / community members in relation to the implementation of all project activities. This includes training on improved governance; management and operation of water supply facilities; improved agriculture; beekeeping.

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (B1.1)

The Project zone mostly falls within the Luangwa Valley and in the hills either side of the valley. Within the project zone there exists a complex mosaic of Mopane, Miombo and Munga woodlands (broadleaved woodlands), with riparian forests occurring along the Luangwa and other major rivers. Seasonally flooded grasslands and oxbow lakes lined with Ebony (*Diospyros* sp.) and *Commiphora* sp. thickets are scattered along the length of the Luangwa river. This is prime habitat for wildlife and many different species of micro and macro fauna are supported. These GMAs along the Eastern and Western banks of the Luangwa river have varying levels of wildlife remaining. Those areas with well managed hunting blocks typically have an abundance of wildlife with stable populations and bell-shaped demographic curves. Unfortunately poorly managed hunting blocks are often sparsely populated with game species which have skewed demographic curves, with few of all age classes existing.

Other activities, besides small scale agriculture and tourism, taking place in the project zone, notably hunting concessions, have left the project zone with varying levels of faunal activity. The project zone falls within the Game Management Areas (GMAs) controlled by the Department of National Parks and Wildlife (DNPW). They also control the hunting concessions that fall within the GMAs. As a result of varying degrees of effective management, some areas of the project zone are all but devoid of the faunal component of the system.

Primary threats to the biodiversity of the project zone are a burgeoning subsistence agricultural industry and population growth. Increased people entering the project zone seeking land for agricultural practice. This directly results in increased deforestation and indirectly in increased illegal harvesting of animal products.

5.1.2 High Conservation Values (B1.2)

High Conservation Value attributes are listed in the table below (Source Timberlake *et al* 2018)

High Conservation Value	Area with endangered large mammals Area of High Plant Diversity (Timberlake <i>et al</i> 2017) Kariba – Luangwa Centre of plant endemism Area of High large mammal diversity
-------------------------	--

	Area of large mammal endemism Area with high or important mammal populations Area important for mammal migration or movement Area of high species diversity Vulnerable bird species Area of high species endemism
Qualifying Attribute	Endangered Large Mammals: <i>Giraffae thornicrofti</i>, <i>Connachaetes cooksoni</i>, <i>Kobus senganus</i>, <i>Cercopithecus moloneyi</i>, <i>Hippotragus equinus</i>, <i>Diceros bicornis</i>, <i>Lycaon pictus</i>, <i>Panthera leo</i> 1350 plant species recorded 8 endemic woody species Endemic large mammals: <i>Giraffae thornicrofti</i>, <i>Connachaetes cooksoni</i>, <i>Kobus senganus</i>, <i>Cercopithecus moloneyi</i> High or important mammal populations: Primates, bovids, carnivores, <i>Hippopotamus amphibius</i>. Locally high concentrations of bats in mopane and miombo woodlands High species diversity: Plants, large mammals, birds Vulnerable bird species: <i>Bucorvus leadbeateri</i> High endemism: Plants, antelope, giraffe, monkeys.
Focal Area	LCFP project area - 12 different REDD zones within Game Management Areas. Private Ranch

5.1.3 Without-project Scenario: Biodiversity (B1.3)

Ongoing uncontrolled deforestation will result in habitat loss coupled with a decrease in species diversity, population decline and possible local extinction of vulnerable species. As deforestation spread, so does poaching and human-wildlife conflict.

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (B2.1)

Biodiversity Element	Habitat connectivity/Wildlife Corridors
Estimated Change	The project will maintain a vital section of intact forest between all along the Luangwa river to the Lower Zambezi ecosystem
Justification of Change	The project area is in excess of 1 million ha (See project area location) and will maintain critical threatened forest habitat in one of Africa's last remaining wilderness areas namely the South Luangwa National Park and Lower Zambezi National Park

Biodiversity Element	Reduction of poaching
Estimated Change	Reduce poaching incidence in the project area resulting in healthier wildlife populations
Justification of Change	Increased anti-poaching patrols, aerial monitoring, employment of scouts, communities signing CFMAs and receiving conservation payments based on decrease in poaching activity.

Biodiversity Element	Increase in wildlife populations
Estimated Change	Currently some areas have denuded wildlife populations due to heavy poaching
Justification of Change	Increased anti-poaching patrols, aerial monitoring, employment of scouts, communities signing CFMAs and receiving conservation payments based on decrease in poaching activity.

Biodiversity Element	Maintenance of endangered species populations
Estimated Change	Reduce poaching incidence in the project area resulting in healthier wildlife populations
Justification of Change	Increased anti-poaching patrols, aerial monitoring, employment of scouts, communities signing CFMAs and receiving conservation payments based on decrease in poaching activity.

5.2.2 Mitigation Measures (B2.3)

There will be no negative impacts on biodiversity, the project is specifically designed and implemented to bolster the protection of the Game Management Areas.

5.2.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

The project aims to protect in excess of 1 million ha of intact forest in one of Africa's last remaining wilderness areas. The HCV of this activity is listed in section 5.1.2 above. The area has exceptional biodiversity as described in Timberlake *et al.* 2018. The without project scenario will see this HCV follow the current trend of GMAs in Zambia – ongoing decline of wildlife populations and habitat loss³⁰.

5.2.4 High Conservation Values Protected (B2.4)

The project is specifically designed to protect the Luangwa and Lower Zambezi ecosystem and will not impact negatively on any HCVs.

5.2.5 Species Used (B2.5)

Potential species that will be used in conservation farming activities are listed in section 5.2.7 below.

5.2.6 Invasive Species (B2.5)

The project will not use any invasive species for project activities.

5.2.7 Impacts of Non-native Species (B2.6)

³⁰Lindsey, P.A., Nyirenda, V.R., Barnes, J.I., Becker, M.S., McRobb, R., Tambling, C.J., Taylor, W.A., Watson, F.G. and t'Sas-Rolfes, M., 2014. Underperformance of African protected area networks and the case for new conservation models: insights from Zambia. *PLoS one*, 9(5), p.e94109.

Species	<i>Zea mays</i>
Justification of Use	<i>Sorghum</i> spp. Is a potential alternative but has been replaced in favour of maize as it produces bigger yields and not prone to bird predation.
Potential Adverse Effect	No negative evidence documented it is used extensively in Zambia

Species	<i>Cajanus cajan</i>
Justification of Use	No known local alternative, it is a nutritious legume.
Potential Adverse Effect	No negative evidence documented it is used extensively in Zambia

Species	<i>Mangifera indica</i>
Justification of Use	<i>Parinari curatellifolia</i> is a potential alternative and is locally utilised but does not have the same production and fruit size and is slow growing.
Potential Adverse Effect	No negative evidence documented it is used extensively in Zambia

Species	<i>Carica papaya</i>
Justification of Use	No known local alternative with fruit that size.
Potential Adverse Effect	No negative evidence documented it is used extensively in Zambia

Species	<i>Moringa oleifera</i>
Justification of Use	There are possibly alternative local species that have the same nutritional values, but the particular species is particularly fast growing.
Potential Adverse Effect	No negative evidence documented it is used extensively in Zambia

5.2.8 GMO Exclusion (B2.7)

The project does not undertake A/R activities that will be used for credit generation and does not use GMOs.

5.2.9 Inputs Justification (B2.8)

Project activities are only involved in conservation farming activities and no fertilizers or pesticides are provided by the project proponent nor is their use encouraged in conservation farming which aims to promote organic farming activities.

5.2.10 Waste Products (B2.9)

Not applicable, project activities will not produce any harmful waste products.

5.2.11 Offsite Biodiversity Impacts**5.2.12 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2)**

There are no anticipated negative biodiversity impacts arising from the implementation of the LCFP REDD+ project. We envisage that the implementation of the LCFP will reinforce and support the conservation status, and hence the biodiversity within the project zone. The LCFP will increase the conservation value of the area and strengthen the project zones status as a “buffer Zone” to the South Luangwa National Park, an area of National Biodiversity significance. As expected the project will assist and facilitate the protection of the natural resources within the LCFP and all the Chiefdoms within the LCFP area. The Chiefdom Operations/Conservation Plans that are developed in conjunction with each respective chiefdom will ensure a positive impact on the forests and hence the biodiversity of the area.

5.2.13 Net Offsite Biodiversity Benefits (B3.3)

No unmitigated negative offsite impacts on biodiversity will take place, the project is specifically designed to enhance biodiversity benefits over an extensive area.

5.3 Biodiversity Impact Monitoring

5.3.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

Monitoring activities are outlined in the Community Forest Management Agreements (Appendix A), species monitored are IUCN listed species particularly Lion, African Wild Dog and Elephant.

Monitoring takes place across the entire project area. Monitoring is done aerially or on foot.

5.3.2 Biodiversity Monitoring Plan Dissemination (B4.3)

The monitoring plan forms part of the CFMA (Appendix A) and as such has been distributed to all relevant stakeholders for endorsement. Aerial and Ground monitoring is tracked and the results made available to stakeholders.

5.4 Optional Criterion: Exceptional Biodiversity Benefits

5.4.1 High Biodiversity Conservation Priority Status (GL3.1)

The project qualifies for Exceptional Biodiversity Benefits with the presence of the African Wild Dog (*Lycaon pictus*) listed as Endangered by the IUCN.

5.4.2 Trigger Species Population Trends (GL3.2, GL3.3)

Trigger Species	<i>Lycaon pictus</i>
Population Trend at Start of Project	The Wild Dog Action Plan (2007) estimated <150 individuals for Eastern Zambia, but recent surveys by Zambia Carnivore Group (2016) identified 170 individuals in the Luangwa system. The trend is currently upwards.
Without-project Scenario	Declining population due to habitat and prey loss and snaring.
With-project Scenario	Maintain existing habitat. Increase in prey population due to reduced poaching thanks to increased monitoring and patrolling, strengthening capacity of local authority. Reduction in mortality due to removal of snares. Ultimately stable to increasing

	population.
--	-------------

6 APPENDICES

Appendix A Community Forest Management Agreements and Communication Agreement. Confidential

Appendix B Approval letters. Confidential

Appendix C Support letters. Confidential.

Appendix D Adaptive Management Plan. Confidential

Appendix E BCP Community Engagement SOPs and Monitoring Systems. Confidential

Appendix F Socio-economic baseline study. Confidential

Appendix G Emissions Reductions Calculations. Confidential

Appendix H VCS Non-permanence risk assessment. Public

Appendix I Spatial Data. Confidential

Appendix J Biodiversity Monitoring Results

Appendix K Analysis of agents, drivers, and underlying causes of deforestation. Internal Report.
Confidential.

Appendix M Human Resource Manual and Grievance Procedure. Confidential

Appendix O Cumulative Deforestation Model. Confidential

Appendix P Biomass and Soil Sampling Protocol. Confidential

Appendix R Leakage Monitoring Protocol. Confidential